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Automated Pill Dispenser

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Team 15
Automated Pill Dispenser

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Professor Nassersharif	Sponsor

University of Rhode Island

THINK BIG WE DO℠

MCE 401: Capstone Design
Department of Mechanical Engineering
Preliminary Design Report
2021-2022

Abstract

Team 15 was given a task from Professor Nassersharif. He tasked us with creating an automated pill dispenser that would improve people's lives who have timed medications. This project was given the past few years but our team has redesigned and improved that dispenser. We look to improve on previous designs and widen the market value of our device for customers on the go or sheltered at home. In the Fall 2021 semester we developed concepts for the problem, started to create a prototype to prove how this device can improve lives, and also have used outside data and literature to improve our design. During the Spring 2022 semester the prototype was completed, then run through several tests which led to some redesigning of certain components until the final showcase came to display and demonstrate to the public.

Our design uses a simple application that will communicate with the dispenser. This will tell the device which pill in storage is needed and to align the vacuum system for pick up. Dropping it down the shoot into the correct timed slot to await the customers' needs. This will fill the timed storage box and allows the customer to take their pills on the go and still enjoy life and not have their pills unorganized while out. Team 15 was very proud of the final design of the dispenser and the advances made on the project even after four years of developement.

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List of Acronyms

1. CAD - Computer Aided Design
2. QFD - Quality Function Deployment
3. CDR - Critical Design Report
4. POC - Proof of Concept
5. ROI - Return on Investment

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1 Introduction

Sixty-six percent of adults worldwide use prescription drugs or some sort of medication. However, not all take this vital component of health care seriously. In 2018, Nihar Patel of USciences conducted research that shows that twenty-five percent of medications aren't refilled by their users, and fifty percent of patients are non-adherent. This non-compliance brings great health risks to people all over the globe. Whether it's high drug costs or the fear of adverse events occurring after taking certain pills, the number of reasons for non-adherence is on the rise.

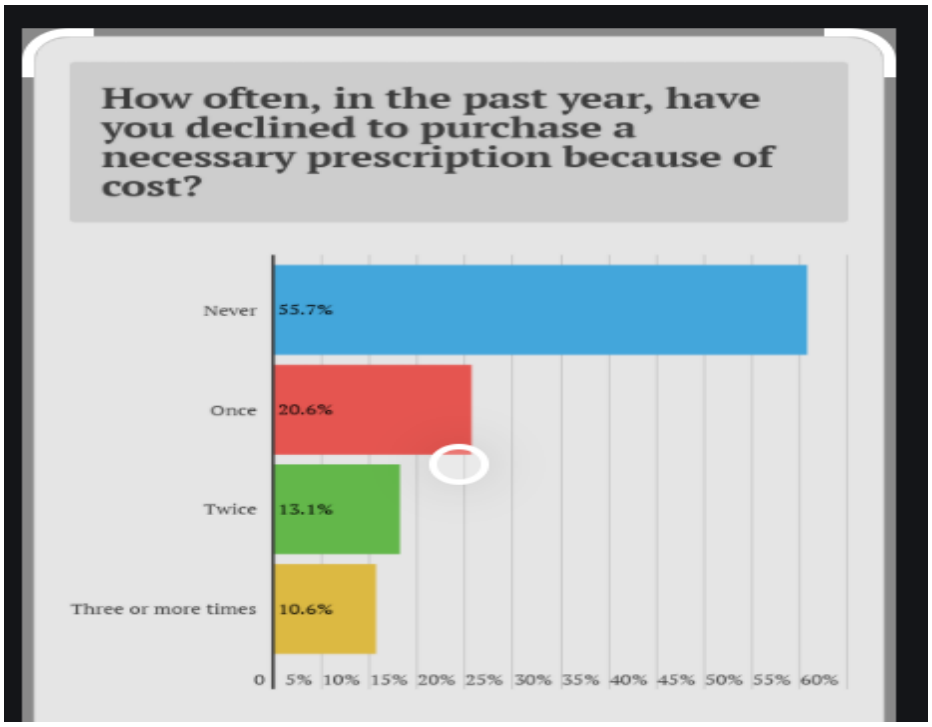

Figure 1: Based on an online survey of over 1,000 U.S. adults conducted by PawnGuru, an online marketplace that regularly surveys its community on a range of topics affecting low-income and under-banked Americans.

Dr. Bahram Nassersharif of the University of Rhode Island has pitched the idea of an automated pill dispenser that will make it easier for prescription drug users to remember and take their pills and stay healthy. Moreover, if the pains in pill taking are eliminated, the health of people across the globe will significantly improve.

1.1 About the Sponsor: Dr. Nassersharif

Dr. Bahram Nassersharif is a distinguished university professor at the University of Rhode Island. He is in the department of Mechanical Engineering as well as Industrial and Systems Engineering. Dr. Nassersharif received his Ph.D. at Oregon State University in 1983. His research interests include engineering design, nuclear systems, safety engineering and simulation and modeling.

1.2 Current Situation

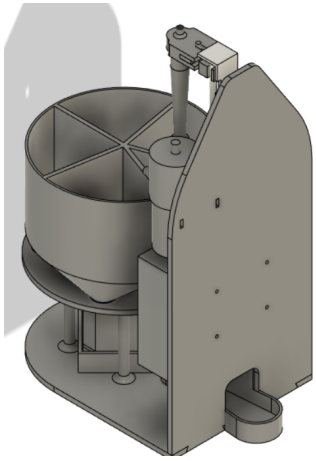

Figure 2: 2020-2021 Pill Dispenser

After looking at last year's group's design, group 15 took on the challenge of making a more advanced design. With last year's group's model as a guide, group 15 will take the next step in making the inconvenience of pill taking cease.

Along with making the device user friendly and operable by the elderly, the price of the device is going to be brought down. While last year's design solved many problems dealing with inconvenience, the biggest problem Team 15 will be trying to face is the trouble of taking medication on the go. Last year's device was mainly a table top design and didn't have an easy way to take medication when not at home.

1.3 Problem Definition

Millions worldwide commit acts of non-adherence when it comes to taking their prescription medicine. Whether it's because of inconvenience, laziness or simply forgetting, people evading taking their pills are putting themselves at serious health risks. Obviously, doctors can't force their patients to take the medicine they prescribe to them. The only thing they can do is encourage taking the necessary dosage of whichever medication is given to them. However, people aren't prescribed medication without reason.

There are many methods and incentives for people to take their pills. Some people set alarms or prepare a schedule so they never miss a dose. The problem is that sometimes there's too much to remember. Those with many disabilities or illnesses are usually required to take multiple medications. This can make it harder for them to remember everything every day.

The goal of designing an automated pill dispenser is to make being healthy not a chore or monotonous. People should not have to struggle day after day in order to maintain a healthy life. With the help of the automated pill dispenser this can be achieved.

1.3.1 Original Problem Definition and Purpose

The purpose of this project was to design an automated pill dispenser. People all across the world struggle daily to remember to take their pills. Additionally, people ignore taking their medicine due to laziness and the inconvenience of having to schedule their pill taking.

To define the problem that the group is working hard on trying to solve is that people with prescription medicine are having difficulty remembering when to take their pills on a daily basis at various time intervals. This problem is exacerbated for the elderly. Moving forward, a better solution is needed to help everyone remember and keep track of their pills and dosages.

1.4 Expectations, Objectives, Scope

Team 15 expects to produce an automated pill dispenser that aids people around the world in taking their medications promptly. The main objective is to make the device as convenient as possible to the point where laziness is no longer an excuse from those unwilling to take their medication. The device is meant to dispense a plethora of pills as well as a variety of medications. Understanding that pills come in all shapes and sizes, Team 15 seeks to accommodate any type of pill that the user throws its way. Focusing efforts on finding a design that can assist with customers' active or sedentary lifestyles. Also, with unstoppable accuracy and precision in dispensation, this device will be desired amongst all who struggle to take their medication on a daily basis.

2 Patent Search

The patent search was conducted for the reason of making sure that the prototype that our group is designing does not infringe on any patented design that is currently recognized. During the development of any product or prototype, performing sufficient research into patents that currently exist is extremely important and detrimental to any product design team. The design team must make sure that any prototype that enters the manufacturing stage, does not infringe on any recognized patent.

2.1 Patent Number: US11,116,699
Date of Publication: 05/03/2016
Inventor(s): Paul Hines

Abstract: A medication dispensing system is provided with several cooperating components which help to accommodate the responsible handling of medications. These systems include a user interface to receive input from the user and to present appropriate information/notices/alarms. Further, physical controls and multiple sensors are provided to detect tampering, misuse, and attempts to inappropriately access medications contained within the dispensing system. This misuse can thus be detected, and appropriate notifications can be provided, where appropriate. Further, by adding network communications capabilities, notices can be widely communicated to others, and operation of the medication dispensing system can be controlled and/or monitored remotely.

Relevance: Medication dispensers often have issues or complications with both security and usability. In addition, very few extend their feature set to be enabled to store timing of events and allow that data to be transferred to the internet or remote server. Additional security features to prevent tampering of medications, undesired access and misuses are a welcome addition to any pillbox. Medication dispensers can be connected to remote servers allowing data (e.g. time when taken, missed medications, relevant statistics) and notifications (e.g. text, emails, phone calls) to be sent to interested parties (e.g. family caregivers, home health care, hospitals, insurance agencies, pharmacies, pharmacy benefit managers). This patent opened our group's eyes to the issues that our early stage prototype had. This also geared us in the direction of allowing caregiver access for an improved safety step as they would be able to help monitor the users activity in taking pills so they are up to date on dosages.

2.2 Patent Number: US11,104,503
Date of Publication: 03/25/2016
Inventor(s): Bradley S. Philip, Mark Woloszyk and James Stelzer

Abstract: A pill dispenser including a first primary ramp and a second primary ramp, which is

opposite to the first primary ramp. Also included are a first secondary ramp and a second secondary ramp, which is opposite to the first secondary ramp. The first and second secondary ramps are both between the first and the second primary ramps. The pill dispenser further has an outlet including outlet sidewalls and an aperture through which pills are dispensed from, or loaded into, the pill dispenser. The outlet side walls extend to the aperture from the first primary ramp, the second primary ramp, the first secondary ramp, and the second secondary ramp.

Relevance: This patent teaches the importance of pill orientation as it moves through the device to the user. This patent talks about how current pill containers and dispensers are suitable for their intended use, but they are subject to improvement. For example, a pill dispenser that controls the speed at which pills are dispensed out of a pill container would be desirable. A pill dispenser that orients pills into a predetermined orientation that best arranges the pills to be counted with a mechanical or electronic counter as they are dispensed or loaded would also be desirable. The present teachings advantageously provide for a pill dispenser exhibiting these advantages, as well as numerous others. This is something our group will take into consideration as we continue to improve our design in the future as this can have serious implications to how our prototype performs. This can affect our dispenser's accuracy and its ability to not deform any pills throughout the dispensing process.

2.3 Patent Number: US11,020,320
Date of Publication: 07/21/2020
Inventor(s): Susheel Panda, **Gaurav Kumar Verma**, **Nithin Naga, Venkata Yalamanchili**, **Srikanth Raavi Venkata, Rafi Shaik Mahammed**, **and Abdel Hamad**

Abstract: A pill dispenser includes a housing and a slidable drawer with a dispenser cup therein. A storage cup is fixed within the housing. The pill dispenser also includes a movable probe, a flexible tip on the probe, and a suction pump in fluid communication with the flexible tip. The pill dispenser may be configured for and/or methods of operating the pill dispenser may include rotating the probe a position above the storage cup, translating the probe downward into the storage cup, and activating the suction pump. While the suction pump is active, the probe is translated to the position above the storage cup, rotated to a position above the dispenser cup,

and translated downward into the dispenser cup. The suction pump is deactivated when the flexible tip is within the dispenser cup.

Relevance: This patent is for a pill dispenser that includes a housing that defines a lateral direction, a transverse direction, and a vertical direction. The lateral direction, the transverse direction, and the vertical direction are mutually perpendicular. A drawer is slidably mounted in the housing. A dispenser cup is in the drawer. A storage cup is disposed of at a fixed location within the housing. A probe is movably mounted within the housing such that the probe is rotatable within a lateral-transverse plane defined by the lateral direction and the transverse direction and the probe is translatable along the vertical direction. The pill dispenser also includes a controller. The controller is also configured for activating the suction pump when the flexible tip is within the storage cup and translating the probe along the vertical direction to the position above the storage cup after activating the suction pump and while the suction pump remains active. The method further includes translating the probe along the vertical direction to the position above the storage cup after activating the suction pump and while the suction pump remains active. The method then includes rotating the probe within the lateral-transverse plane to a position above the dispenser cup along the vertical direction after translating the probe along the vertical direction to the position above the storage cup and translating the probe along the vertical direction from the position above the dispenser cup until the flexible tip is within the dispenser cup. The suction pump is deactivated when the flexible tip is within the dispenser cup. This patent proves to be useful to our group because it uses a similar concept, the patent using a suction system to dispense pills into the pill cup, and yours using the vacuum system to do something similar. This is important because we as a group will need to make sure we do not infringe on any of the patented ideas of this prototype when we enter the manufacturing phase of our design process.

2.4 Patent Number: US11,135,135
Date of Publication: 09/26/2017
Inventor(s): Yesudasan Sujith

Abstract: A pill dispensing canister mountable onto a pill packaging device comprising a

hopper, a bulk dispenser, and a discrete dispenser. The hopper preferably includes a reservoir and an initial passage. The bulk dispenser preferably includes a bulk dispensing base, a bulk dispensing disc, a bulk dispensing motor, a pill exit chute, and bulk dispensing sensor. The discrete dispenser preferably includes a discrete dispensing base, a pill sorter, a discrete dispensing disc, a discrete dispensing motor, a pill drop chute, a pill jam sensor, a discrete pill sensor, a pill gate, a gate motor, and a control board.

Relevance: This patent features a pill dispenser that is directed to methods and apparatuses that contains a canister that preferably comprises a hopper, connected to a bulk dispenser, connected to a discrete dispenser, to carry pills from the hopper, through the bulk dispenser, and through the discrete dispenser to dispense pills, one at a time, in a controlled manner. The present invention may address one or more of the problems and deficiencies of the current technology on the market. However, it is contemplated that the invention may prove useful in addressing other problems and deficiencies in a number of technical areas. This patent outlines a clear and concise approach on how to build a sound pill dispenser with many different parts and components that overall satisfy many of the mandatory design requirements that our prototype needs to meet. This Patent does an excellent job of pointing our group in the right direction, as we first started the design process for our prototype.

2.5 Patent Number: US11,117,728
Date of Publication:
Inventor(s): Michael H. Weston and Frank Edward Smith

Abstract: A blister and package system made of a blister card and an outer package with unique interacting locking mechanisms. The blister card or strip is inserted into the outer package or sleeve and locks immediately without having to further process the outer package or the blister card or strip. The user is able to expose the blisters from the outer package by disengaging the lock. The blister card or strip includes a formable, pliable sheet material, into which one or more individual cavities or blisters are formed. The separated cavities are loaded with product and sealed with one or more layers of puncturable or tearable sheet material, such as aluminum foil. The blister card or strip contains a plastic rivet assembly that forms an interference mechanism.

The rivet is trapped by an internally formed panel and stopped by a catch flap included in the outer package.

Relevance: This invention relates to the field of packaging, and more particularly, to a dispenser package having a blister strip encapsulated in an outer sleeve. The package, preferably, has a locking function designed to provide senior citizens easy access to the contained drugs while preventing a child from gaining access to the contents of the blister strip. Generally, the invention provides an improved child-resistant and senior-friendly pill dispenser package particularly suited for the distribution of pharmaceutical drugs for public or clinical trial use. Specifically, the pill dispenser package of the invention achieves federally mandated child-resistant and senior-friendly guidelines through a locking function, while also providing a pill dispenser package that is easy to use for all adults including those with impaired physical abilities. Additionally, the pill dispenser package of the invention allows an individual pharmaceutical drug to be removed from the pill dispenser multiple times without degrading the child resistance rating of the pill dispenser packaging enclosing the remaining pharmaceutical drugs. This is something that is very important that our group did not previously think about. The fact is that in today's world, many of the common medications that people take look like candy especially in the eyes of children. This patent cause our group to have a serious discussion on how to childproof our prototype so that children can not take any of the medication from our dispenser and eat, as if they thought it was a piece of candy, as this would be proven to be extremely dangerous.

2.6 Patent Number: US11,083,850
Date of Publication: 08/21/2018
Inventor(s): Mehdi Hatamian and Paul Edalat

Abstract: A smart dosing device includes: a flow pathway having a cartridge receptacle that is able to house a cartridge; at least one sensor that captures identifying information related to the cartridge; a wireless communication module; and at least one security feature that restricts use of the smart dosing device. A method of providing a measured dose using a smart dosing device includes: enabling dosing; measuring a provided dose; and disabling dosing. A smart dosing

system includes: a smart dosing device having a unique identifier, the smart dosing device including a cartridge housing a substance, the cartridge having a unique cartridge identifier; and a server including a central database that stores usage information related to the smart dosing device, the usage information including: the unique identifier and the unique cartridge identifier.

Relevance: This patent mentions some embodiments may include a security module that may capture data such as a fingerprint scan and/or facial photograph in order to compare the captured data to reference data and determine whether a user is authorized to use the dosing device. Some embodiments may combine security features or otherwise capture usage information for later analysis. For instance, during dosing, a fingerprint scan may enable dosing (and also be captured and stored) while a camera captures a photo of the current device user or usage environment for later verification or analysis. In some embodiments, the dosing device may include various user interface (UI) elements. Such UI elements may include, for instance, buttons, displays, indicator lights, touchscreens, etc. In addition, the dosing device may be able to utilize inputs received from an external device (e.g., a smartphone) and/or provide received inputs to such an external device. The dosing device may be able to communicate with various user devices (e.g., smartphones, tablets, personal computers, etc.) across wired or wireless channels that may include one or more networks. Such user devices may execute an application associated with some embodiments that may allow a user to register a device, download usage information, adjust device settings, and/or otherwise interact with the dosing device. This patent leads us down the path of allowing UI elements as well as security features to our prototype in the near future. One thing we will be implementing to our prototype in the future is the factor of authentication of the user. This makes sure that the only person taking the medication is the person that it is prescribed to. As the dangers of opioid use have increased exponentially in the last decade and a half, this is a huge area of concern for the near future of our group.

2.7 Patent Number: US8,985,388
Date of Publication: 12/11/2004
Inventor(s): Nitesh Ratnakar

Abstract: Present invention discloses a dispenser having means to dispense desired number of pills from a bulk supply of pills contained in the dispenser. The dispenser comprises a storage compartment having bulk supply of pills and having a discharge port emptying into the counting compartment. The counting compartment contains first and second conveyors moving at first and second speed; wherein the second speed is greater than the first speed thereby enabling pill separation; the second conveyor discharges pills into the dispensing compartment. Sensors are strategically placed along the conveyors to count pills discharged into the dispensing compartment. A pill recovery system and apparatus is disposed inside the dispenser having means to recover pills remaining on first and second conveyors upon completion of a dispensation cycle and thereafter deposit recovered pills back into the storage compartment for use in future dispensation cycles.

Relevance: This patent works by having the means to dispense desired number of pills from a bulk supply of pills contained in the dispenser. The dispenser consists of the first storage compartment having bulk supply of pills and having a discharge port that empties into the second counting compartment. The counting compartment contains first and second conveyors moving at first and second speed; wherein the second speed is greater than the first speed thereby enabling pill separation; the second conveyor discharging pills into the third dispensing compartment. Sensors are strategically placed along the conveyors to count pills discharged into the dispensing compartment. A pill recovery system and apparatus is disposed inside the dispenser having means to recover pills remaining on first and second conveyors upon completion of a dispensation cycle and thereafter deposit recovered pills back into the storage compartment for use in future dispensation cycles. This patent was relevant to our group because the means of dispensing of conveyor belts is something that our group previously did not think of. Another aspect of this patented design that came as a shock to our group was the fact that the dispenser has a function that is a pill recovery system and apparatus is disposed inside the dispenser having means to recover pills remaining on first and second conveyors upon completion of a dispensation cycle and thereafter deposit recovered pills back into the storage compartment for use in future dispensation cycles. This is a brilliant idea because it fixes any machine sensor error that the machine encounters and is a backup plan that fixes any error that would risk the dispenser not being able to operate under the mandatory design requirement of

100% accuracy.

2.8 Patent Number: US9,597,261
Date of Publication: 04/25/2012
Inventor(s): David W. Baarman, **Sean T. Eurich**, **Scott A. Mollema**, **Cody D. Dean**, **Neil W. KUYVENHOVEN**, **Matthew K. Runyon**, **Joseph C. Van Den Brink and Ryan D. Schamper**

Abstract: A pill dispensing system that includes pill packages that can be used to dispense pills manually or with a dispenser system to provide enhanced functionality. The packages can be provided with information relating to the packaged pills or to the use of the packaged pills. By reading the information from the package, the dispenser system can know what is in the package, when it is to be taken and can understand and track inventory. The dispenser system provides reminders of when the pills should be taken. The dispenser system may have the ability to key a specific electronic device, such as a cell phone, to a specific user and the dispenser system may require the electronic device to be within proximity of the dispenser system before dispensing pills for that user.

Relevance: This patented pill dispenser works by using multiple different embodiments. One specific embodiment that would benefit our group in its use would be the one that controls the dispenser system that is connected to a network and may be configured to provide information via the network. For example, in the context of medication, the dispenser system may communicate pill usage information to a doctor, physician, pharmacist or other health specialist and in the context of food supplements, the dispenser system may communicate pill usage information to a food supplement representative. The dispenser system may use inventory information to reorder pills. For example, the dispenser system may warn the user that inventory is getting low and may request authorization from the user to restock the inventory. Upon authorization, the dispenser system may place an order over the network. As another example of a reordering system, the dispenser system may be configured to automatically reorder pills when the inventory is sufficiently depleted. This is a very useful feature that would benefit consumers immensely. This is a feature that the group could integrate into the application relatively easily.

2.9 Patent Number: US7,016,766
Date of Publication: 12/07/1995
Inventor(s): Jeffrey P. William, Galina Potepalov, Allan Dolores and Michael Bergeron

Abstract: A pill dispensing system includes a shelving unit in array form that holds a number of bulk containers, each holding a bulk amount of a pill to be dispensed. A computer controlled robot removes a selected bulk container and places it on a counter that also dispenses pills. The robot has an arm with a free end portion that can grip a bulk container or a single pill bottle to be filled. The robot is computer controlled to retrieve an empty pill bottle, place it on a label printing and applying unit, then place it next to the counter/dispenser to receive the selected number of selected prescription pills, then place the filled, labeled bottle on a conveyor. In an alternate embodiment, pill bottles are dispensed from a dispenser that holds bottles on inclined plates and feeds them to a vertical dispensing channel. Gates prevent the flow of bottles from an inclined position until the plate above has been emptied.

Relevance: This patent is for a prototype that provides a method and apparatus for automatically dispensing containers such as bottles for prescription medication. The system produces a label and fills prescription vials with a specific drug for a specific patient. The controlling computer directs a robotic manipulator to retrieve a drug storage unit which includes a queuing and separating means (i.e., cassette). While the pills are being counted the robotic manipulator arm retrieves an empty vial from a vial dispenser. The manipulator then places the vial on a vial rotating device which is positioned in a manner so that the prescription label is applied to the label to the vial as it is printed. The robotic manipulator then retrieves the labeled vial from the labeler and maneuvers the vial to the temporary pill holding volume. Then a robotic manipulator lifts a gate allowing the pills to fall from the holding volume into the vial. A robotic, computer controlled arm is used to grasp a selected one of the containers that has the correct pills for the prescription to be filled. With the method of the present invention, the robotic arm then removes the container from the counter/dispenser and returns it to its receptacle. This patent is very unique and useful for our group because it is the first design for an automatic pill dispenser that we've encountered that uses a robotic arm controlled by a micro computer. As of the time of this report, the group is currently using a vacuum system and not a computerized robotic arm system,

but this is something that will definitely be in the back of our minds in terms of future improvements to our design if the vacuum system does not meet the design requirements.

3 Evaluation of the Competition

3.1 Quality Function Deployment

In this section of the report, a competitive analysis of the top automated pill dispensers on the market will be completed. In this analysis, our team's top designs will be compared to these other pill dispensers that are currently on the market, under a multiplitude of factors. From the 150 total designs that were depicted and discussed in the conceptual design section of the report, the team completed three total pugh analysis and narrowed down the 150 total concepts for the teams pill dispenser to three final conceptual designs. Once the three final conceptual designs were established for our group, a QFD (quality, function and deployment) analysis was completed against three of the market's top competitors. These competitors include Hero Health, MedaCube and E-pill Voice Pro. The purpose of comparing our team's final three concepts is to investigate how our concepts would hold up in the market for automated pill dispensers. By doing this investigation our team realized some of the weak spots in our concepts in terms of design flaws and allowed us to make improvements to our prototype in order to be as competitive as possible on such a large scale market in the pharmaceutical industry.

3.2 Market Analysis

The first competitor on the market that will be analyzed is the automatic pill dispenser known as the Hero Automatic Pill Dispenser. The Hero automatic pill dispenser is in the middle of the pack when looking at the cost analysis of the three competitors on the market. Hero Automatic Pill Dispenser is a dispenser manufactured by The company "Hero Health", who is also partnered with "AdhereRx" and "Adherehealth Company". This dispenser is marketed as a smart service that takes the hassle out of medication management. It is a three in one commodity that includes a smart device, phone application and a refill and delivery service. Hero health allows the user to set up a medication schedule in the provided phone application in order to alert the user and dispense the medication at the correct time. This dispenser allows the user to save

time, as instead of sorting and organizing medication, Hero Health can hold a 90 day supply of medication for 10 different prescriptions.The phone application also tracks what pills the user has already taken and at what time, all while refilling the medications for the user and alerting the caregiver if the user forgets to take their medication. This pill dispenser also offers a 90 day risk free trial. To date the Hero Health pill dispenser has dispensed approximately 70,900,000 pills to its users. When analyzing the cost of owning a Hero Health, the user must pay a monthly subscription of $29.99 per month, as well as a one time $99.99 initiation fee that requires a 12 month commitment plan. Once the subscription is bought, the user can expect a “Hero automatic pill dispenser” as well as mobile app functionality, 24/7 support via email, phone or chat, caregiver accounts, notifications when the user is low on pills, late or missed dose notifications and progress statistics reporting. The Hero pill dispenser is very well evaluated by consumers. With over 1,200 ratings the dispenser received a 4.8 out of 5 stars.

Figure 3: Hero Pill Dispenser

The second competitor on the market that will be analyzed is the automatic pill dispenser known as MedaCube. MedaCube is an automatic pill dispenser that is at the top of the pack when analyzing its cost and comparing it to the other three competitors listed in the QFD analysis. The MedaCube is a pill dispenser manufactured by PharmAdva. MedaCube is the only automatic pill dispenser on the market that is clinically proven to significantly increase medication adherence. An interesting advantage that this competitor has is that due to the Covid-19 pandemic, since it only needs infrequent filling, given its large 90-day capacity for up to 16 different medications, it can keep people safe by helping to maintain social distancing and even quarantine. Text, email, and voice notifications, ensure all is well, doses are not going missed, and there is an ample supply of pills. This is especially important for anyone who is elderly or has a medical condition that can make them vulnerable to the more severe consequences of Covid-19. This dispenser helps the user by preparing and delivering the right dosing at the right time, alerting the patient when their dose is ready, as well as re-alerting via call or text, when late and if dose is missed. MedaCube also gives the caregiver peace of mind. It does this by alerting the caregiver if any dose is missed via text, email and call, having an easy setup, as well as providing "Need to refill" alerts when any prescription is low. Some of the key features and benefits of MedaCube is its easy setup and load. With MedaCube there is no pre-sorting of pills and no hardware to assemble, saving time and reducing errors. MedaCube houses Up to 16 different medicines, with up to 90-day supply and can refill a medication in 2-minutes, even if the brand changes. MedaCube is a pill dispenser with the ultimate reliability & safety, it does this by picking each dose as set (never picks the wrong pill – many millions of doses tracked). It also allows the user to view a picture of the pills for each dose given. It even has a feature that recognizes tilt sensor alerts for possible tampering or incorrect dosing/spillage, and is even lockable. Not only is this dispenser simple and reliable, but it is also highly effective. MedaCube is the only medication dispenser tested to significantly improve adherence in an FDA approved clinical trial, noting that patients took the correct dose 97% of the time versus only 49% prior to the MedaCube. When looking at the cost of MedaCube, the starting cost to purchase the dispenser is $1849.00.

Figure 4: MedaCube Pill Dispenser

The last competitor on the market that will be analyzed is the automatic pill dispenser known as E-pill Voice Pro. E-pill Voice Pro is an automatic pill dispenser that is at the bottom of the pack when analyzing its cost and comparing it to the other three competitors listed in the QFD analysis. The E-pill Voice Pro is a pill dispenser manufactured by E-pill Medication Reminders. The E-pill Voice Pro is a locked automatic pill dispenser that allows patients to take the correct medications at the correct time. This significantly improves medication adherence. Its Bluetooth capabilities allows the patient or caregiver to set up the dispenser and review dosage records via a free iOS (iPhone) or Android app. This dispenser is great for patients who want to live independently or are residing in an assisted living facility. It also benefits caregivers who want a solution that provides an easy and accurate way for their loved ones to remember their medications. The E-pill Voice Pro features a key locked lid, up to 9 daily alarms with voice notification. An example of a voice notification for one of the 9 daily alarms could be“It's time

for your medications”. This pill dispenser also utilizes blinking lights when it is time for the user to take their medication. The dispenser is AC powered with 2 months of battery backup. When the batteries are low, the dispenser will prompt the user with a low battery warning. This dispenser is also bluetooth compatible. When bluetooth is enabled, the user can use their smartphone to set up the dispenser and to review dosage records. The dispenser will always dispense at the set times, but due to the technical limitations of Bluetooth, the smartphone must be within 33 feet of the dispenser to take advantage of the app’s capabilities. The dispenser has a 28 compartment medication tray, as well as filling guides and two keys for the locking feature. depending on the amount of medication the user needs they can refer to the schedule below.

- 1 medication time per day - refill tray every 28 days (4 weeks)
- 2 medication times per day - refill tray every 14 days (2 weeks)
- 3 medication times per day - refill tray every 9 days
- 4 medication times per day - refill tray every 7 days (refill once per week)
- 5 medication times per day - refill tray every 5 days
- 6 - 7 medication times per day - refill tray every 4 days
- 8 - 9 medication times per day -refill tray every 3 days

When looking at the cost of the E-pill Voice Pro, it will cost the consumer $549.95.

Figure 5: E-Pill Voice Pro Pill Dispenser

Our team's plan to gain market/customer advantage is to purely put all of our effort into marketing and advertising. There are many things that we could focus the marketing and advertisement on, but we believe that the biggest factors of our pill dispenser that we should focus on are the price, the limited user responsibility and the inclusiveness of primary caregivers. When looking at the cost factor of our pill dispenser when compared to the three competitors listed on the market, there is a very large difference between ours and the competition. Our pill dispenser will cost the consumer a A one time only fee of $100 to buy our entire pill dispenser. There is no monthly subscription or hidden fees. There is no once a year initiation fee with a yearly commitment plan. This is the opposite of the Hero pill dispensers, monthly subscription of $29.99 per month, as well as a one time $99.99 initiation fee that requires a 12 month commitment plan. Also when looking at the price of the MedaCube and the E-pill Voice Pro, our price per unit blows these two competitors out of the water. When looking at the cost of MedaCube, the starting cost to purchase the dispenser is $1849.00 and then when looking at the

cost of the E-pill Voice Pro, it will cost the consumer $549.95. This shows that as a marketed pill dispenser when looking at the cost analysis of our pill dispenser compared to the three current market competitors, we are light years ahead of the competition which we think should be the most marked factor of our product. With this large price difference, it needs to be said that just because we are offering a pill dispenser for a fraction of the price, it isn't because our dispenser is cheap and does not offer the features that the current market competitors do. This could not be farther from the truth, as we provided almost all the same features that these competitors offer with the exception of a customer support system. In terms of limited user responsibility, our pill dispenser is unique in that the user only has two quick and easy responsibilities to complete before they are free of any tasks to make our machine work. Our customer is responsible for two quick and easy steps to start this process. The first step is the customer will use our designed phone application and input all their medical prescription information such as the name of the prescription, and dosage information. Their other responsibility is to pour their prescriptions into the corresponding pill storage box and load that pill storage box into the main storage. Once these two steps are completed, the customer is free of any responsibilities in how our automatic pill dispenser works. The last factor we will focus the marketing and advertising on in order to gain market/customer advantage is that our pill dispenser provides so many options for the inclusiveness of primary caregivers. This is such an underrated factor that many of the current on the market pill dispensers do not offer. One of the biggest parts of the pill dispenser market is for geriatrics. Old people make up the biggest portion of consumers for the product that we are currently working on delivering. This is because as we age, so do our brains. It is not a secret that as we age the majority of people's ability to remember things enter a rapid decline, this is not to mention the serious diseases that affect this ability such as Dementia and Alzheimer's. So the fact that our product will allow a mostly unrestricted access to primary caregivers, allows for a more safe environment for our consumers in the market. We have two extra safety steps that include primary caregivers such as the concept that if the user has yet to take their dispensed medication, the machine will send a signal to the application that the dispensed medication has not been taken. When this happens the application will immediately send a sms text message to the designated primary caregiver. The other safety step includes the primary caregiver being given full access to the application, only with permission of the user. This will be an option in the app as a possible third safety step for the user. In this situation the caregiver would have the

exact same access as the user. With these three factors in mind we can successfully market and advertise our product in order to gain market/customer advantage. In terms of the marketing and advertising itself, we believe the best place to establish our campaign is in magazines, pamphlets and television commercials. This may seem like a step back from the modern day campaigns that take place all over the internet and social media, but what must be kept in mind is what our target audience is. Older people do not spend their days surfing the internet on popular social media platforms, they instead watch television or read magazines/newspapers. Also, we believe that it would be a great market strategy to place advertisements in the pamphlets and magazines found in doctor's offices and hospitals. As stated earlier, the elderly do not surf social media, so they are in these waiting rooms at doctor's offices and hospitals (which they visit disproportionately more than younger generations) reading magazines and pamphlets that the buildings have to offer. This is why we feel these forms of advertising and marketing in these specific locations will allow our product to quickly and exponentially perform well when introduced into the marketplace.

4 Design Specifications

4.1 Original Design Specifications

The requirements for this automated pill dispenser is that it must have a sale price between fifty and one-hundred dollars, it also needs the capability to dispense at least four different types of pills. The device must have some sort of reminder system. The reminder system would be taken into account within the application that was made by the team. The application would be able to remind the user when to take their pills at various points in the day, as well as be user friendly. It must have the ability to charge or have some battery life. Another factor was that it had to be portable. Lastly the most important factor was that it has to be 100% accurate in dispensing pills.

4.2 Team 15's Design Specifications

The specifications are not only a challenge but a way that the outcome design can be made to fit the needs of the potential consumers. Taking into account the needs of the consumer

is important to how the overall model is designed. Having the pill dispenser be automated is the main portion of how it works in conjunction to the design specifications.

Pill dispensers are often only aimed towards elderly people, when in fact there is a whole group of people that could also benefit from this type of product. Making it portable for people on the go could be an attraction to Team 15s' automated pill dispensers. There are various forms of pill dispensers out on the market such as CVS, Hero, Amazon, Livi and LiveFine. With the added feature of portability, it can be the product that stands out.

To describe the chosen design, it has a compact shape. It has the dimensions of 1ft x1ft x1ft. An addition to the preliminary design, is the outside casing that makes the dispenser look more sleek and appealing to keep on the counter. On the front of the design there is a linear actuator that contains a pill box with four different compartments. This pill box allows the design to be portable and accessible to those who are not in the reach of the pill dispenser. The 3-D printed design allows it to be light-wieght and have a minimum weight of five pounds. With the added electrical components of the automated pill dispenser it adds to the overall weight. The aim of this product was to also be inexpensive. The material that was provided is more expensive than the cost of mass production of the item would be. The financial analysis is provided in section 15 in which the final product's price is between fifty and one-hundred dollars. The plan for warranty is not fully developed; however if there is a warranty it would cover any technical problems that the dispenser may have. This device is specified to last about thirteen years. With tested excellent accuracy in dispensing pills, this product is accurate. Team 15 understands that there is no room for error in dispensing pills, which could lead to serious injury or even death to the user. The product is made out of components that are entirely recyclable. Team 15 has looked into legal requirements which include the certification by the Food and Drug Administration (FDA). Also, with the help of the patents researched, the team's final product is planned to be unique and original. It is able to dispense at least four types of pills of most shapes and sizes excluding antihistamines which are about 3mm. A companion phone application is designed to cooperate with the device. It will remind the user when to take their pills as well as administer the correct dosage with extreme accuracy. Unfortunately after many attempts the design specification for the required application was not successful. The 3-D designed prototype exhibits that the design specifications are mostly met.

Design Specifications	Numerical Parameter
Sale price	$50-$100
Pills	4 different types
Life Expectancy	13 years
Portability	4 compartments

Table 1: Design specifications

5 Conceptual Design

5.1 List of Concepts Generated

Part of the design process is to brainstorm a wide list of ideas and concepts as a team, using the problem definition, design specifications and literature we researched. Each team member was tasked with developing thirty unique concepts that could be feasible options for the solution to our problem. Combining all our designs into a total of one hundred and fifty concepts that we systematically reduced down to a top three to five designs. To determine the best concepts, Team 15 organized a quality function deployment (QFD) analysis using design specifications listed in the prior section.

5.2 Jared Brule's Concepts

1. Concept 1 is a 4-compartment design, one for each of the four pills. Each pill section will have its own picker that allows the pills to flow to one central cup. There will be a display on the machine that can be programmed to what pills need to be dispensed at each time in the day. This is a larger size sorter with lots of extra pill space but the downsize is it is not a portable design. Also making it a bit complicated to sort because there are 4 separators for the pills increasing the price of the product.

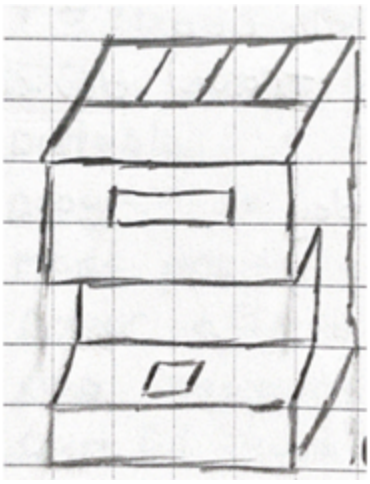

Figure 6: JB1

2. This concept is like number 1 with 4 separate pill compartments, as well as individual pill pickers for each pill, one change is that a small speaker can be added for a sound tone to be played when pills are dispensed. Thinking about the main customer base, which is mainly elderly, some of the customers could have a visual disability and having a sound tone tell them when the pills are ready can be a selling point.

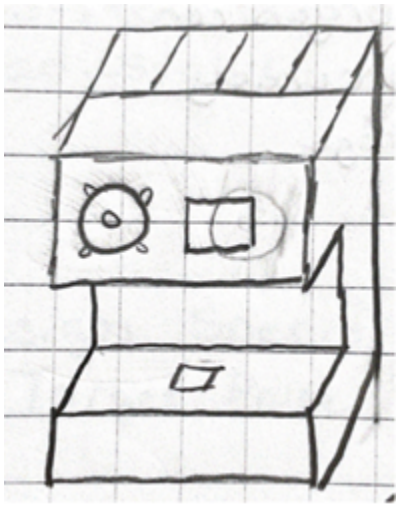

Figure 7: JB2

3. This concept is like number 1 with 4 separate pill compartments, as well as individual pill pickers for each pill, one change is that a visual light is set off when it's time for the elderly to

take their pills. Having a visual alarm allows for people with hearing disabilities to benefit from this product as well increasing our potential market.

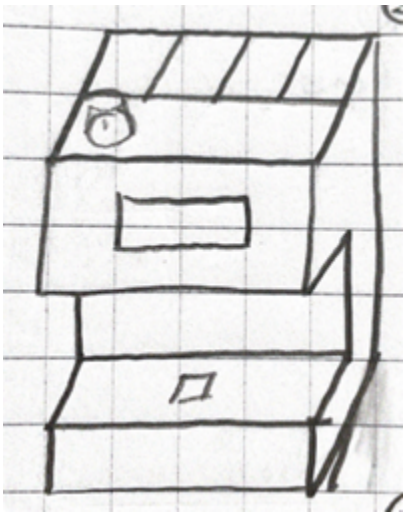

Figure 8: JB3

4. Concept 4 is a coffee machine size device with similar storage to concepts 1-3 allowing for a large amount of pill storage. This also has 4 individual pill pickers to separate one pill at a time from storage. This design has a central storage of 4 cups, one for each time in the day to take pills. This allows the machine to sort the pills at one point in the day and then sets off an alarm at each time pills need to be taken, then restocked at the end of each day.

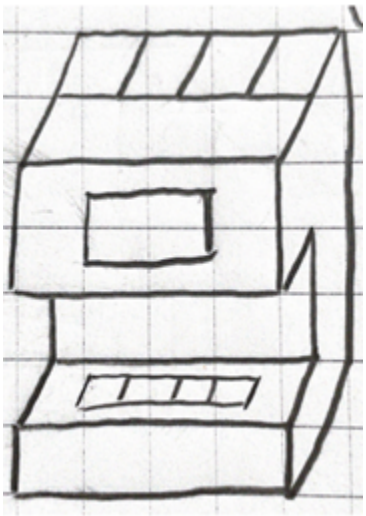

Figure 9: JB4

5. The fifth concept is a single modular pill storage and sorter. This product would be sold individually, and customers would need to purchase multiple if they have more than one pill that needs sorting. It will be programmed using a smartphone application to pick the times of the day that pill is needed. They would be a small design but that would limit the number of pills in storage.

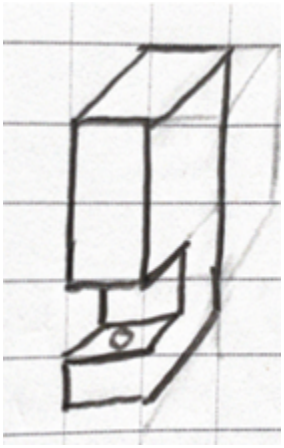

Figure 10: JB5

6. Design concept 6 would be a small portable design. The main purpose would allow for pill storage and dispensing on the go. There would be 4 small sections with very limited space for pills. The pills being dispensed would be held behind a door until the customer is ready to take them. The device would be programmed by a separate downloadable application that would remind the customer to take their pills.

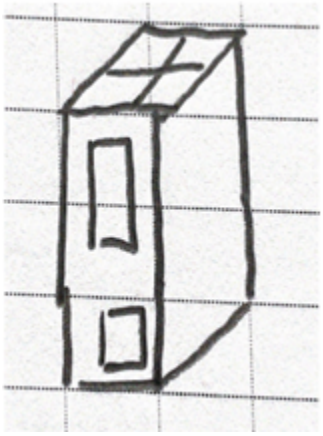

Figure 11: JB6

7. Design 7 would be another portable design that is like the plastic pill boxes people currently own. This concept would be manually filled daily and would allow the customer to have alarms that would go off and allow the sections to unlock at specific times in the day. These times would be set on their smartphones as well as the reminders.

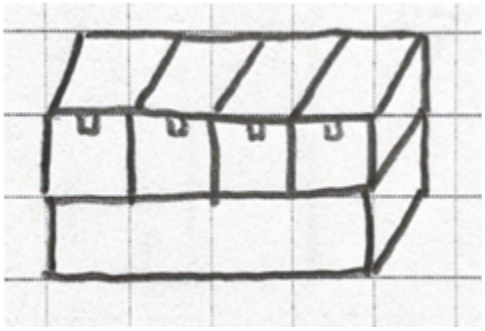

Figure 12: JB7

8. Concept 8 returns to the coffee machine sized device with individual pill storage as well as individual pill picks. This concept added a more portable aspect to the device in the form of a bag. The device drops the pills into bags that can be taken on the go with you. This homebased model has multiple timing for each time in the day. The connected application would still allow the customer to schedule their pills and remind them on the go when to open their pill bag.

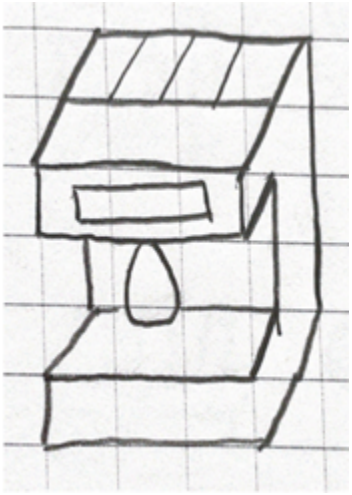

Figure 13: JB8

9. The idea behind concept 9 was a larger pill sorter for people with more than 4 pills in their daily cycle. This dispenser has 8 different sections with one output which is programmed on the small screen included with the device. The concept wouldn't hold as many pills as the 4 pill home models but more variety. Each pill having its own dispenser is tough to include in the octagon frame, but it does have a sleek design.

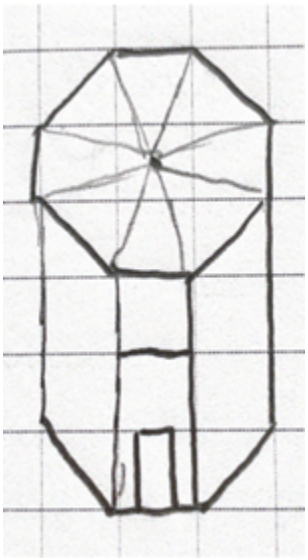

Figure 14: JB9

10. Concept 10 is a simple pill box design that has 28 individual slots labeled with a day of the week and time in the day. Each slot is programmed on the smart device to unlock at certain times when the pills inside are needed. They can also set reminders to go off on their phone. The device doesn't automatically separate the pills into each slot so they will manually need to be filled by the customer every week.

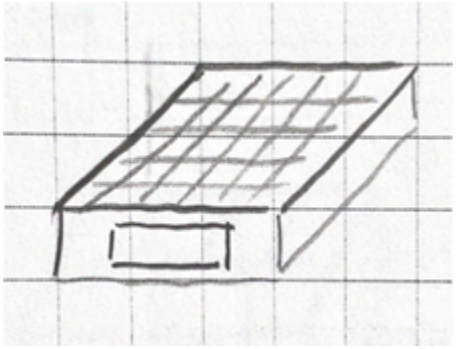

Figure 15: JB10

11. Concept 11 is a simple pill box design that has 8-12 individual slots labeled with a time in the day. Each slot is programmed on the smart device to unlock at certain times when the pills inside are needed. They can also set reminders to go off on their phone. The device doesn't automatically separate the pills into each slot so they will manually need to be filled by the customer every 2-3 days, this option is more portable than concept 10 but takes a lot more time to keep up with the sorting.

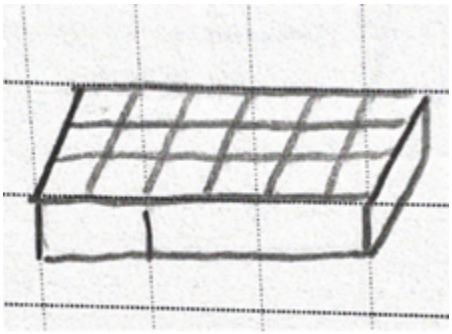

Figure 16: JB11

12. The idea behind concept 12 is a simpler form factor by having a cylindrical shape to the pill holder. There are 8 slots to each of the 4 holders so that 8 days can be sorted at once. Having these sorters allows people to bring a quality of 2 days of pills with them that are sorted by time in the days. Keeping it very simple on this design in terms of technology. Only producing an application that allows for scheduling of pill times so the customer can be notified when to take each section.

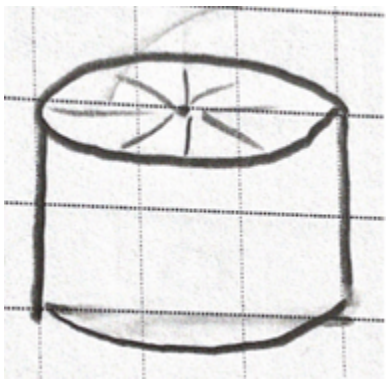

Figure 17: JB12

13. Concept 13 is a design that has 4 small pill sorters all with their own storage sections. This design is a lot smaller than most to allow for easy portability. With this being portable it will be difficult to fit all the dispensers into the device while allowing for a lot of pills to be stored. This design has a lot of difficulties but could benefit people on the go a lot.

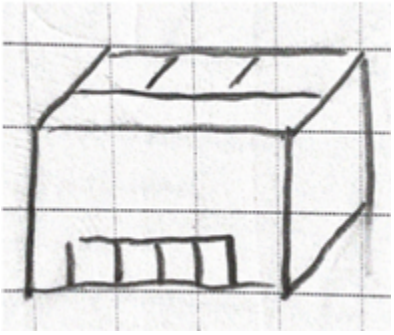

Figure 18: JB13

14. Design 14 is a simple design for one day storage, with 4 individual time slots. Each compartment has a time of the day on it, to allow the customer to carry this on them during the day when they go out and still be able to take their medication on time. This will simplify people's day when needing to carry a lot of pills on them, but it would need to be refilled everyday which could get annoying to deal with constantly.

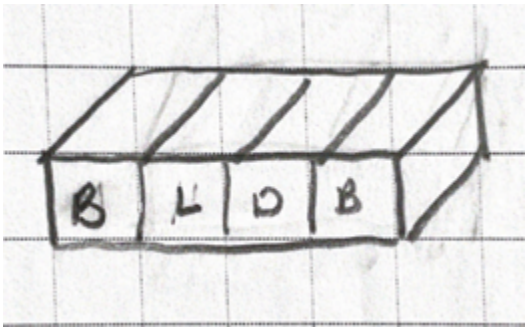

Figure 19: JB14

15. The 15th concept is a similar design to number 14 but it's for a weeks' worth of storage. It has 28 refillable slots to allow for the customer to fill a full week 4 times in a day. The pills would be refilled every week, but the sorter would be able to be taken anywhere but it is a lot

bulkier than design 14. This design has its flaws because one the customer must fill the compartment, secondly the design is tough to be portable unless you could carry the packet around.

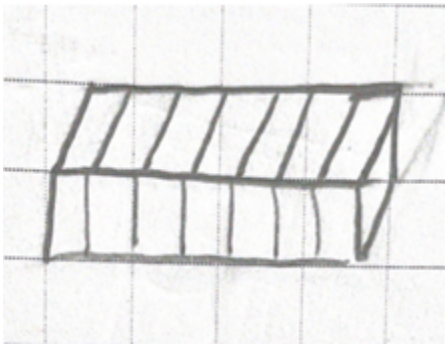

Figure 20: JB15

16. The concept number 16 is a stackable design with 7 different packets, each having 4 sections for separate times in the day. This design allows a customer to sort a week's work of pills in daily packs allowing each to be portable. Once the week is over the customer refills all the packs for the next week. This design allows for a lot of portability, but automation is limited due to how the packs are stacked.

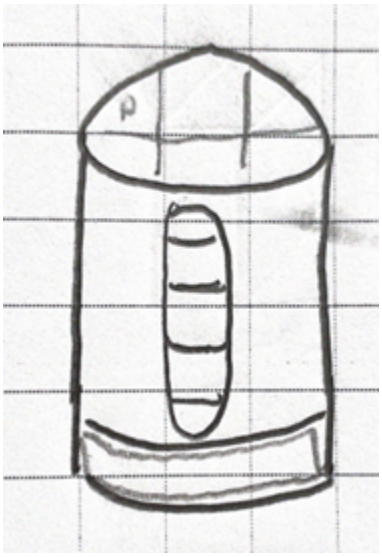

Figure 21: JB16

17. Design 17 is a cap to customers' pill bottles. The caps have a timer that can schedule the number of doses a day the person needs to take. This simple solution allows them to each be time separately and contain the full amount of medication the customer has. The cap can also be programmed on their phone to have a reminder system through an application. The customer can have a lot more than 4 medications and keep them in order.

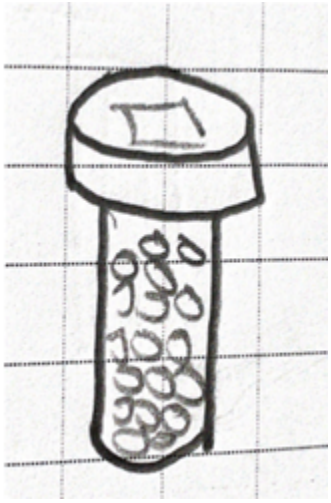

Figure 22: JB17

18. Design number 18 would be a small phone sided sorter that can store 4 different medications in locked compartments. This design allows for an on-board computer to program when each compartment opens during the day. The customer is then told how many pills from the compartment that unlocked they should take, once closed again goes to the next section and repeats the order until the next time interval is hit.

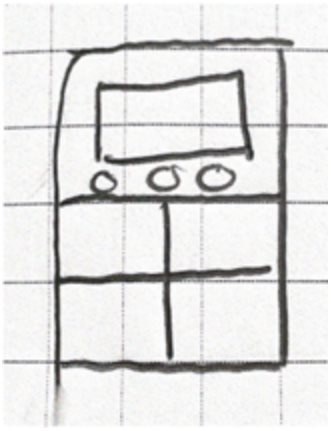

Figure 23: JB18

19. Concept 19 is not very complex or connected to the pills; it is mainly a reminder system allowing the customer to not forget their medication time. The timer allows the customer to specify the times in a day that they need to take certain meds and alert them. This design doesn't allow for the sorting of the pills but covers the reminder parts of the project.

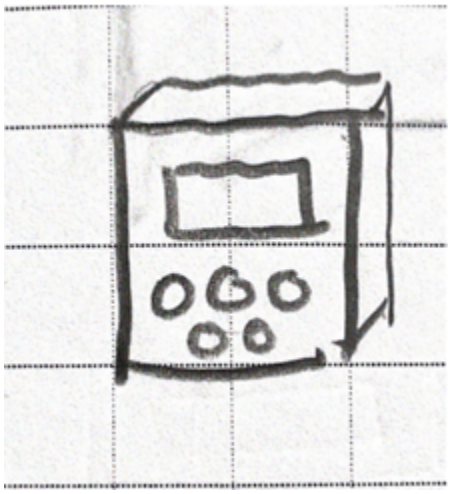

Figure 24: JB19

20. Design concept 20 is a design that utilizes the customers orange medications bottles. This design has the customer flip the prescription bottle upside down and insert them into one of the four designated locations. This turns the orange bottles into the storage compartments, then each section has a shorter underneath to allow the pills to be separated at the specific times needed. This allows for quick changes of the pills once the customer runs low.

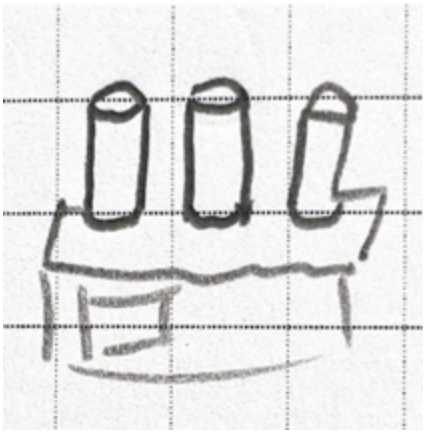

Figure 25: JB20

21. Design 21 is a simple storage solution for the orange medication bottles, there is a slotted design that can be labeled with the times of the day which each medication needs to be taken. The simplicity of the design also is the reason this design doesn't encompass a lot of the design requirements. It is missing the automated potion to separate the pills for the customer, but it does solve the organization for them instead.

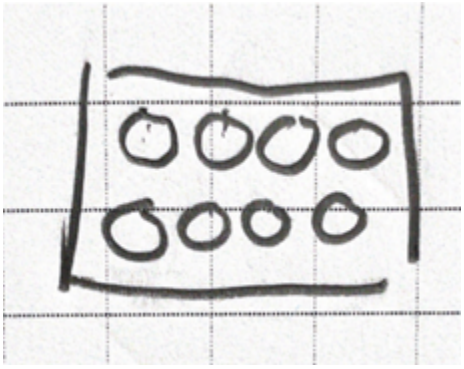

Figure 26: JB21

22. Concept 22 was a failed design overall. It has no organization system, no sorting system for the medication and no simpler way for the customer to take their pills. It is the baseline where the customer just has their medication on the counter and takes them as the label says. No work on our part is needed but allows for a comparison for all other products to ensure a better product was designed.

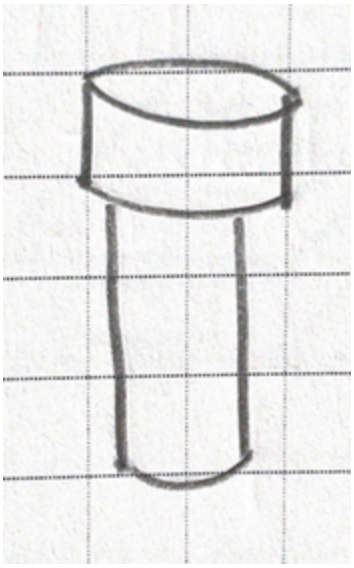

Figure 27: JB22

23. The design number 23 is just a jar that allows the customer to keep all their pills in one place, but this makes it hard for them to keep track of what pill they need to take at what time other than the shape and design of the pill. This was another comparison concept to improve other models. No automation of any sorts would be included in this design.

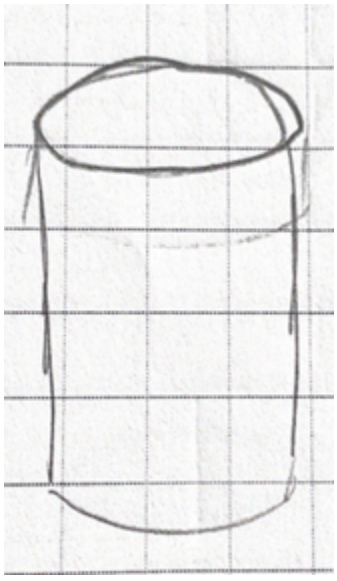

Figure 28: JB23

24. Model design 24 focuses on a nice cylinder design that is a better form factor then the other designs. It has 4 storage compartments that allow each pill to be stored separately. Each compartment has its own singular pill dispenser that when the button for the pill is pressed one pill corresponding to the pill is given to the customer. This design doesn't have a huge storage compartment for the pills but allows for each travel with multiple medications.

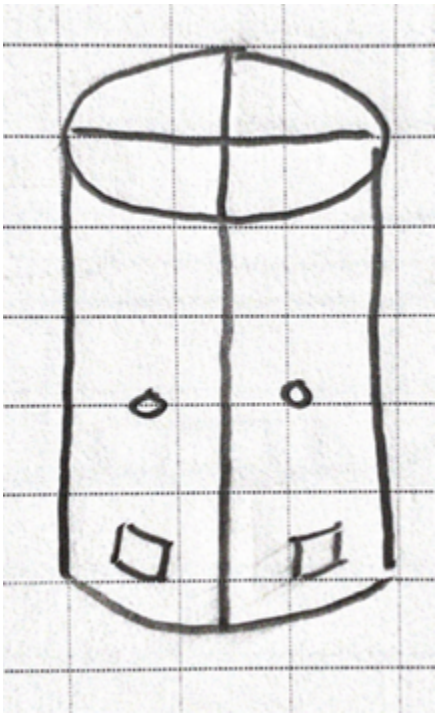

Figure 29: JB24

25. Design 25 is like a quarter candy dispenser, allowing for individual storage and the use of a turn dial to allow a single pill at a time to drop into the shoot. This device doesn't keep track of the times for the pills to be taken but does have a clock in it for easy checking of the time. Is a smaller design with less electronics saving counter space for the customer, could also be wall mounted, limiting the portability.

Figure 30: JB25

26. Concept 26 is a simple application that allows for customers to organize when their medication needs to be taken and reminds them. It can update them with the dosage and number of pills they should take at that time in the day. Also, by inserting the number of pills prescribed it can update you when to call for a refill to not run out of medication.

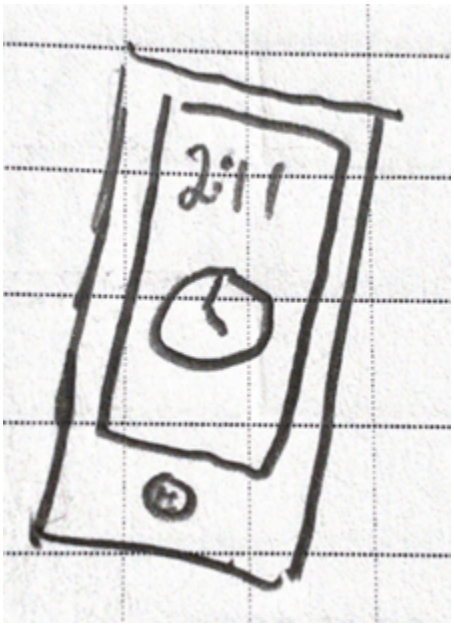

Figure 31: JB26

27. Concept 27 is designed for the customer on the go, it is a pill bottle storage container that can hold up to 4 different medications. Storing the pills in a case that keeps the organized and travel read. Simple design with multiple straps holding the bottles in place. There is also space to add in a schedule to keep track of pill taking.

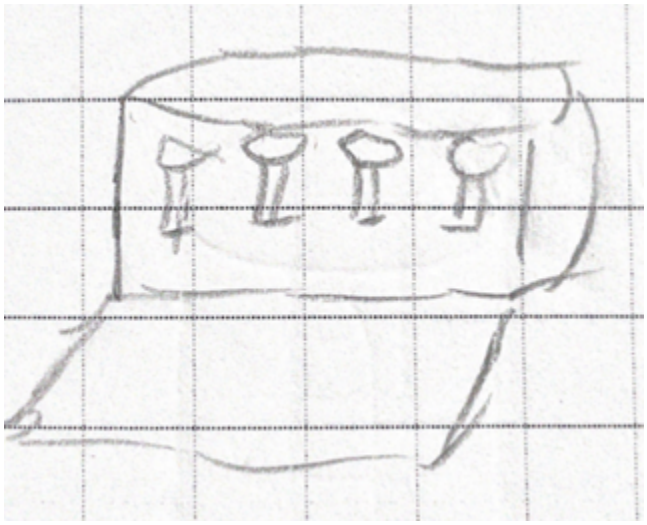

Figure 32: JB27

28. This concept number 28 is a design like a google home with a lower pill draw. This design has a screen on it that allows for easy scheduling, reminders, and alerts about your medication. Also allows the individual draws to be unlocked at certain intervals in the day when needed. Very high tech for a simple system, but design is limited on the number of pills that can be stored.

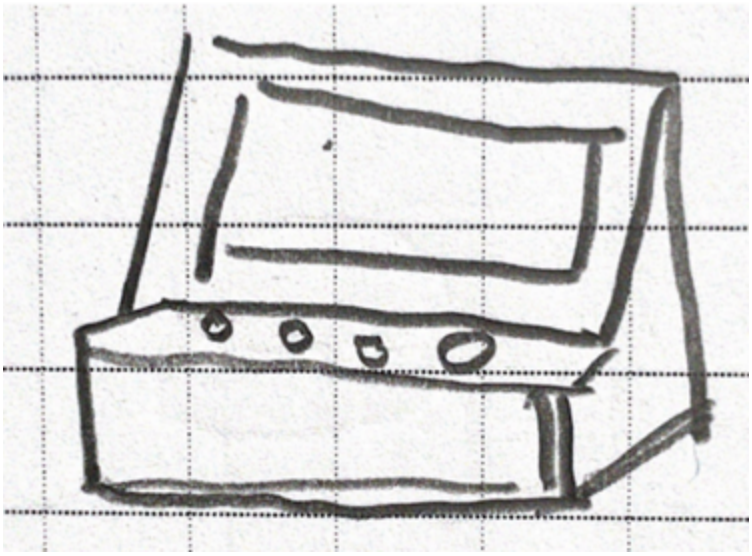

Figure 33: JB28

29. Design concept 29 is a concept like 24 where a push of a button is how the pills are dispensed, but this is a linear design instead of a cylindrical design. Four different pill storage areas that each have a dispenser on them. The customer will keep track of the timing for each pill that should be dispensed. The customer doesn't need to mess with the pill bottles and caps, just need to push the button as many times as they need.

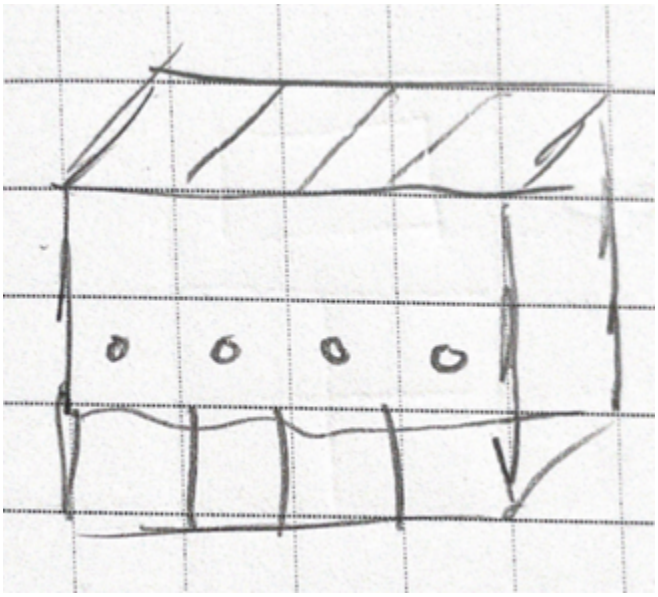

Figure 34: JB29

30. Concept 30 is a package design like the Amazon prescription bags. The customer can individually pack their bags that can be sealed. After sealing the label, the bags with the time and date for the medication inside. These bags can be stored in their cabinet or taken on the go if traveling but allows for one pack to be opened at a time when needed. Packs are easy to have but tough to organize at the beginning of the season.

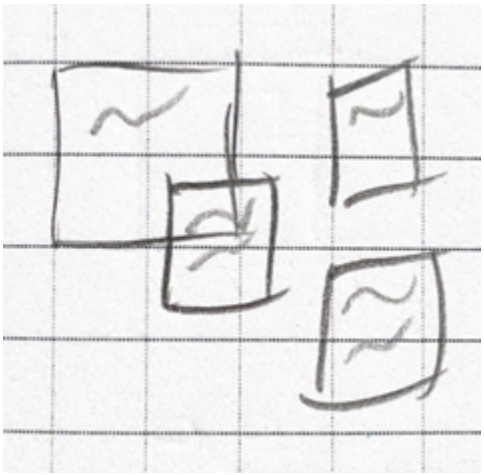

Figure 35: JB30

5.3 Cullen Burns's Concepts

1.) This concept is the first screen of the application design that will show the different types of pills, what time of the day they will be taken and the total quantity.

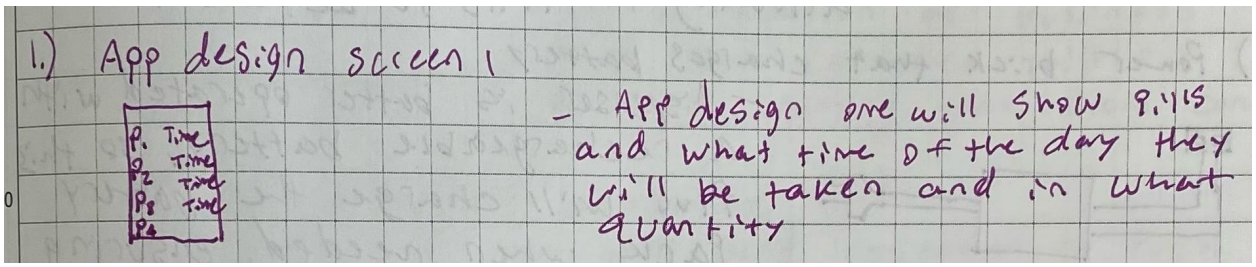

Figure 36: CB 1

2.) This concept is the second screen of the application that will show the types of pills, the amount of pills that are left in the main storage for each prescription as well as the date the prescription needs to be refilled. This will be a great feature for the user because it ensures that the user never runs out of medication

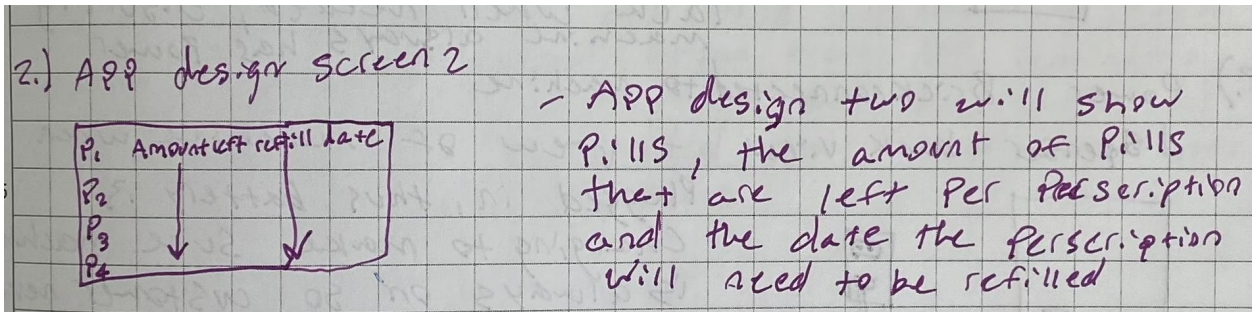

Figure 37: CB 2

3.) This concept is for the loading chamber of the pill dispenser. In this design there will be one loading column for each of the four different types of medication. As the user gets a new bottle of their medication from the pharmacy, they will load the entire prescription into the corresponding loading chamber.

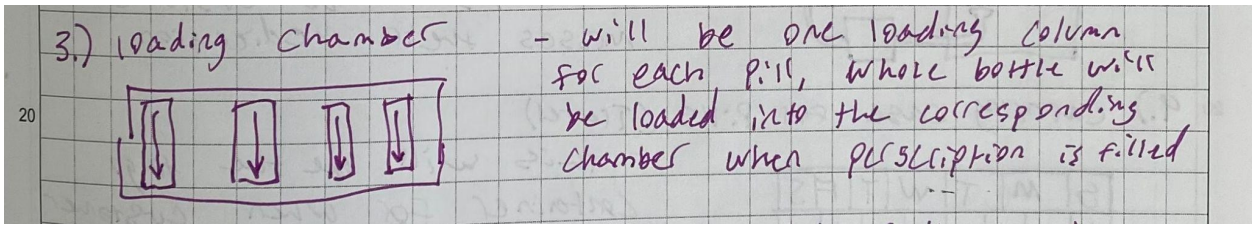

Figure 38: CB 3

4.) This concept is for the dispensing tunnel of the prototype. This works by the pills that were loaded in the loading chamber from the previous concept, and will fall into this tunnel when dispensed. It will feature a ramp with a steep slope, allowing the pills to slide down into the dispensed cavity.

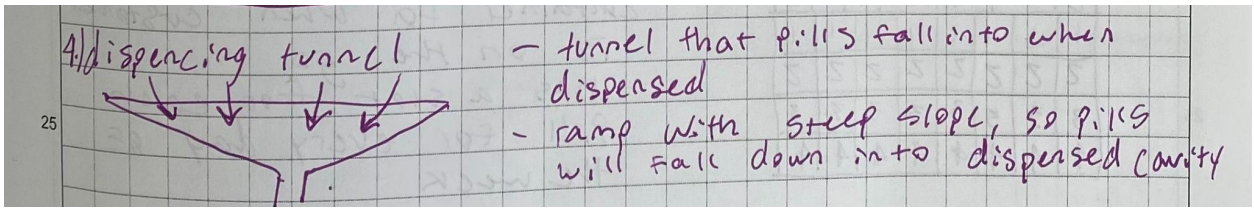

Figure 39: CB 4

5.) This concept is for the magnetic door to the cavity containing the dispensed pills. In this design, a magnetic door in front of the dispensed pills locks the cavity shut until it is time to take the medication ensuring that the user only takes their medication at the right time.

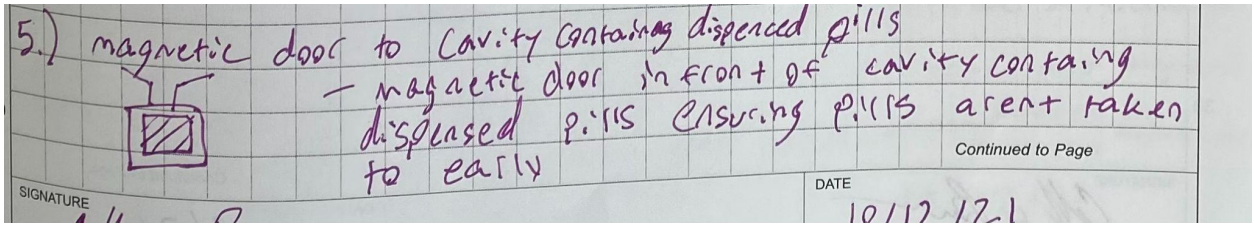

Figure 40: CB 5

6.) This concept is for the timer that controls the magnetic door. In this design the timer counts to the exact time that the pills need to be taken as dictated from the application. When the correct time arrives the timer goes off, and the magnetic door will open, which will allow the user to take their medication.

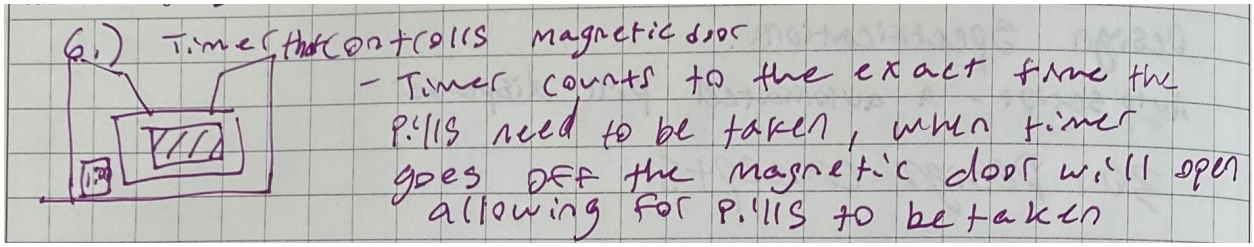

Figure 41: CB 6

7.) This concept is for the power brick that charges the battery that powers the machine. In this design the dispenser is battery powered with a rechargeable battery, so this pluggable

power brick will charge the battery pack when needed. This is done to ensure that the machine always has power.

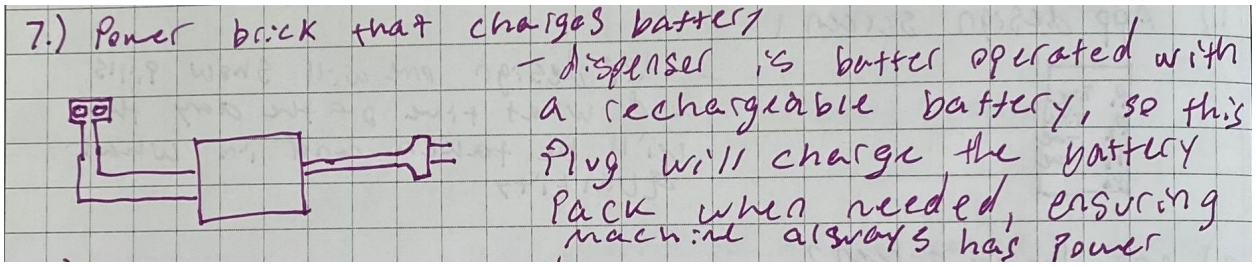

Figure 42: CB 7

8.) This concept is the power brick connected to the dispenser while plugged into a power outlet. This design is drawn out from the back view of the dispenser to show visually what the prototype would look like when plugged in and charging. This is done to make sure the machine is always charged and has power so the user never misses a dose of their medication.

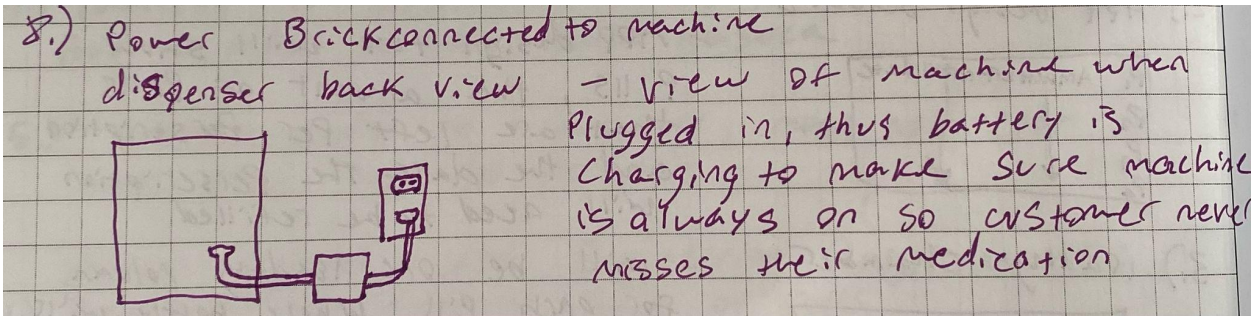

Figure 43: CB 8

9.) This concept is for the travel case for bringing the users medication on the go. This design is of a to-go container for when the customer is on the go. This travel case has a container slot for each of the four pills for all seven days of the week.

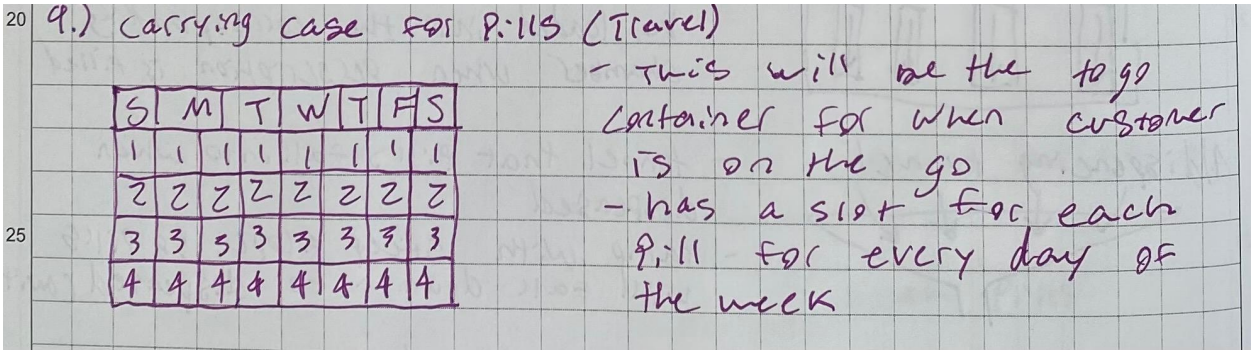

Figure 44: CB 9

10.) This concept is the screen of the application when the user is traveling. In this design the user will set the application on “go mode” , which will allow for the user to still be reminded to take their medication when they are away from the machine itself and are traveling with the travel case. This will allow the user to dispense how many days worth of medication they will need while they are away from home. This concept is paired with concept 9.

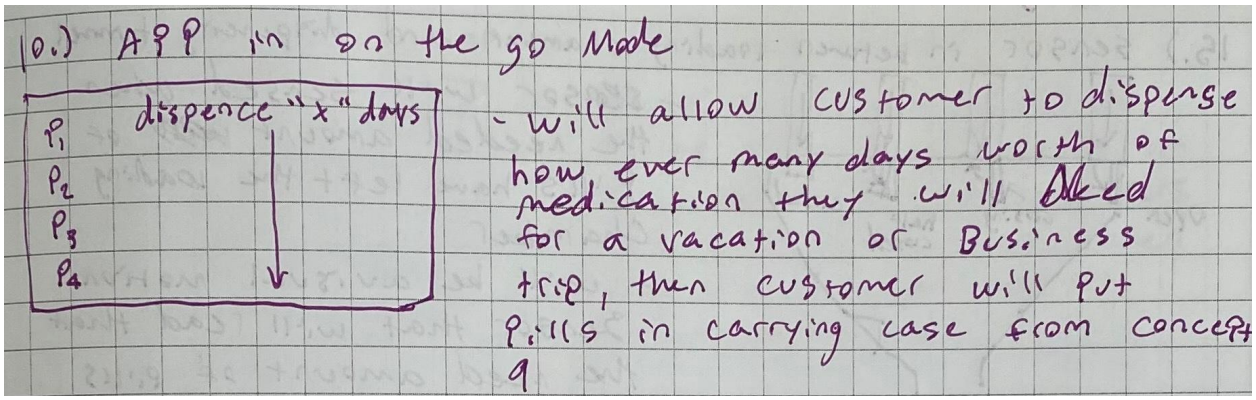

Figure 45: CB 10

11.) This concept is similar to the previous concept of the application in “go mode” when the user is traveling, but what makes this concept different is that this concept is the application sending push notifications to the user. This design works by sending the user a push notification through the application, which will remind them to take their

medication at the needed time with any special instructions such as quantity and if they need to take the medication with food to avoid nausea.

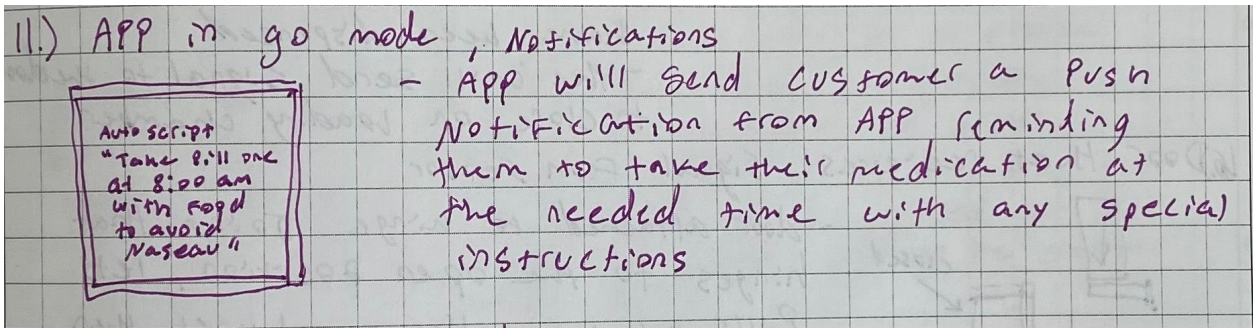

Figure 46: CB 11

12.) This concept is for a touch screen on the front of the dispenser. In this design the touch screen will be in sync with the application in order to provide visual information and instructions to the user on the dispenser. The information displayed on the touch screen could be for example, any special instructions such as quantity and if they need to take the medication with food to avoid nausea.

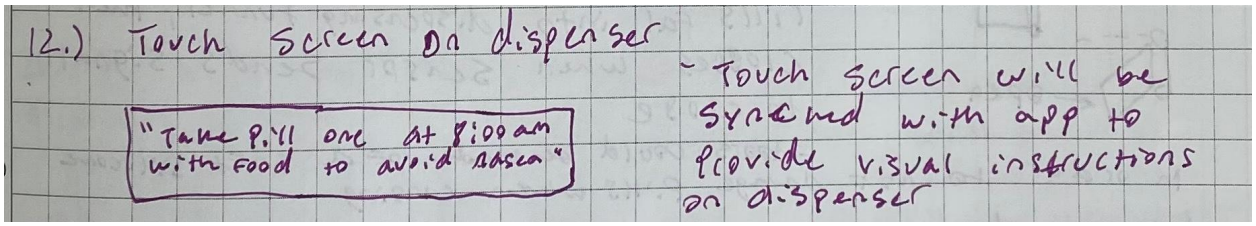

Figure 47: CB 12

13.) This concept is what the dispenser looks like from a frontal view with all previous design components.

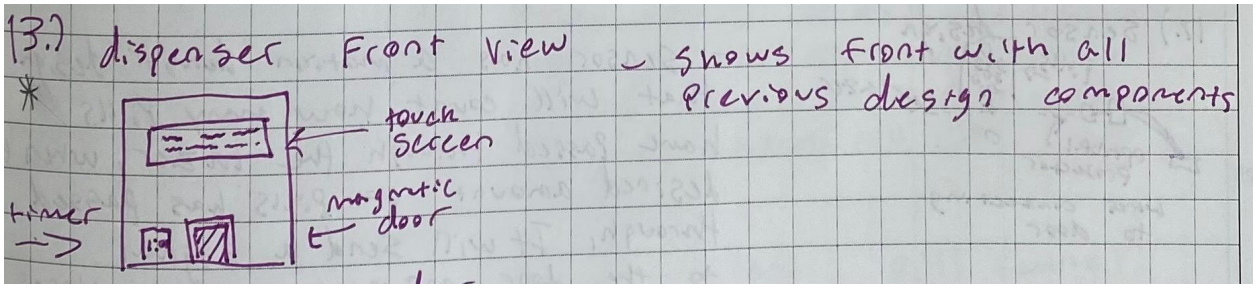

Figure 48: CB 13

14.) This concept is for an alarm reminder on the back of the dispenser. This design works by having an alarm that will produce a loud audible sound similar to that of a microwave. This alarm will be in sync with the timer as well as the magnetic door. This concept was made in order to provide a secondary reminder of the user in case their phone was off, as they would not receive the push notification from the application. So, the alarm is a secondary emergency reminder.

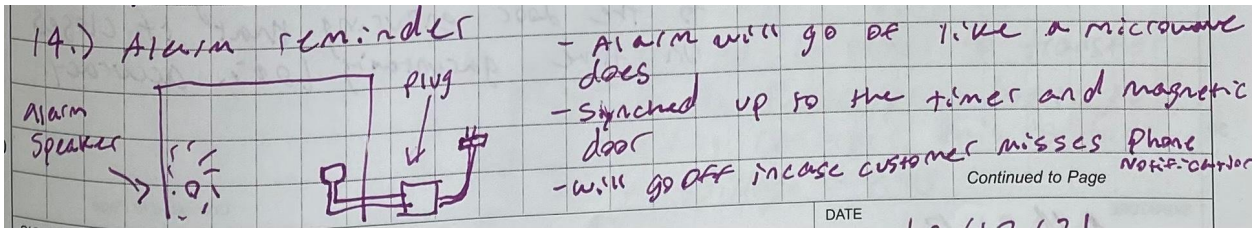

Figure 49: CB 14

15.) This concept is for a sensor found in between the loading chamber and the dispensing tunnel. This design works by using a sensor that will sense when the needed amount of pills have fallen from the loading chamber and into the dispensing tunnel. This sensor will be a visual motion sensor that will read the motion and analyze that the needed amount of pills have been dispensed. Once the sensor reads that the correct amount has been dispensed that the sensor will send a signal to close the loading chamber.

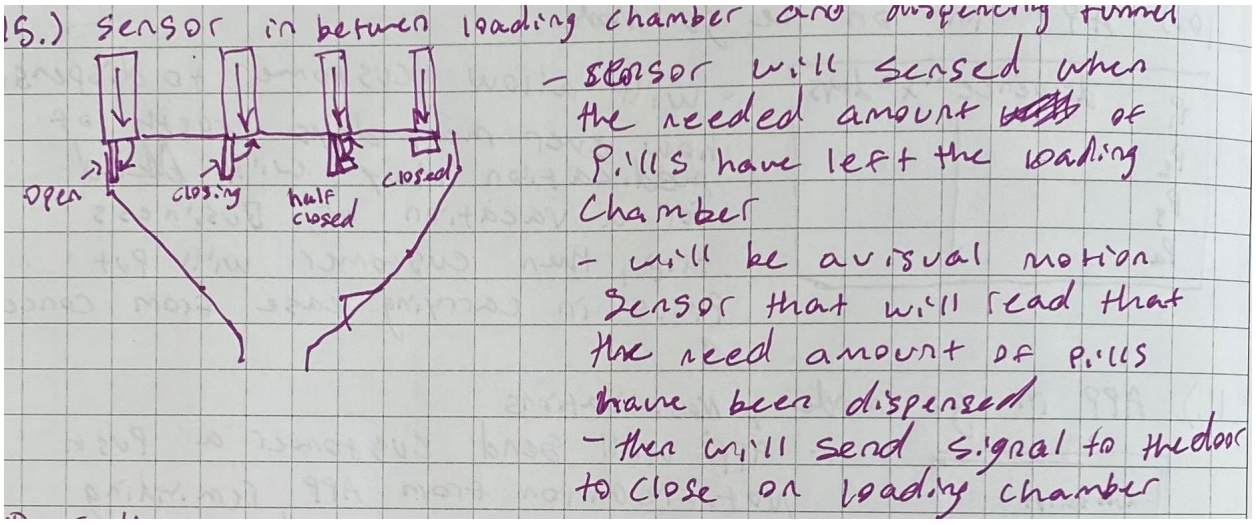

Figure 48: CB 15

16.) This concept is for the doors that receive the signal from the motion sensor. In this design the doors are attached to a hinge joint that hinges to the open position shown. When the door is hinged at the open position, the pills will fall into the dispensing tunel. When the sensor reads that the correct amount has passed through the opening provided by the door being hinged into the open position, the motion sensor will send a signal back to the door which will then hinge back to the closed position. These doors will be manufactured out of a soft silicone in order to not damage the remaining pills in the loading chamber during the closing function.

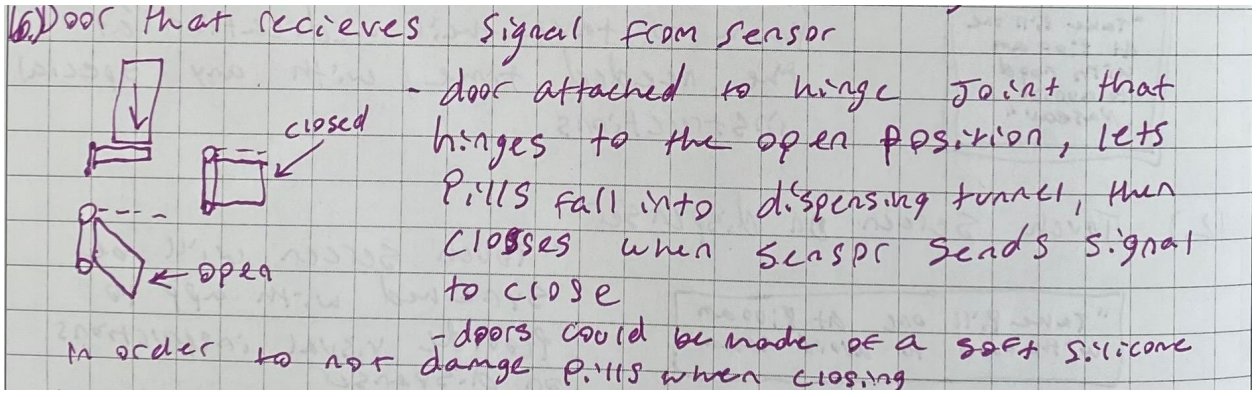

Figure 49: CB 16

17.) This concept is an expansion on the sensor design from concept 15. In this design the sensor has a motion sensor design that will count how many pills have passed through the lazer that is projected by the optical projector. When the desired amount of pills have passed through the lazer the sensor sends a signal to the silicone door to close ensuring the user 100% accuracy.

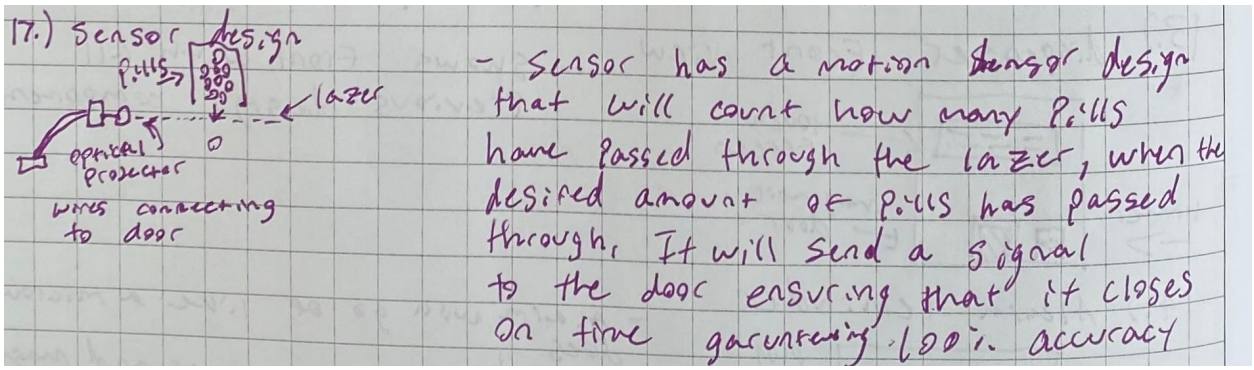

Figure 50: CB 17

18.) This concept is for a loading door on the top of each of the four loading chambers. This design features a simple hinge door placed on the top of each pill column. This will be opened when the user is dumping their new prescription into the chamber when they have received it from the pharmacy.

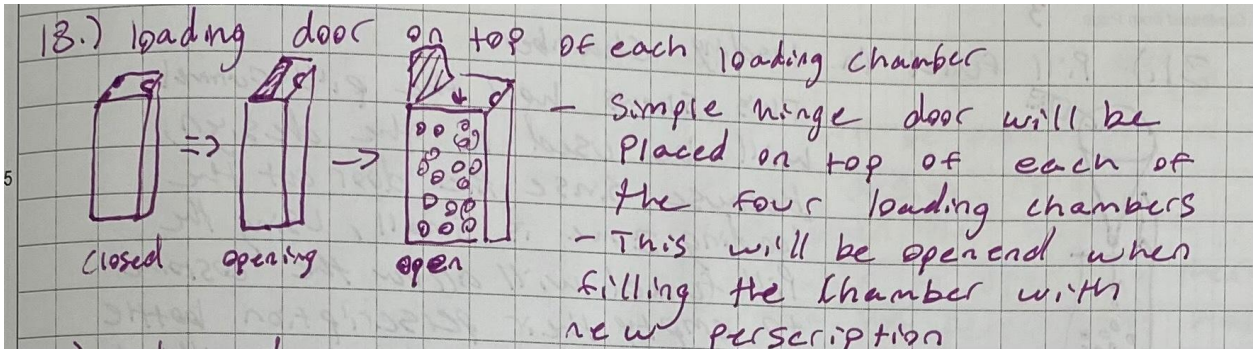

Figure 51: CB 18

19.) This concept is a further concept of the door design featured in the previous concept. In this design the door will be in place so that no foreign substances can enter the loading chamber, as it will remain closed at all times other than when the user is loading the pill columns with new

prescriptions.

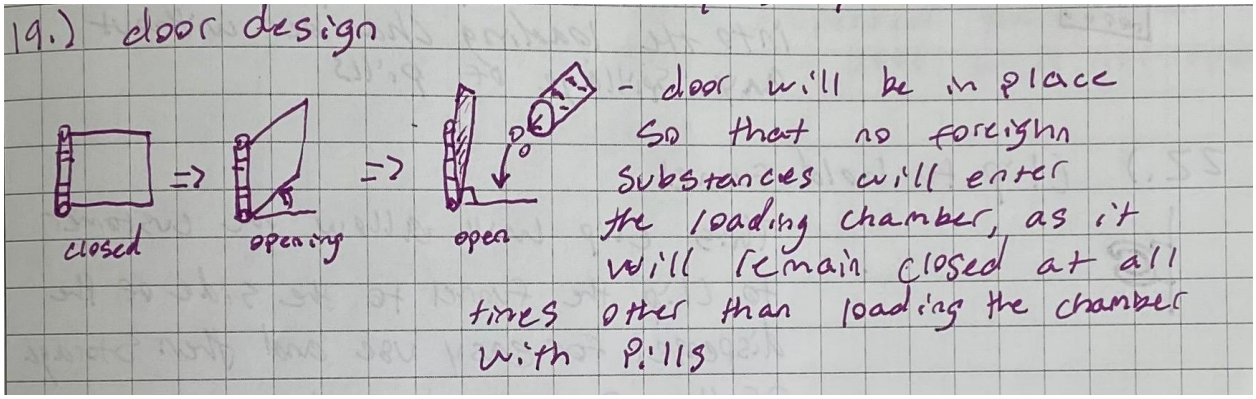

Figure 52: CB 19

20.) This concept is for a carrying case for the hub machine and travel case. This design featuring a carrying case will be the case that is used for the transport of the whole hub, if for example you are moving homes or taking an extended vacation where this would be necessary. This carrying case will be lightweight and made out of aluminum to be strong enough to protect the hub, but not be too heavy for the user to carry or transport for an extended period of time. In this carrying case there will be room for the hub, travel pill case and the power brick charger.

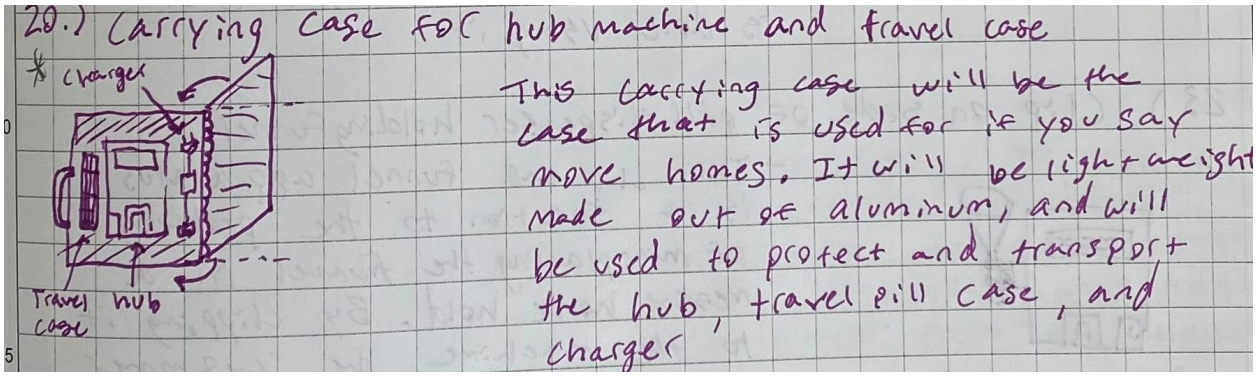

Figure 53: CB 20

21.) This concept is for a pill funnel. The design features a simple plastic funnel that will be used in order to transfer pills from the prescription bottle to the loading chamber. This is done to avoid spilling pills into unwanted places.

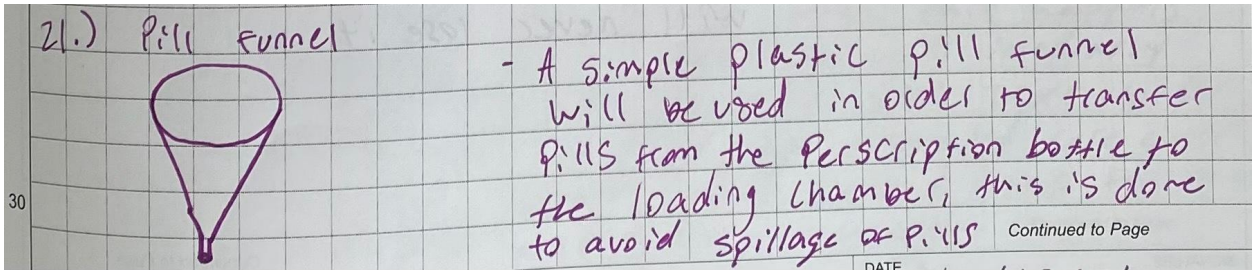

Figure 54: CB 21

21 part 2.) This concept shows how the pill funnel works when the user is pouring their pills into the loading chamber when they receive a new prescription bottle. This visual shows how the pill funnel will be used in the design since the door above each loading chamber is small, the pill funnel will allow the user to empty their prescription into the loading chamber without the spillage of any of the user's medication.

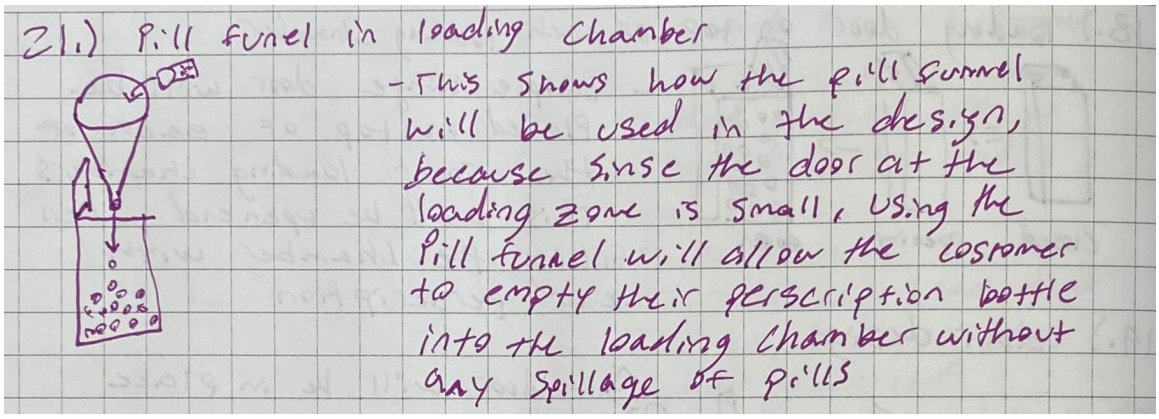

Figure 55: CB 21 part 2

22.) This concept is for a clip on the side of the machine that will house the pill funnel when it is not in use. This design of a clip will allow the customer to clip the funnel to the side of the machine for easy use and then storage of the funnel when the user is done using it.

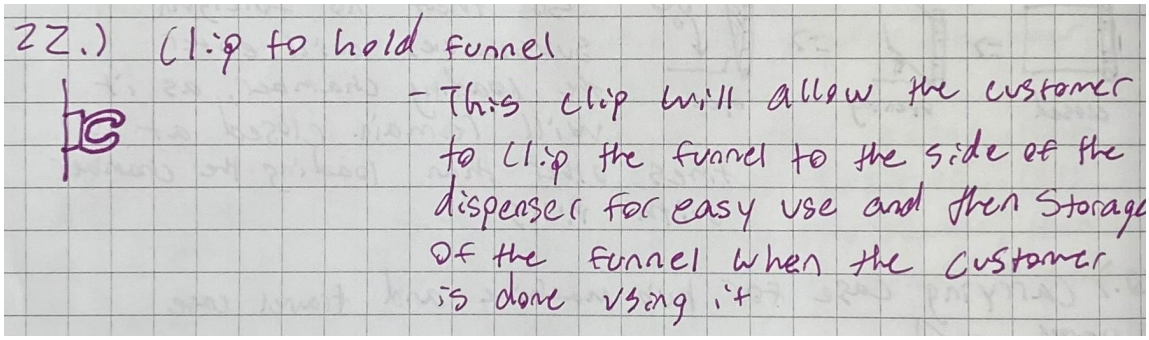

Figure 56: CB 22

23.) This concept is a visual that depicts what the funnel will look like when clipped in from a frontal view of the machine. This clip and funnel apparatus is a solution to the problem of misplacing the funnel in a potentially messy house hold. By clipping it to the side of the machine, it guarantees that the user will never misplace it.

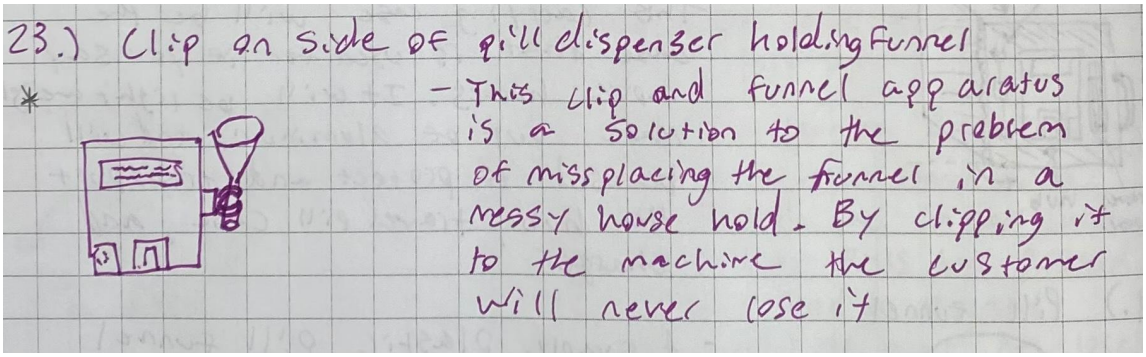

Figure 57: CB 23

24.) This concept is for a signal light that will show the user when the magnetic door is opened. This design of the signal light will be featured on the front of the machine, in which it will be illuminated when the magnetic door will be opened. If the light is not illuminated, then it will let the user know that the prescription has not been dispensed yet, therefore the magnetic door is closed.

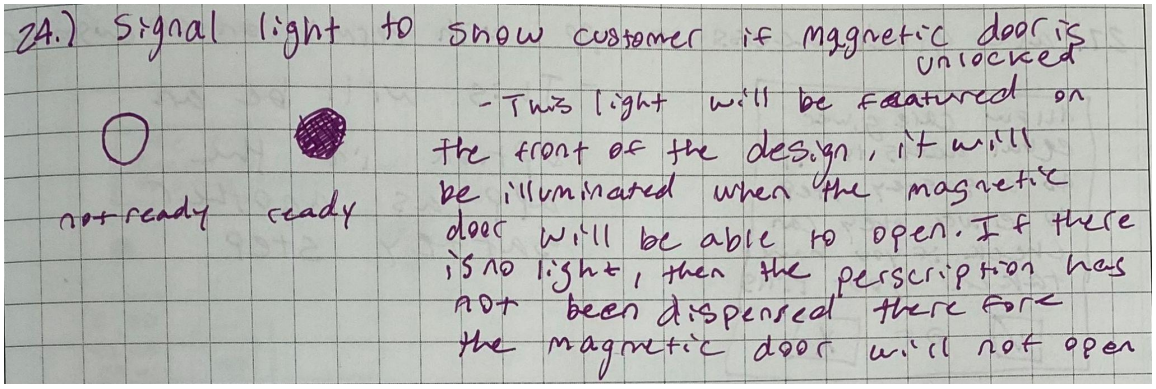

Figure 58: CB 24

25.) This concept is visual depicting the frontal view of the machine with the signal light illuminated.

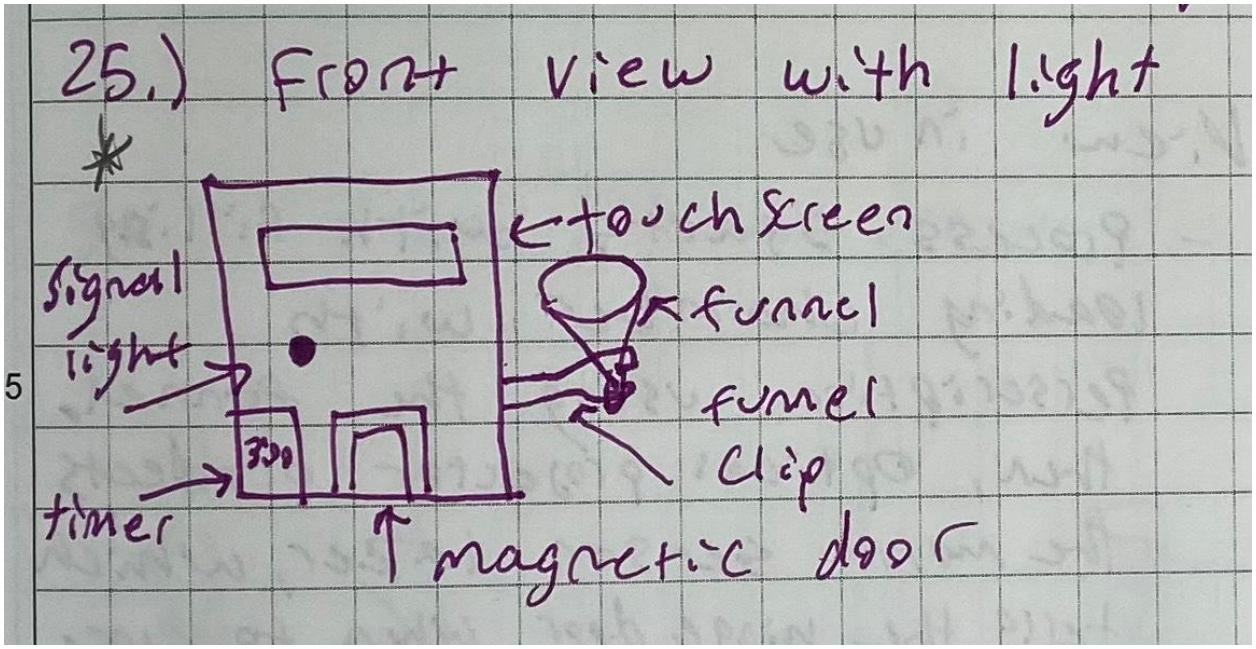

Figure 59: CB 25

26.) This concept is for the application design in relation to the user's primary caregiver. As a safety measure tailored to the user, there will be a backup plan. In the case that the user has yet to take their dispensed medication, the machine will send a signal to the application that the dispensed medication has not been taken. When this happens the application will immediately

send a sms text message to the designated primary caregiver. This is a designed secondary security step for the user's health and safety that could possibly be implemented in this prototype.

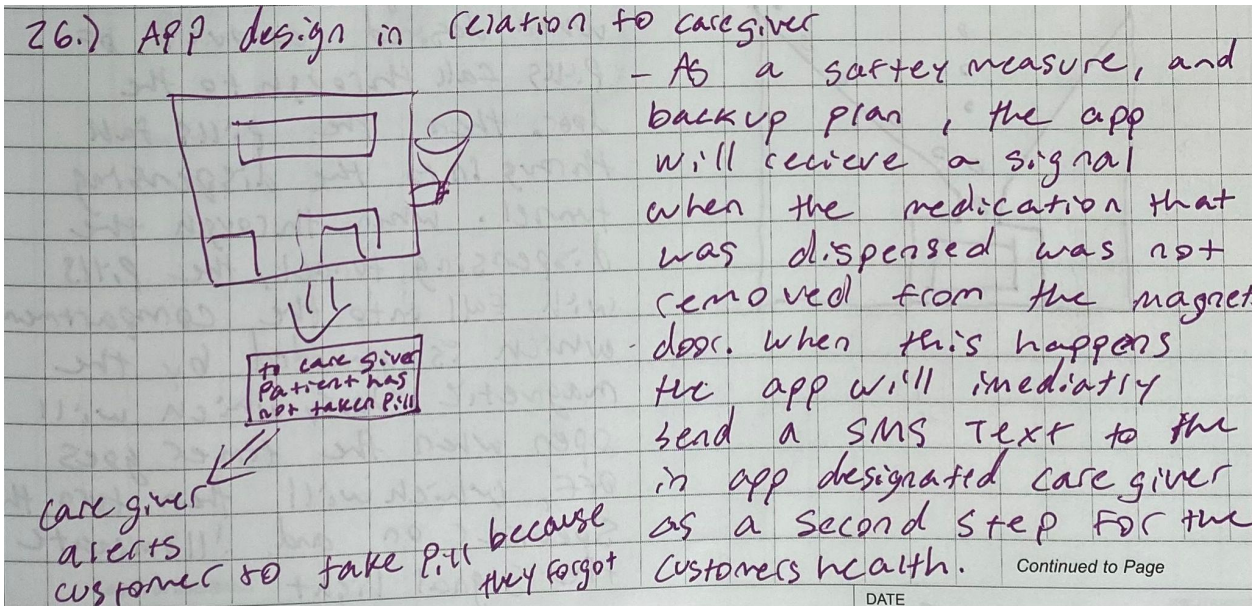

Figure 60: CB 26

27.) This concept is for giving the designated primary caregiver access to the application, only with permission of the user. This will be an option in the app as a possible third safety step for the user. In this situation the caregiver would have the exact same access as the user.

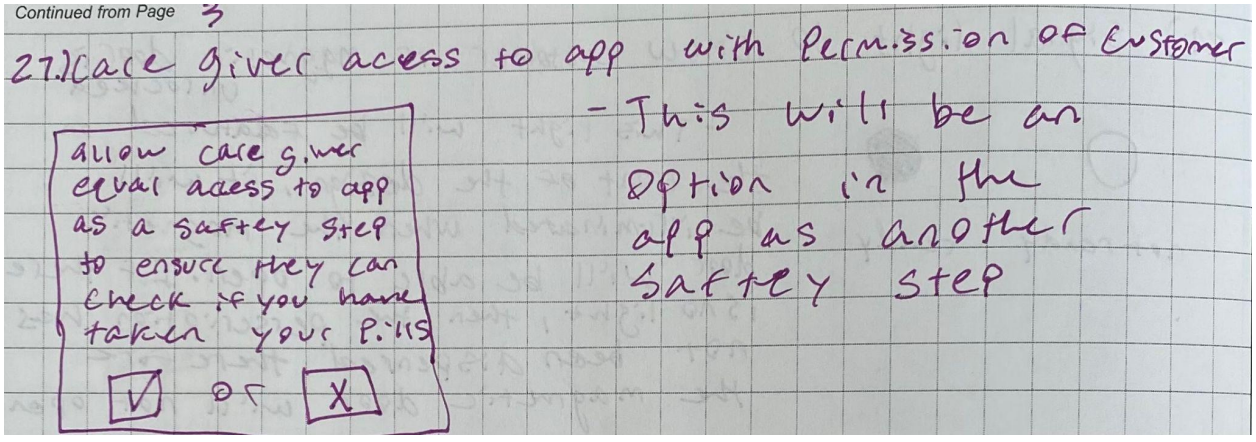

Figure 61: CB 27

28.) This concept is the final internal view of this prototype in use. This process starts with filling the loading chamber with the prescription using the funnel. Then, the application will communicate with the machine to open the door attached to the pill column by a hinge. This door will swing open when a signal is received to do so, at the time where the user needs to take their medication. When the door opens a motion sensor turns on and an optical projector will project a laser. This motion sensor will use the projected laser and sense the motion of each individual pill as they fall. The sensor will then send a signal to the hinge door, which will then swing closed, when the correct amount of pills have fallen from the pill column, through the door and the laser. This is how our team will guarantee 100% accuracy to the consumer. These doors will also be produced to be made out of a soft silicon inorder to not damage pills as they fall down the pill column when the hinge doors close. Once the hinge doors have closed, the pills will fall from the dispensing tunnel into the compartment which is guarded by the magnetic door. When this happens, the magnetic door will open when the timer goes off , which triggers the speaker to produce an audible alarm and illuminate the signal light.

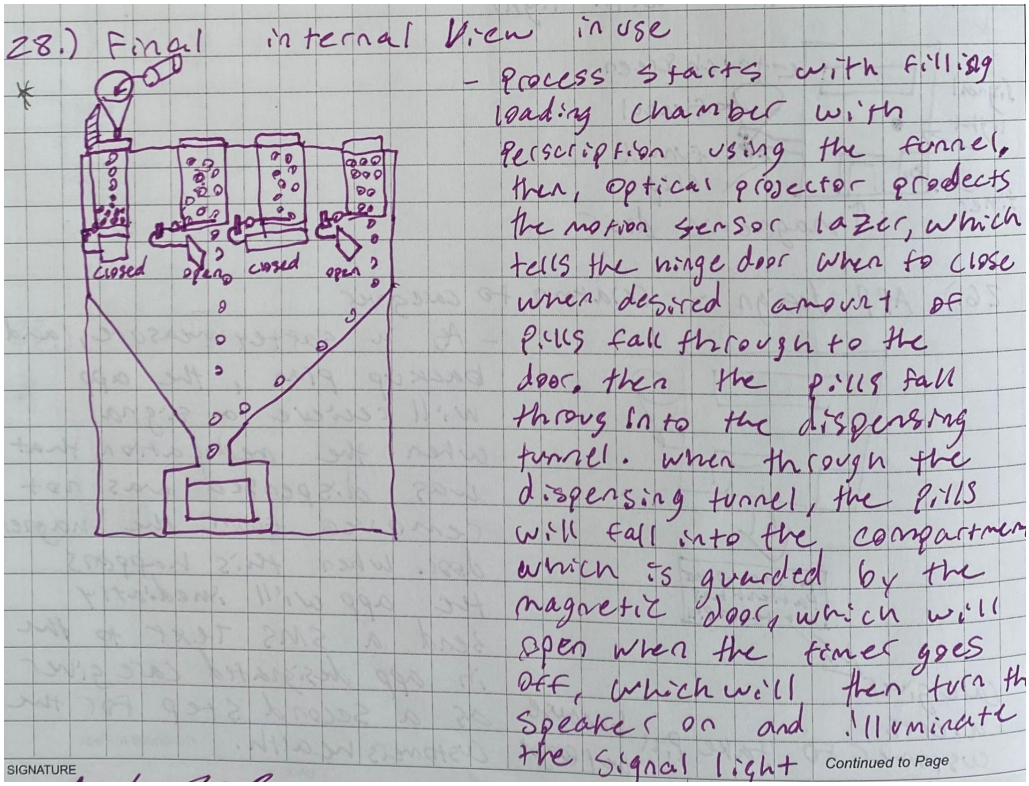

Figure 62: CB 28

29.) This concept is a visual depicting the final frontal view of the machine in use with all the design components that have been shown previously.

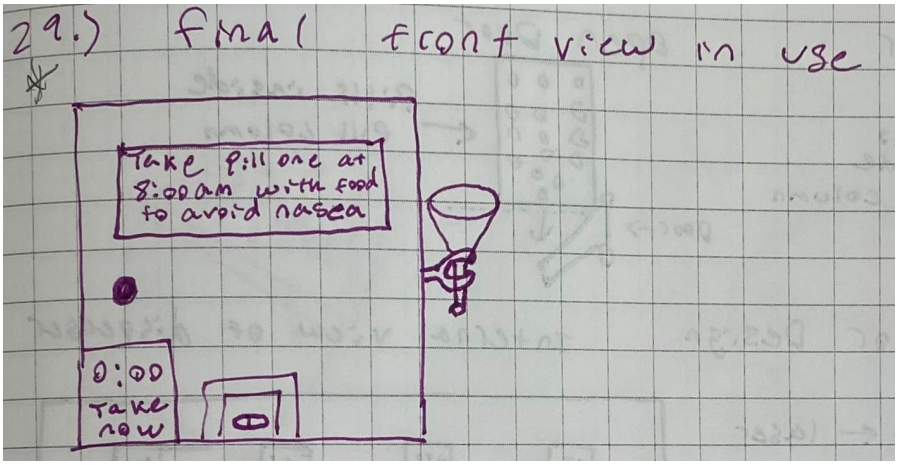

Figure 63: CB 29

30.) This concept is a visual depicting the final back view of the machine in use with all the design components that have been shown previously.

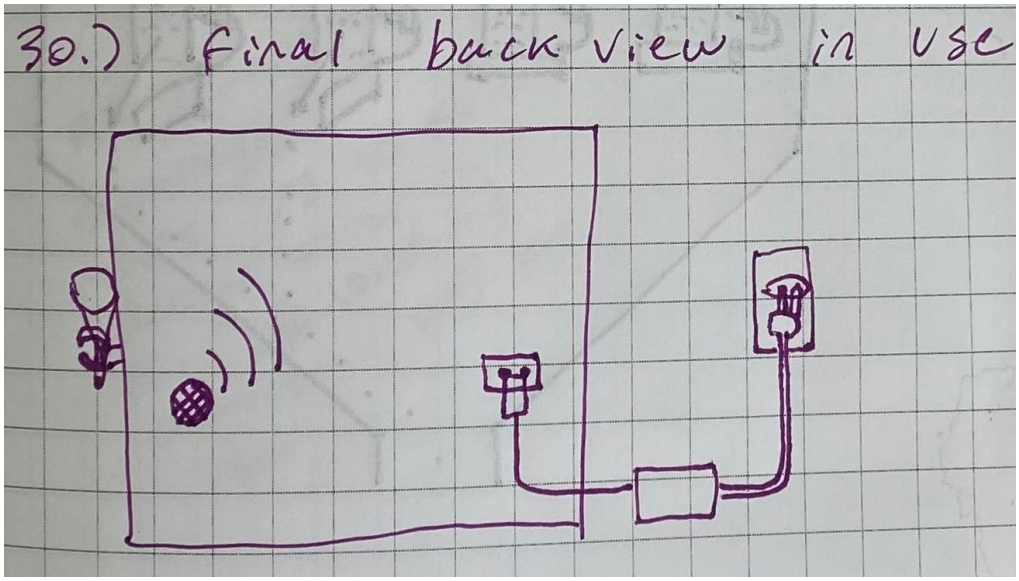

Figure 64: CB 30

5.4 Amy Chicaiza's Concepts:

1. This concept is the form in which the pill dispenser will be manufactured. Injection molding is a must, because it is easier to make products and the price for making injection moldings is less expensive than other processes. It would optimize time and multiple can be produced at one time.

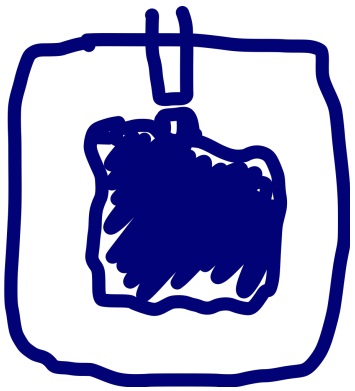

Figure 65: AC 1

2. This concept ties into concept 1, and is about waste management. With injection molding we want to ensure that we don't have a lot of waste. Opting for a plant based plastic or bio-plastic could not only reduce waste but could be remelted and used again. This would not only benefit the environment but also save money on manufacturing.

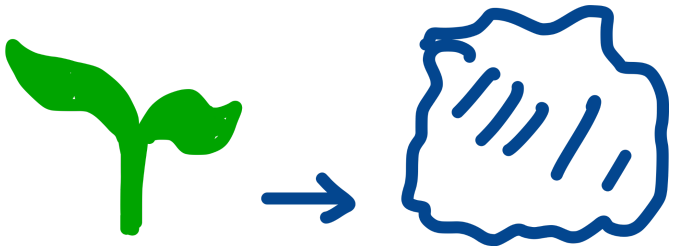

Figure 66: AC 2

3. This concept is a further look into bio-plastic. Although it may cost more to use bioplastic and to manufacture bio-plastic, we still want to ensure quality and functionality for the long term. There are health hazards that can come from oil-derived plastics. These health hazards should be avoided especially if this is being marketed to a group that requires taking medicine by mouth that has been sitting in a health hazardous receptacle.

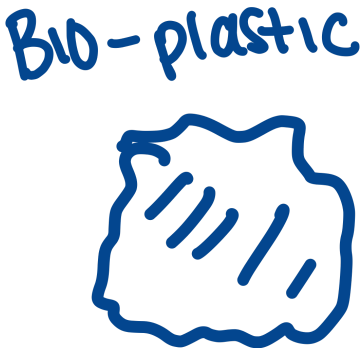

Figure 67: AC 3

4. This concept evaluates the potential of adding a screen to the automated pill dispenser. The screen on the hub machine can be a touch screen, to make it easier to use. It can also allow the user to set up the pills and time of day they need to be taken. This would also

allow the user to input how many pill “towers there are and confirm what drugs are in each.

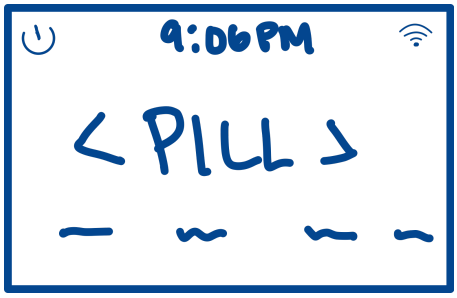

Figure 68: AC 4

5. This concept adds on to concept 4. The screen could also display a “⚠” warning symbol when supply is running low. It will also send a message to the app and can even go as far as to reach out to the pharmacy to refill your prescription.

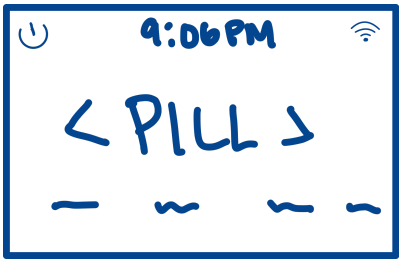

Figure 69: AC 5

6. This concept also continues to add on to concept 4 and 5. The screen will display pills you are about to take to remind you which ones exactly are you taking so there is no miscommunication. It can also remind you how to take it and can read you the description of the pill(s) you are about to ingest.

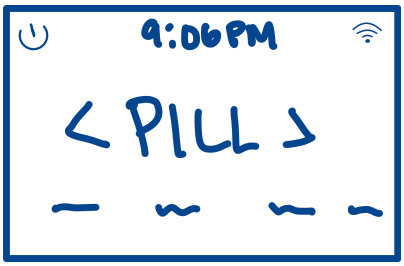

Figure 70: AC 6

7. This concept is part of the actual machine hub. Each pill will have its own little “tower” or section to separate them and when it's time for your pill it will dispense them into a communal cone. The pills can then be dropped into a receptacle and can be taken with ease.

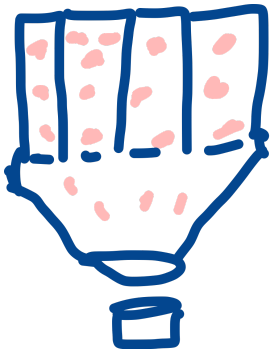

Figure 71: AC 7

8. This concept is an addition to concept 7. The main hub should have a feature where you can add extra compartments / towers. This would be ideal to those that have more than 4 medications that they take a day.

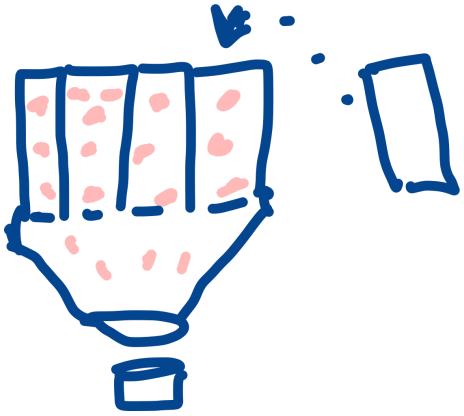

Figure 72: AC 8

9. This concept is an addition to the main hub; it is similar to the days of the week containers. In order to broaden the spectrum to who would be able to use this pill

dispenser, there should be a detachable compartment from the main hub. This would allow people with active lifestyles to be able to use it as well as people with sedentary lifestyles. The hub will still dispense the pills and remind you to take them with you. it would only have three compartments for the day labeled “D” for day, “A” for afternoon and “N” for night.

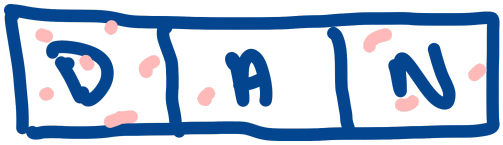

Figure 73: AC 9

10. This concept is another part of the main hub. there should be a sensor added to the platform in which the pills are dispensed into a cup/receptacle. The sensor would alert the user if the cup/receptacle hasn't been lifted by feeling the weight and send another reminder to the user. This is extremely helpful for people that have trouble remembering to take their pills even after notification has been sent the first time.

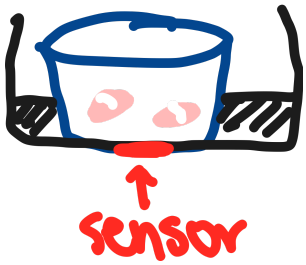

Figure 74: AC 10

11. This concept is part of the towers/compartments attached to the machine. There should be a lining on the inside of the towers that is made out of silicone. It would make it easier to be able to dispense gel pills without popping them. The silicon lining would also add extra cushion and protection to any sort of pills and would prevent them from sticking on to the towers.

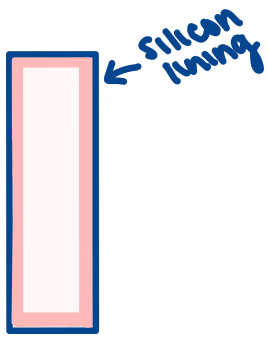

Figure 75: AC 11

12. This concept is also an addition to the towers/compartments attached to the machine. A tight air sealed cap would be beneficial for preserving pills. Specifically for pills with softer outer casing like gel capsules, and it would keep them from sticking together and going stale.

Figure 76: AC 12

13. This concept is another addition to the towers/compartments. Having a soft gate made of material like silicon at the bottom of the towers to let out the pills would make it easier for the pills to fall down. It would also be beneficial for gel capsules as it would provide a safe exit for the pill without having it first.

Figure 77: AC 13

14. This concept is also another addition to the towers/compartments. Being able to keep each Tower at one temperature would allow for the pills kept inside and getting sticky in warmer weather. It would also prevent any sort of pills from going stale and keep them in excellent conditions.

Figure 78: AC 14

15. This concept is for the long term of the already manufactured machine, machine warranty. It would provide customers with the reassurance that if it were to break down they would be able to get a replacement or have it fixed. This would include a monthly fee and would provide a monthly inflow of money to the company to keep it running.

Figure 79: AC 15

16. This concept adds to the sensor factor from concept 10. The sensors would be able to help people with dementia because it would beep when it's ready to be taken/ the pills have been dispensed. The sensor would also provide a safety aspect because it would log time and date the pill was taken. If you forget you could go into the app and double-check to make sure you are not getting confused about the time you took your pill or what pill you took.

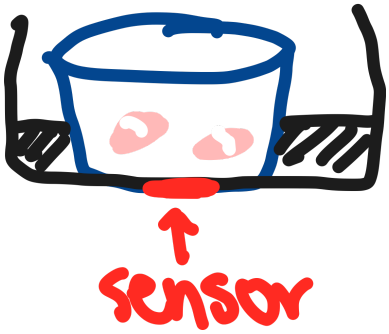

Figure 80: AC 16

17. This concept adds to the dispenser part of the machine. The dispenser can detect if you are running low on a certain medication using a Scale based system. It would alert you or be set to automatically fill up another prescription at your pharmacy of choice.

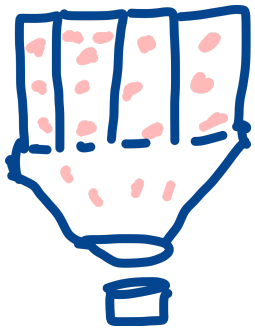

Figure 81: AC 17

18. This concept adds to the main hub for bluetooth. It would be able to connect wirelessly and / or with Bluetooth. If not technologically or app inclined this would benefit the person where it can be set up so that the person can receive reminders via Alexa or Google home devices.

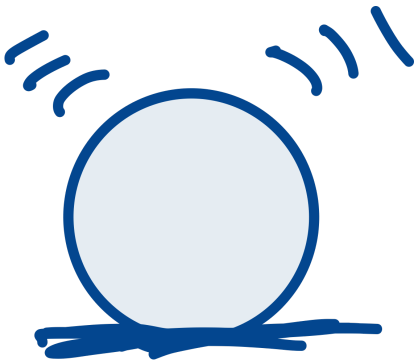

Figure 82: AC 18

19. This concept is an addition to the towers/compartments. If by any chance the pills do become stuck together then the main hub would set out an alert, warning the user the incident. The Hub would show which pill in particular was dispensed stuck to one another.

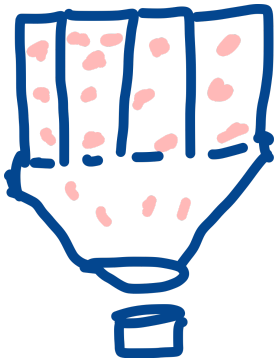

Figure 83: AC 19

20. This concept is part of the app that will be connected to the pill dispenser. Having an app is an easy and simple way to have people connect to the machine. That being said, the app should be easy and have a simple design so it can ensure everyone can use it without an issue. The app will be able to send reminders to users of when to take their medication / when their pills have been dispensed.

Figure 84: AC 20

21. Following concept 20 this concept is also part of the app. The marketed audience is usually people with vision. Instead of just setting up reminders that pop up on the screen it can also have audible reminders. It doesn't only let you know that it's time to take your pill but, it also allows you to have a choice to have the app read the description of the medication.

Figure 85: AC 21

22. This concept is also part of the app design.The app could be designed to scan prescriptions and automatically add it to the main Hub machine. You can add it to the app and it will set it up for you on the machine. Theoretically, it would know how many pills you need to take per day and instructions of how it should be taken (i.e. with food and water or without anything).

Figure 86: AC 22

23. This concept is the second to final addition to the app. After a certain amount of time the app can be set to send reminders to primary emergency contact. This would allow any caregiver or family member to know that the consumer has not yet taken their pill and time has now passed (could be set to 30 minutes or an hr). This would provide a safety aspect and reassurance to the loved ones family and, or caretaker.

Figure 87: AC 23

24. This concept adds to the screen on the hub machine. The buttons on the screen that are attached to the hub machine can be made so they are sensitive. The sensitivity would allow you to use the buttons with ease, this is made more for people that have difficulty with things like this, like people with arthritis.

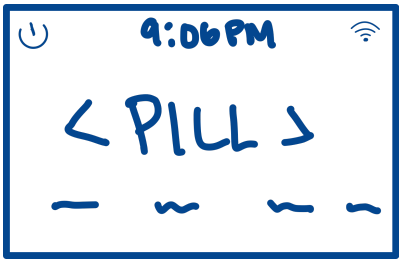

Figure 88: AC 24

25. This concept is an addition to the dispenser portion of the hub machine. The towers/compartments would be able to hold up to a 60 day supply. This information can also be sent from the pharmacy to the user's app/hub machine to inform them the correct amount of pills the prescription contains.

Figure 89: AC 25

26. This concept is an addition to concept number 14. The temperature regulation of the hub, will keep it at one temperature. This would be obtained by having a mini Hub on top of the regular Hub machine that would regulate it. This way pills wouldn't become moist with humidity or even melt with warmer weather.

Figure 90: AC 26

27. This concept is the final addition to the app. The app will be watch compatible for people on the go. The phone alerts would transfer to the watch in case your phone is not on you and will send the same reminders and alerts to take your medication if you are on the go and have your portable hub with you.

Figure 91: AC 27

28. This concept is the overall design of the hub machine. The design will be sleek so it can be placed anywhere in your home. Medicine “Towers” can be interchangeable so that it fits better with your space and can be made wider or longer.

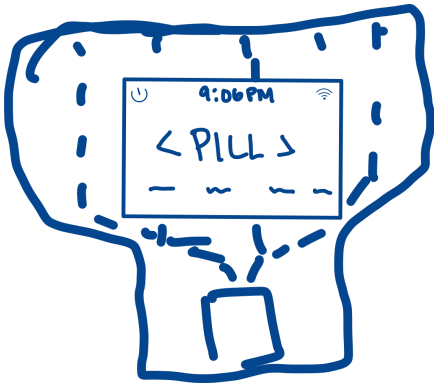

Figure 92: AC 28

29. Concept is in correlation to the receptacles in which the pills will fall into. The receptacles that are used in the pill dispenser should be reusable. The machine hub can self slide them under the “shoot” where pills are being dispensed.

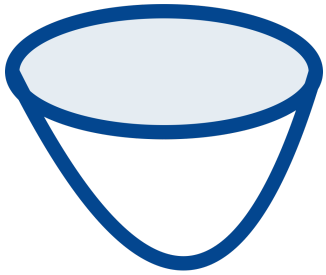

Figure 93: AC 29

30. This concept is an addition to concept 29. The receptacles can be bright colored so they are hard to miss. If they are still sitting out on the counter, it would be easier to catch the users eye apart from being sent the notifications to the app.

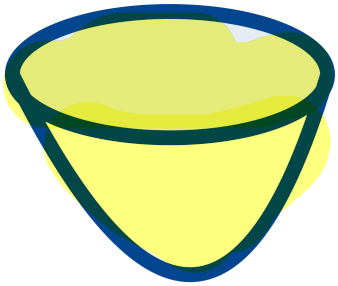

Figure 94: AC 30

5.5 Alex Nocera's Concepts:

1. This concept is designed to remind the user. It contains an integrated timer that tells the user when to take their medication. This device would have a weekly calendar as well.

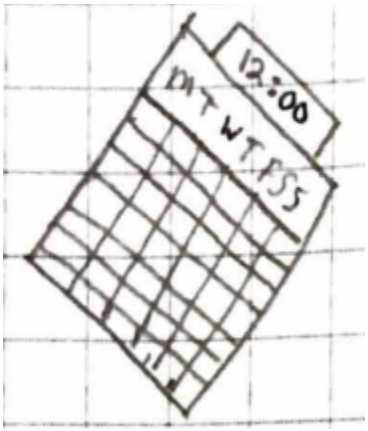

Figure 95: AN 1

2. This concept has a box-like shape that can accommodate different types of pills. With the help of its companion phone application, it has an alarm clock to alert the user. At the top there are four openings to satisfy the design specification of dispensing at least four different types of pills.

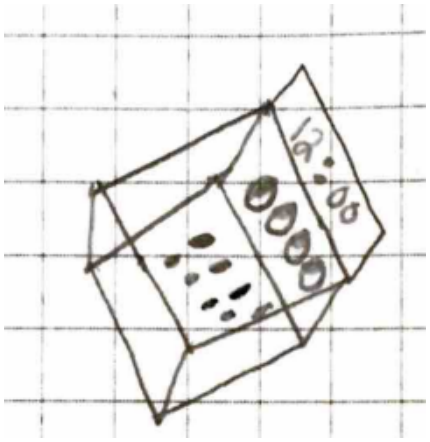

Figure 96: AN 2

3. This concept uses the structure of four tubes with different diameters. Like concept two, it has an alarm clock that cooperates with its companion phone application. The different sized tubes allow for different sized pills in case the user is taking multiple prescriptions at once. The pill gets ejected from one of the tubes depending on its size.

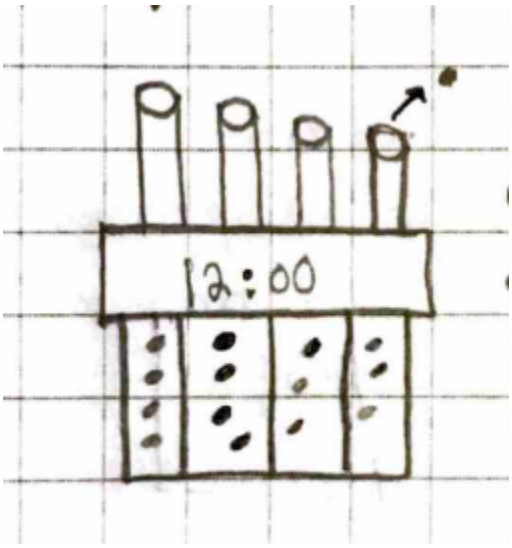

Figure 97: AN 3

4. This concept showcases the importance and use of a plug-in style of machine. The user will most likely rely on a rechargeable battery to keep it going for whenever they need it. Like concept three, it involves a tubing system. However, the difference here is that the

four different tubes all eject into a more dilated tube and are then ejected for the user to take. It also has an integrated alarm clock like previous concepts.

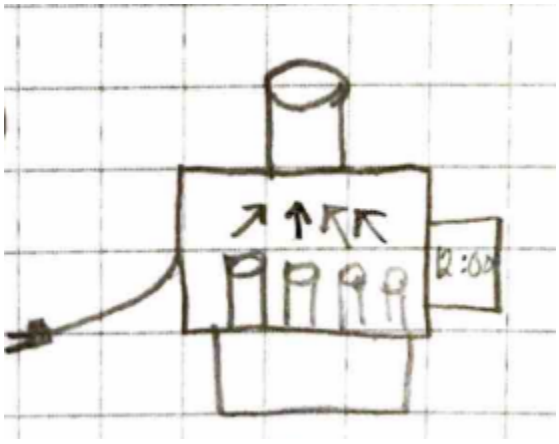

Figure 98: AN 4

5. This concept utilizes a vacuum system. With the integrated alarm clock still in place, as well as being rechargeable, a pill gets lifted into the tubing system and is dispensed into a tray. The tube that dispenses the pill is dilated to contain any sized pill.

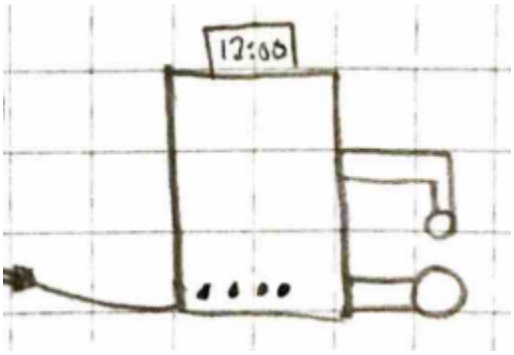

Figure 99: AN 5

6. This concept showcases how the automated pill dispenser's application communicates with the device. When it's time to time a pill, the user goes to the application and accepts the request to dispense a pill. The given side view shows how the pill can end up in one of four tubes. It is then dispensed into a tray for the user.

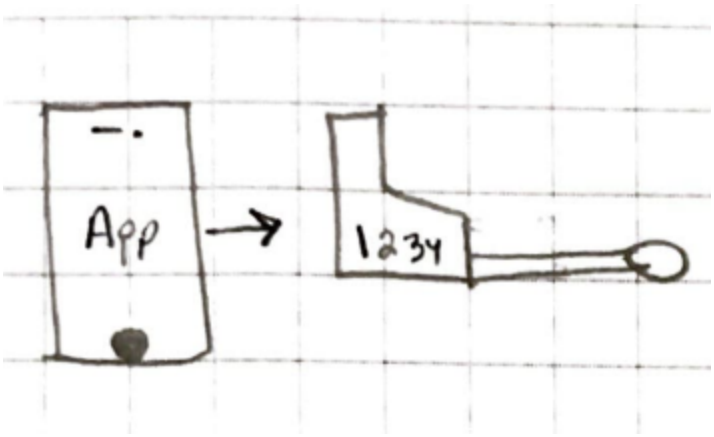

Figure 100: AN 6

7. This concept involves a box design meant to hold pills of all sizes. On each side, there's a tray that can catch a pill when it is dispensed through one of the four holes. It is meant to be a lightweight design and portable. The battery compartment would be at the bottom of the box.

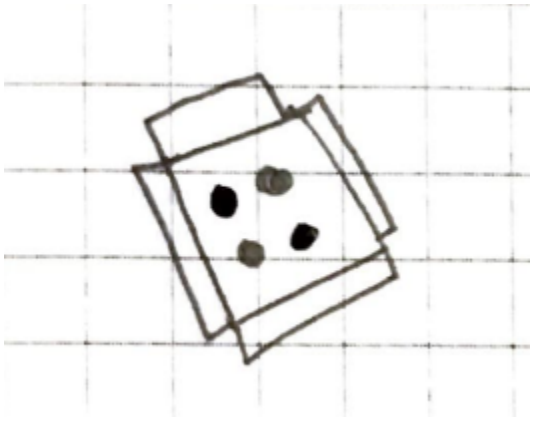

Figure 101: AN 7

8. In this concept, there is an integrated alarm system that cooperates with the device's companion phone application. The long rectangular box contains the pills. When the pills get dispensed, they are brought up through one of the four tubes.

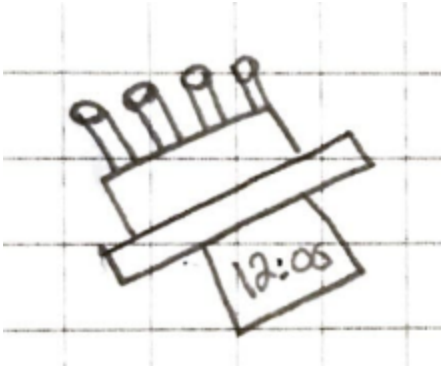

Figure 102: AN 8

9. This concept resembles a traffic light design. Four openings are connected to the compartment where the pills are held. When a pill is dispensed, it comes out of one of the four openings depending on pill type and size.

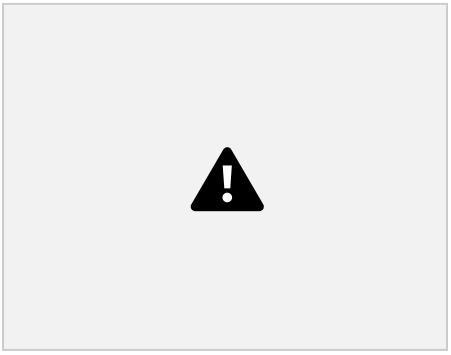

Figure 103: AN 9

10. Concept ten is meant to resemble an octopus design. It has an integrated alarm clock operable from the phone application. It involves four different tubes. When a pill is dispensed from a tube, it lands in its corresponding tray to be received by the user.

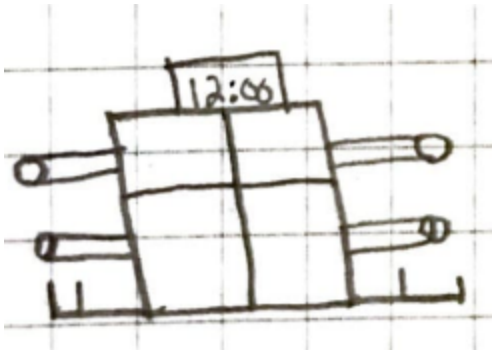

Figure 104: AN 10

11. This concept involves pills getting lifted from the stationary bowl at the bottom of the device. Inside this bowl, pills are contained. The larger tube receives a pill when instructed by the phone application. Four more tubes are connected to the largest tube. Depending on the type and size of the pill, the medication gets dispensed to the user through one of the four pills.

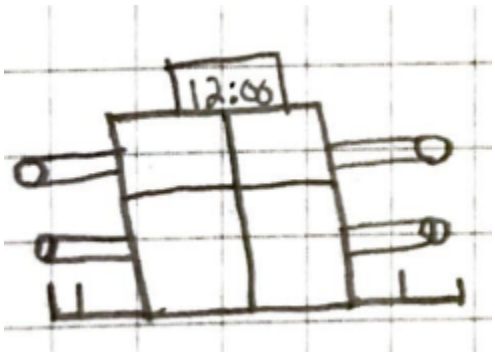

Figure 105: AN 11

12. This concept involves the use of tabs. These tabs are controlled by the companion phone application. When a pill is dispensed, the application activates a tab depending on the pill. Once a tab is pressed at the bottom, a pill is dispensed through one of the four tubes. It has a box-like structure, allowing for storage of many different types of pills.

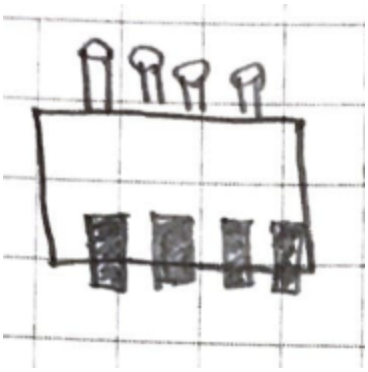

Figure 106: AN 12

13. Concept 13 is similar to concept seven. Instead of a box-like design, it has a rectangular prism that contains the pills. It also has four openings where the pills get dispensed from. There are four trays the pills can be dispensed onto so the user doesn't have to worry about dropping a pill onto the floor. It is a lightweight design and still portable, but may take up more space.

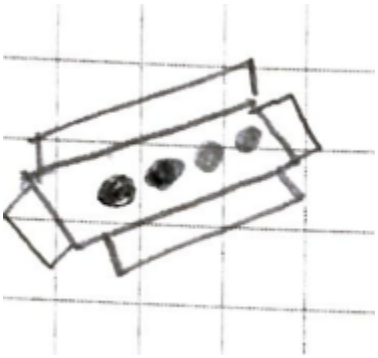

Figure 107: AN 13

14. This concept allows the user to drop their pills into a tube with a wide opening at the top. Once the pills enter the device, they land in a box-shaped compartment where they are stored. On the bottom there are four tubes. Once a pill is selected from the application it gets dispensed through one of the four tubes depending on its type and size.

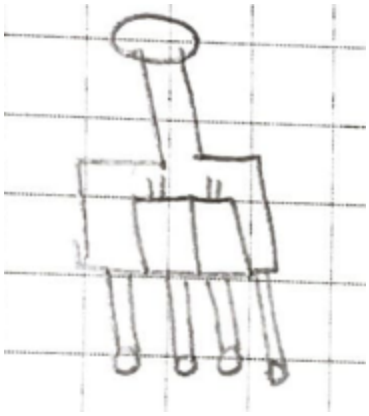

Figure 108: AN 14

15. This concept uses a design of a circular tube where pills are stored. Inside this circular tube there are four different compartments located ninety degrees from each other. Each compartment is meant to hold a different type of pill. These tubes would be made of rubber. Lastly, depending on the pill selected from the application, it is then ejected through one of the four outside tubes.

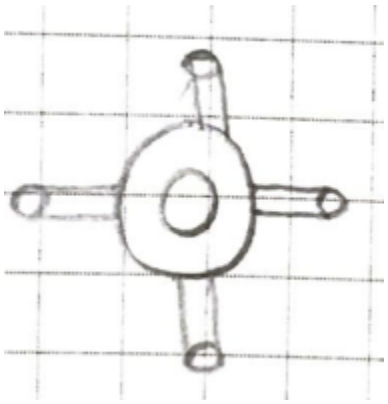

Figure 109: AN 15

16. This concept resembles an ice cream cone. Pills are stored in the cone shaped compartment. A pill gets dispensed through one of the holes at the top. Each hole is has a different diameter to accommodate different sized pills. This concept is also plug in friendly allowing for an extended battery life.

Figure 110: AN 16

17. This concept is reminder friendly. Pills are stored in the box-shaped storage compartment. Pills enter the machine via the four tubes on the back of the box rather than dispensed through them. However, the pills get dispensed through the frontal tube and land onto a tray at the front of the device. The tray has a back piece that acts as a backboard in case the pill bounces after coming out of the frontal tube.

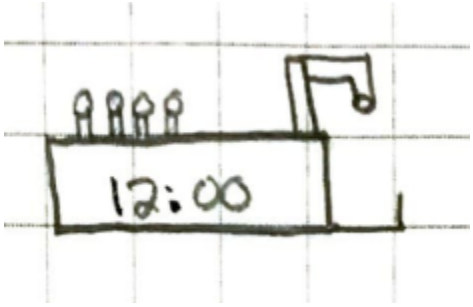

Figure 111: AN 17

18. This concept resembles a disc. The disc acts as the heart of the design. Here, the pills are stored and are also initiated to be dispensed. At the bottom of the disc the battery is kept. At the top of the disc there are four different sized holes. Each hole is connected to a tube

where the pill finally comes out. This design is meant to be portable and extremely lightweight.

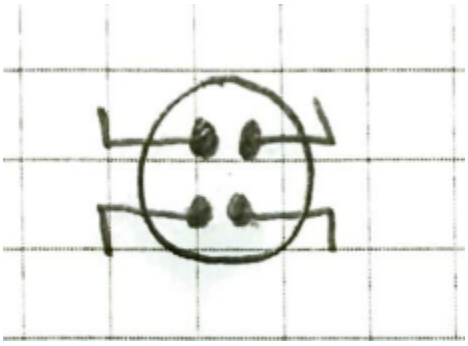

Figure 112: AN 18

19. This concept has a back compartment where pills are put into the machine and are stored. Next to this compartment there is a tubing system where the pill is brought up. At the tip of the triangle there is another connected tube which dispenses the pill. The companion application choses a specific pill to be dispensed. It is designed to have a long battery life and be accurate in dispensing pills. The pill lands into a tray to be received by the user.

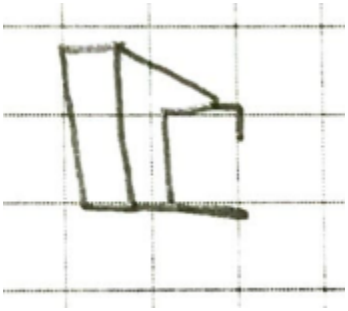

Figure 113: AN 19

20. This concept involves a disc connected to a large tube. Pills are stored in the large tube and are dispensed through the top disc. The disc contains four different sized holes to accommodate for a variety of pills. Once a pill is dispensed through the disc, it is dispensed again through the additional tube at the front. The pill then lands on the tray at the bottom.

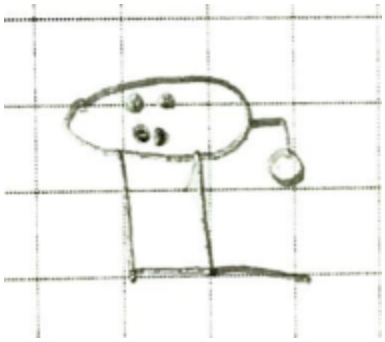

Figure 114: AN 20

21. This concept has a heavier design, however it allows for a much larger storage. Pills are dropped down into one of the tubes depending on their size. They are then dispensed through the tube at the front of the device for the user. The battery for this machine is stored at the bottom of the storage compartment.

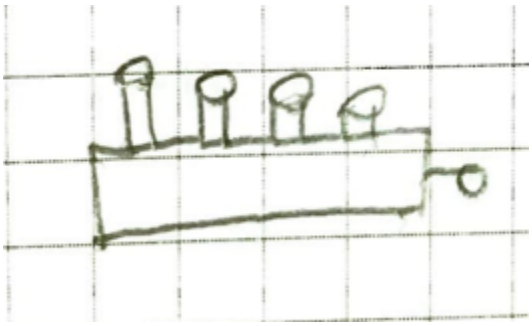

Figure 115: AN 21

22. This design is meant to be as lightweight as possible. However, there is not much room for storage as the pills are delivered and stored inside the same pill. It has four tubes on the back where pills can be stored. The pills travel through the middle of the device when they are being dispensed. Finally, the pills come out of the frontal tube and land into a cup for the user. This design is also plug in friendly.

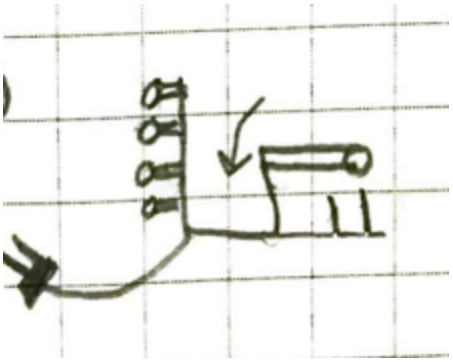

Figure 116: AN 22

23. This design showcases the hexagonal disc. The disc has four holes at the top where the pills come out and can be retrieved by the user. Each hole is of different diameter. This allows for different types of pills to be stored and dispensed. The battery pack sits at the bottom of the hexagon.

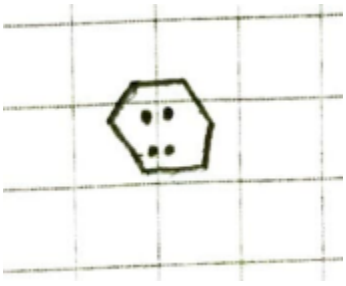

Figure 117: AN 23

24. This design involves many tubes. It is plug in friendly allowing for an extended battery life. Pills enter the top of the machine through the large tubing system at the top. When they are being dispensed, they come through one of the tubes at the bottom. Each tube at the bottom has a different diameter allowing for different pills to be stored and dispensed.

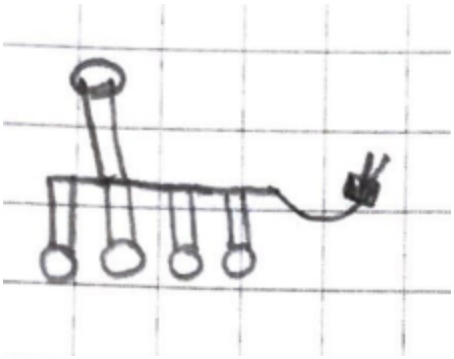

Figure 118: AN 24

25. This concept involves an automated pill dispenser that operates via its companion phone application. Pills are stored inside the ball-shaped compartment. A tube is connected to this ball where the pills are dispensed. The ball is designed to have a good amount of volume allowing for many pills of different shapes and sizes to be stored. The pill then lands into a curved tray to be received by its user. This machine is battery operated and is estimated to weigh only five pounds.

Figure 119: AN 25

26. This concept is shaped similar to concept thirteen. However, the battery pack is located at the back of the machine. Pills stored in this machine enter through one of the four holes. Each hole is of different diameter allowing for a variety of pills to be stored. Pills are dispensed through the frontal tube. This design has a heavier weight but allows for more storage.

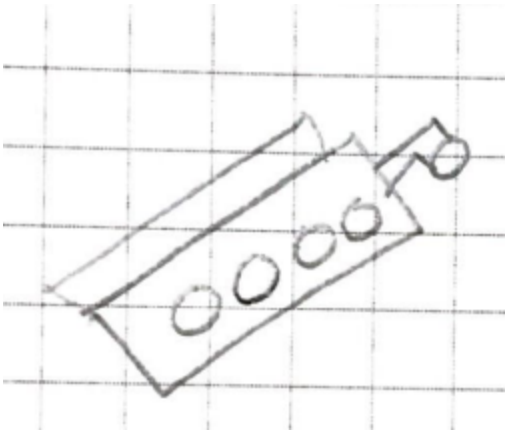

Figure 120: AN 26

27. This concept is plug in friendly so its users never have to worry about the device dying on them in time of need. Pills enter the large tubing system at the top and are stored in the larger compartment. Smaller pills are able to fall down into the smaller compartment. By separating the different types of pills, the device is able to be more accurate. The pill gets dispensed from the tube at the bottom right.

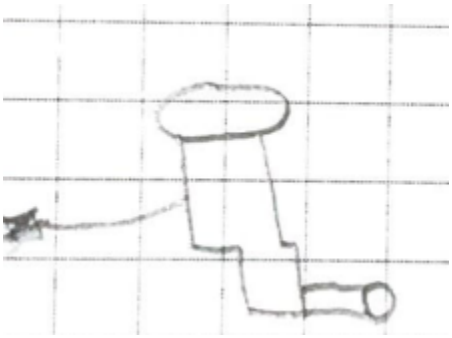

Figure 121: AN 27

28. This concept is similar to concept twenty-five. It uses a ball-shaped storage compartment allowing for pills of various sizes to fit. It can be plugged into a wall for an extended battery life. This concept has quite reliable storage for larger prescriptions.

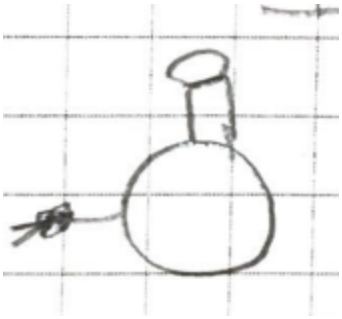

Figure 122: AN 28

29. This concept has a machine connected to a pill selector (seen on left) that cooperates with its companion phone application. On the right, pills enter the machine through the top tube and are stored in the box below it. The pill lands in the tray at the front of the machine.

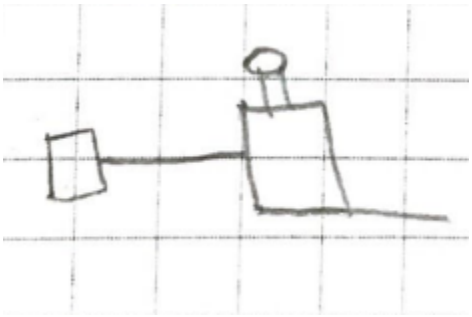

Figure 123: AN 29

30. This concept is a visual representation of four tubes connected to the pills' storage compartment. Each tube has a different diameter to accommodate for at least four different types of pills. This design is meant to be lightweight and accurate.

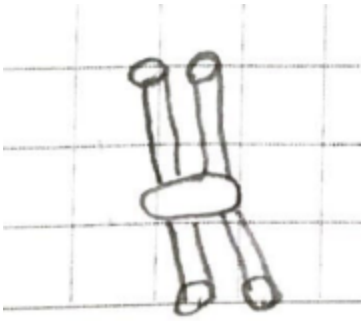

Figure 124: AN 30

5.6 Akash Shah's Concepts

1. This concept drew inspiration from a coin sorting machine. The pills are inserted from above and based on shape and size are sorted into a container. A sensor on both sides are used to identify pills in real time so they can be appropriately sorted.

Figure 125: AS 1

2. This concept was designed with the portable aspect in mind. The pills are all sorted into individual containers that can be configured to customizable categories; morning, afternoon, and night for example.

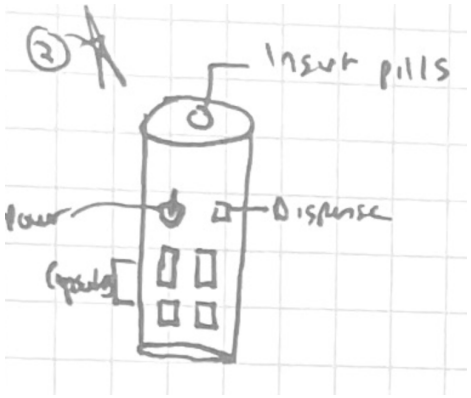

Figure 126: AS 2

3. This concept attempts to combine both concepts 1 and 2 to offer solutions for sedentary and active lifestyles. Pills can be dispensed from 4 different chambers into a cup, or each chamber can be used to contain a collection of pills for a certain time of day.

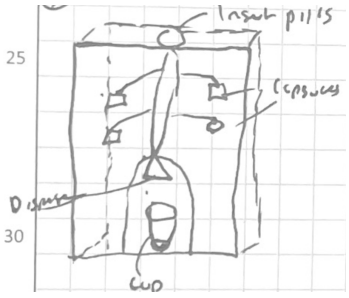

Figure 127: AS 3

4. This concept focuses on the application that can interact with the machine. It shows the main menu where the user can set up schedules, dispense pills and input data for their prescriptions. Another function of this application is that multiple users are supported so

that multiple people can synchronize their prescriptions and schedules on one account. This can be useful for a caretaker in a hospice or hospital with multiple patients.

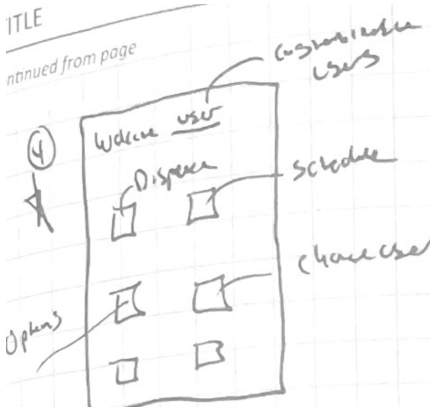

Figure 128: AS 4

5. This concept was to illustrate a machine that supports multiple users on site instead of via application. Each block is reserved for a separate user and each user can receive their pills from their block.

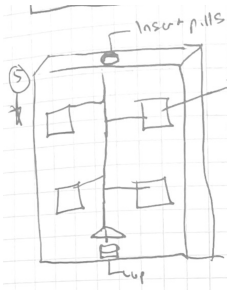

Figure 129: AS 5

6. This concept is the same as concept 5 except that it satisfies the portability needs instead. Each block contains a user's pills for a given period (such as the morning, or afternoon). The user can remove the entire block from the machine.

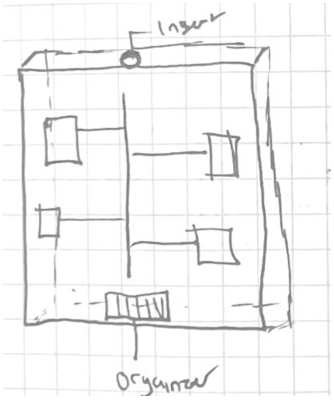

Figure 130: AS 6

7. This concept allows the typical pill bottle received from pharmacies to be inserted directly into the machine and it will be stored internally into a chamber. Up to 4 prescriptions can be stored and then it can be dispensed from each individual chamber as needed.

Figure 131: AS 7

8. This concept is to maximize the portability by manually inserting your pills into this bottle which contains a configurable timer that will ring via speaker to alert you when it is time to take your pill. This design only supports 1 prescription.

Figure 132: AS 8

9. This concept was merely the typical Sunday through Saturday pill organizer. This was conceptualized in order to illustrate its shortcomings against other designs and the needs of a reminder, an automated sorter and application.

Figure 133: AS 9

10. This concept builds off the S-S pill organizer by being able to insert that organizer into the machine. Pills will be dispensed from 4 chambers into each day in the organizer as needed.

Figure 134: AS 10

11. As concept 1 did, this concept builds off a different type of mechanism. A gumball machine simply lets its contents fall into the dispenser through gravity. This design only supports 1 prescription.

Figure 135: AS 11

12. This concept is similar to concept 10 except that the insertable organizer is in disk form. This is because the disk is slightly more compact than the box.

Figure 136: AS 12

13. This design will dispense water for you to consume your pills with. It does support multiple pills but does not support portability.

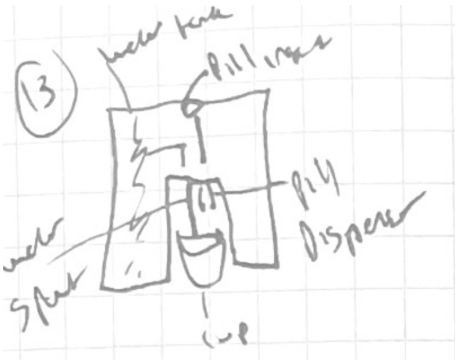

Figure 137: AS 13

14. This concept caters the issue of the elderly having issues with smartphone applications. So this concept is for an application that is extremely simple and only has 1 function, which is to dispense.

Figure 138: AS 14

15. This concept dispenses pills straight into the established pill bottle.

Figure 139: AS 15

16. This concept has a speaker so that the user can receive an audible reminder whilst near the device. This dispenses 1 pill on a timer.

Figure 140: AS 16

17. This design differs from others where the user inserts the pills in the bottom inside a drawer. The pills are removed from the drawer via vacuum pump and appear on the top of the device to be removed by the user.

Figure 141: AS 17

18. This is a handheld concept where it's possible to pocket the device. The pills are dispensed 1 at a time via button press. This only supports 1 prescription.

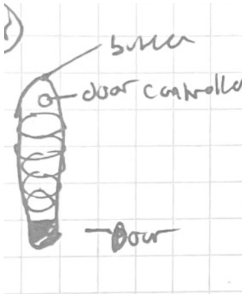

Figure 142: AS 18

19. This concept is inspired by the rotary phone. The pills are preloaded by the user and then presented like buttons on a rotary phone where they can be removed.

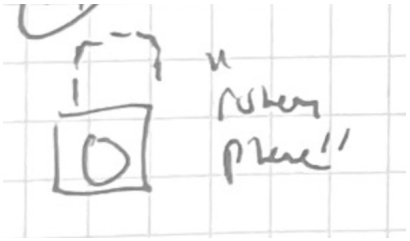

Figure 143: AS 19

5.7 Pugh Analysis

Pugh Chart for Automatic Pill Dispenser

| | Datum | Concepts | | |

Figure 144: Team Pugh Analysis Chart

After completing the three iterations of the pugh analysis for the groups total of 150 individual designs, the group ended up deciding that the best design concept available was “AC 10”. We narrowed it down to this design after three total iterations of the pugh analysis, ranking each individual design against each other. As a group the design “AC 10” was chosen because it performed higher than the rest in the categories of elederly usability, reliability, retail price, the use of a phone application, compactness, weight, possibility of a warranty and differentiation of other pill dispensers on the market. This is the design that the group moved forward with, when the team entered the build, test and redesign phase.

6 QFD

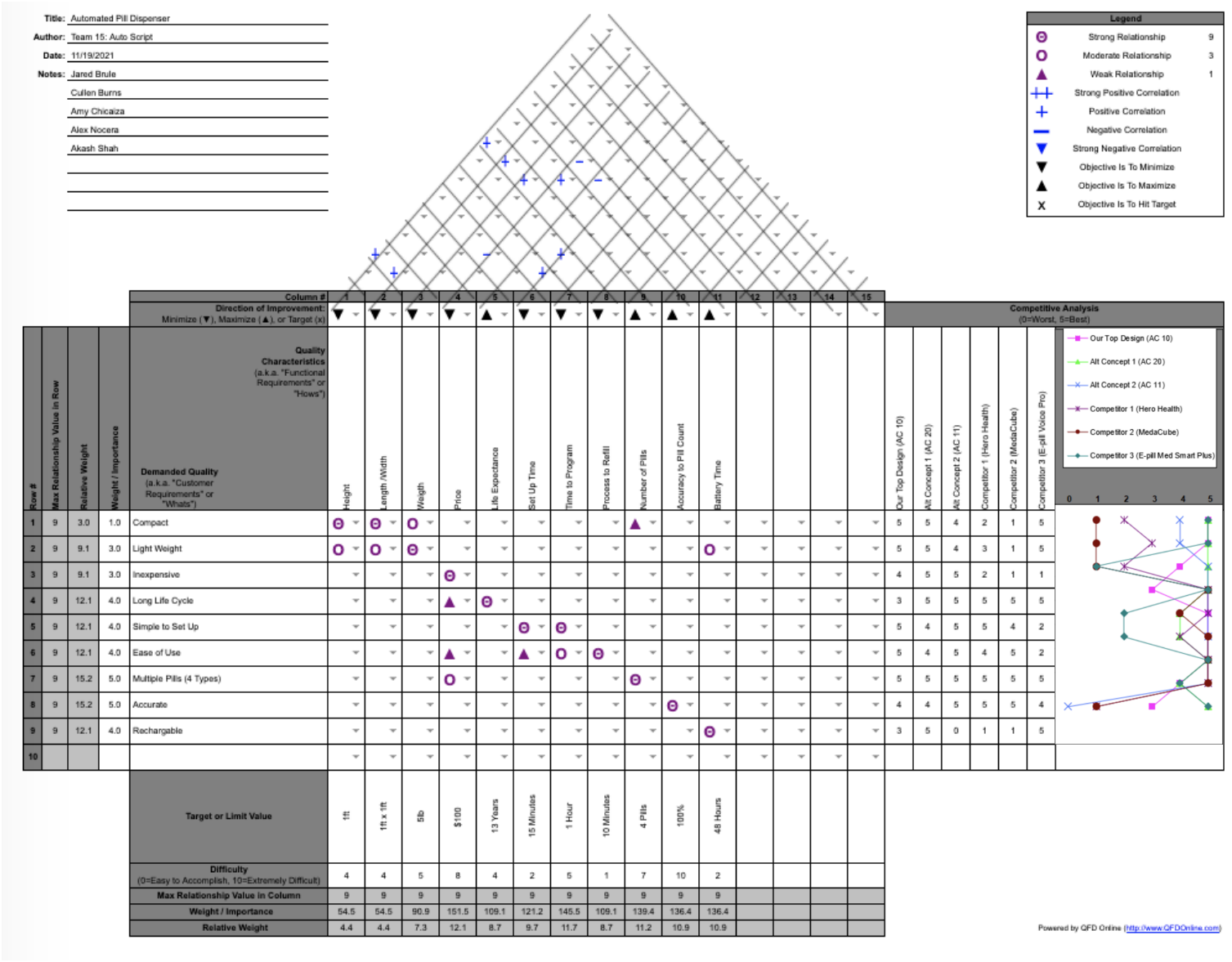

Figure 144: Quality Deployment Function Analysis

In this section, it will be explained how our team performed a Quality Function Deployment to analyze where our team's future prototype would rank among other pill dispensers that are already developed and on the market. When examining the QFD analysis the

“Demanded Quality” section must be first looked at. It is in this section that we look at what factors the six total competitors will be ranked by. It must be noted that the six competitors are our team's top design, alternate concept one, alternate concept two, Hero pill dispenser, MedaCube pill dispenser and E-pill Voice Pro pill dispenser. The factors at which these six pill dispensers will be analyzed are if the dispenser is lightweight, compact, inexpensive, long life cycle, simple to set up, ease of use, multiple pills (four types), accurate and rechargeable. These rankings were given on a scale ranging from 1-5. This scale shows how competitively each of the six competitors reach the design requirements in the “Demanded Quality” section. A competitor that received a score of 5, means that they perfectly met this factor. If the competitor received a grad of a 1, then they did not remotely come close to meeting the required factor.

Demanded Quality (a.k.a. "Customer Requirements" or "Whats")
Compact
Light Weight
Inexpensive
Long Life Cycle
Simple to Set Up
Ease of Use
Multiple Pills (4 Types)
Accurate
Rechargable

Figure 145: Demanded Qualities of QFD Analysis

The second section of this analysis of the columns at the top of the QFD chart, the characteristics of the quality of the dispensers were listed. It is in this list of qualities that the team can determine the quality of our prototype that is needed before we finalize this product to be displayed in the market and purchased by consumers. These qualities are denoted with marks that can be found in the legend. These qualities are desired to be either maximized or minimized depending on the scenario the team is found in. It is the goal of the team to make sure that the

quality matches the desired value. The qualities that our team felt were important to be analyzed are height, length/width, weight, price, life expectance, set up time, time to program, process to refill, number of pills, accuracy to pill count and battery time.

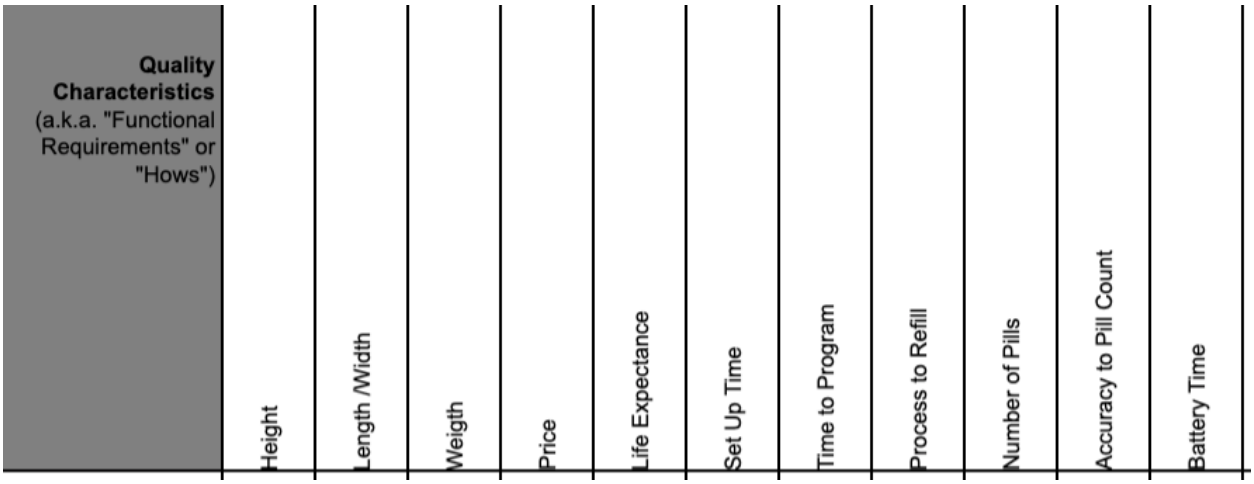

Figure 146: Quality characteristics of QFD Analysis

Given both of the sections described above, a few of the factors and qualities are very similar. This is due to the fact that a few of them are correlated for one reason or another. To show this relationship, the correlation between two coinciding factors and qualities were noted. These correlations were given ranking. These rankings were marked as a 1, 3 or 9. These denotations show that the two parameters are correlated but also show the degree that the two parameters are dependent on one another. Thus the more correlations between the parameters and the products, the more the consumer favors this certain parameter. When analyzing our group's QFD analysis it can be seen that the parameters that the consumer values the most in our product is the price, ability to dispense four different medications at different frequencies, accuracy of machine and ease of programming/use.

				Column #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15							
				Direction of Improvement: Minimize (▼), Maximize (▲), or Target (x)	▼	▼	▼	▼	▲	▼	▼	▼	▲	▲	▲					Competitive (0=Worst,						
Row #	Max Relationship Value in Row	Relative Weight	Weight / Importance	Demanded Quality (a.k.a. "Customer Requirements" or "Whats") / Quality Characteristics (a.k.a. "Functional Requirements" or "Hows")	Height	Length /Width	Weigth	Price	Life Expectance	Set Up Time	Time to Program	Process to Refill	Number of Pills	Accuracy to Pill Count	Battery Time					Our Top Design (AC 10)	Alt Concept 1 (AC 20)	Alt Concept 2 (AC 11)	Competitor 1 (Hero Health)	Competitor 2 (MedaCube)	Competitor 3 (E-pill Voice Pro)
1	9	3.0	1.0	Compact	⊖	⊖	O						▲							5	5	4	2	1	5
2	9	9.1	3.0	Light Weight	O	O	⊖								O					5	5	4	3	1	5
3	9	9.1	3.0	Inexpensive				⊖												4	5	5	2	1	1
4	9	12.1	4.0	Long Life Cycle				▲	⊖											3	5	5	5	5	5
5	9	12.1	4.0	Simple to Set Up						⊖	⊖									5	4	5	5	4	2
6	9	12.1	4.0	Ease of Use				▲		▲	O	⊖								5	4	5	4	5	2
7	9	15.2	5.0	Multiple Pills (4 Types)				O					⊖							5	5	5	5	5	5
8	9	15.2	5.0	Accurate										⊖						4	4	5	5	5	4
9	9	12.1	4.0	Rechargable											⊖					3	5	0	1	1	5
10																									

Figure 147: The relationships between the Quality Characteristic and the Demanded Quality of the QFD Analysis

In the top of the QFD Analysis, there is a visual representation of the correlations that were previously talked about and whether they were positive or negative correlations. These positive or negative correlations were denoted using a specific mark that can be found in the legend of the QFD Analysis.

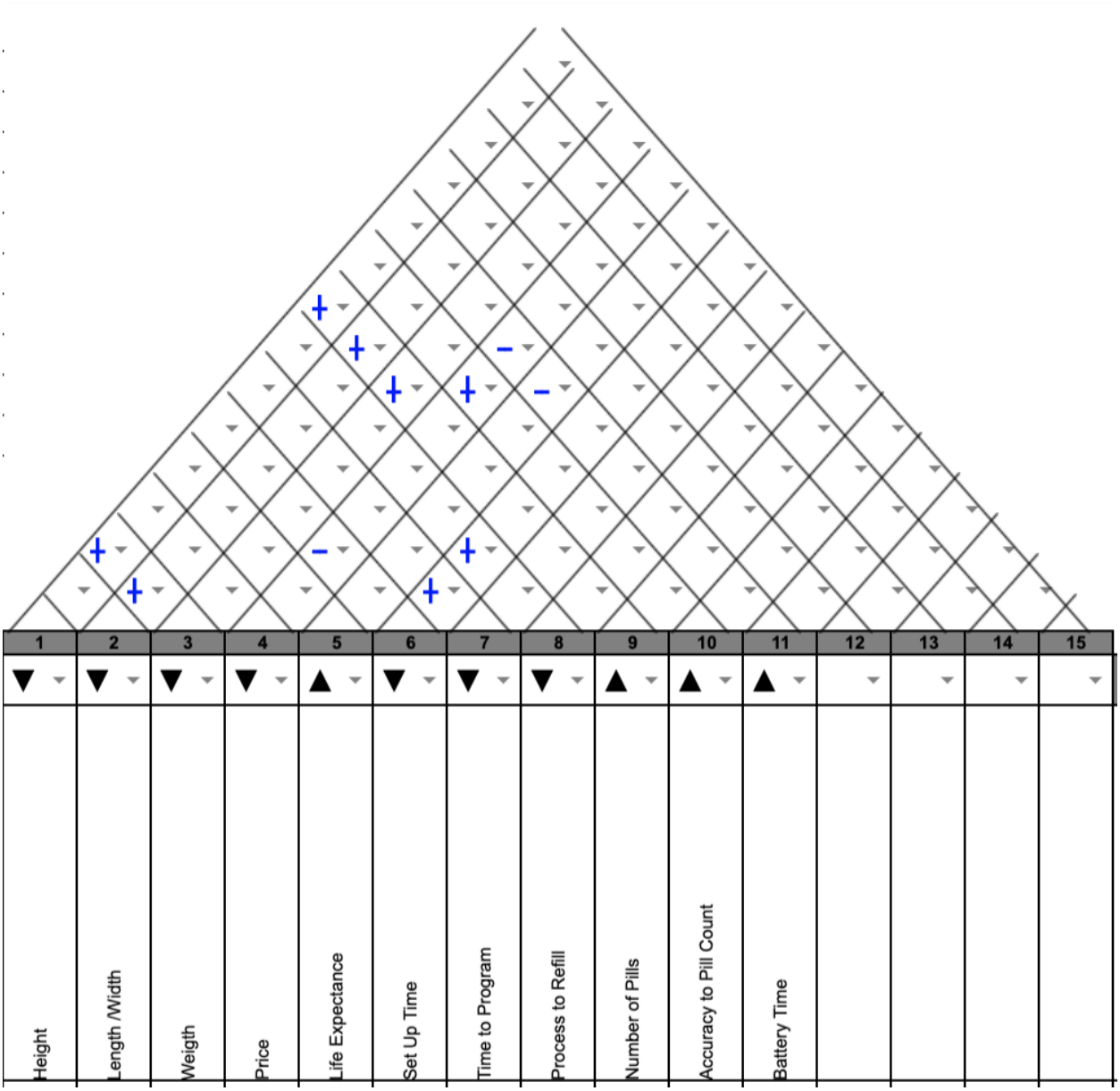

Figure 148: The Relations Between Each Quality Characteristic

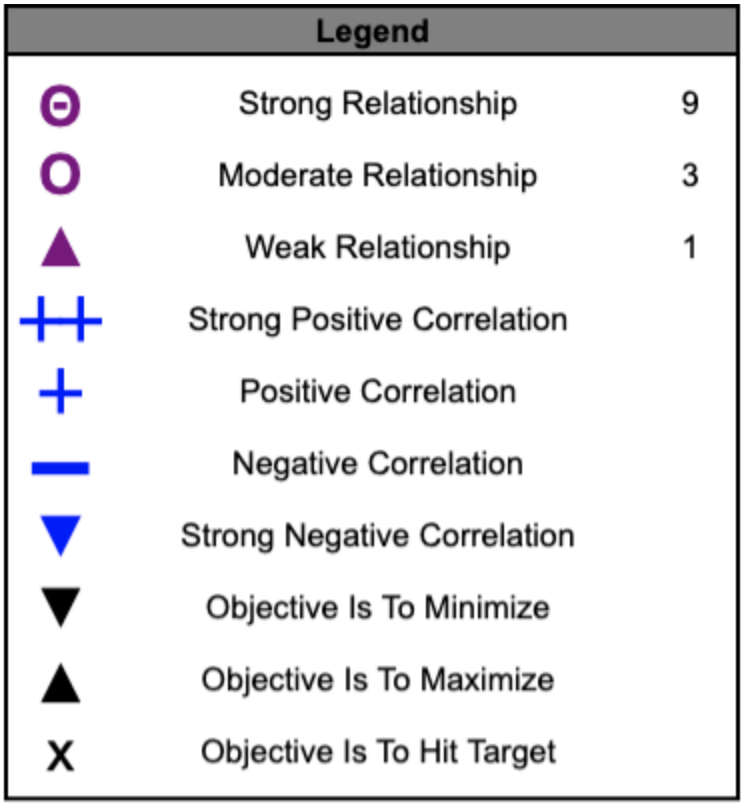

Figure 149: Legend of the QFD Analysis

Demanded Quality (a.k.a. "Customer Requirements" or "Whats")	Our Top Design (AC 10)	Alt Concept 1 (AC 20)	Alt Concept 2 (AC 11)	Competitor 1 (Hero Health)	Competitor 2 (MedaCube)	Competitor 3 (E-pill Med Smart Plus)
	Competitive Analysis (0=Worst, 5=Best)					
Compact	5	5	4	2	1	5
Light Weight	5	5	4	3	1	5
Inexpensive	4	5	5	2	1	1
Long Life Cycle	3	5	5	5	5	5
Simple to Set Up	5	4	5	5	4	2
Ease of Use	5	4	5	4	5	2
Multiple Pills (4 Types)	5	5	5	5	5	5
Accurate	4	4	5	5	5	4
Rechargable	3	5	0	1	1	5

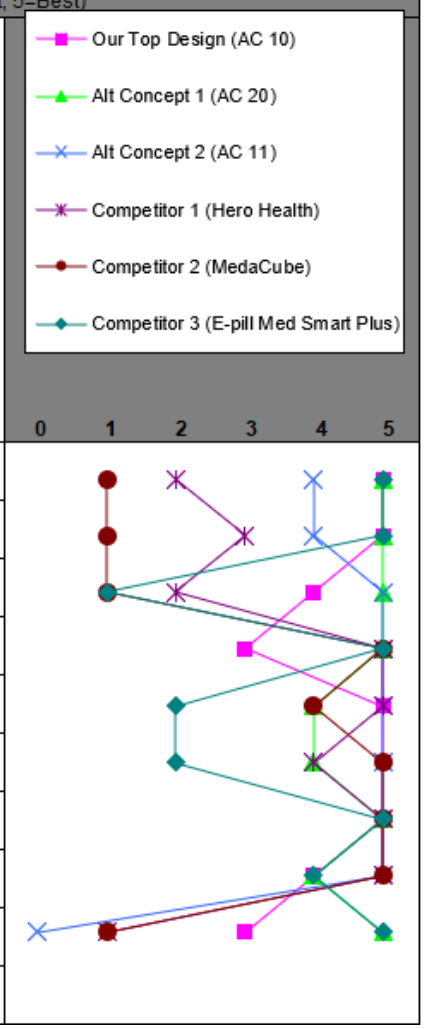

Figure 150: QFD Competition Analysis

Based on the QFD competition analysis, there are a few conclusions that can be drawn regarding the performances of the three competing products as well as our group's top design. This being said, our group's top concept performed very well in comparison to the competition, but was lacking in respect to two of the demanded qualities. These two qualities that our top concept under performs is in life cycle and the strength of its ability to be rechargeable. While our top concept lacks these two demanded qualities, it trades off performing well with the fact

that it is both compact and lightweight, as well as its ease of use and setup and its ability to hold four different types of medication. When analyzing the performance of Hero Health, it performs poorly for the demanded qualities of compactness of the machine, cost of product and its ability to be rechargeable. While it lacks these demanded qualities it performs well in accuracy, ability to hold four different medications, simple set up and long life cycle. When analyzing the performance of MedaCube, it performs poorly for the demanded qualities of compactness, weight, cost of product and rechargeability. While it lacks these demanded qualities it performs well in accuracy, ability to hold four different medications, ease of use and long life cycle. When analyzing the performance of E- pill Voice Pro, it performs poorly for the demanded qualities of cost of product and ease of use/set up . While it lacks these demanded qualities it performs well in compactness, weight of product, life cycle, ability to hold four different medications and rechargeability.

As the team moved forward and worked on and completed a final prototype, there were a few updates on the QFD competition analysis. Compared to the preliminary design report that was completed last semester that provided a QFD competition analysis of the team's proof of concept. As major changes were made to the circuitry, support system and outer shell of the device, the QFD competition analysis of our device compared to other competitors on the market has changed. Some of the largest changes found in the QFD came in the categories of weight, height, length, width and process to refill the medication.

7 Design for X

7.1 Design for Versatility

To make an effective pill dispenser, it needs to be able to work with a variety of pills. This means that size, shape and even texture need to be a factor. The team made use of a carousel stepper motor and housed a five compartment chamber. There are four equally sized chambers and one smaller funnel shaped chamber with a hole leading to the pill shoot. With this design, each of the larger chambers can contain one type of pill, meaning a total of four different pills can be housed within this device. The team tested to make sure different textures can be dispensed with different types of candies ranging from hard to gel with a variation of sizes. Each chamber housed a different candy and each successfully dispensed through the pill shoot. The

design itself is also versatile as the team also saw no reason to believe the number of chambers couldn't be increased if needed. The concept of the design is simple in nature and it's hypothesized that all that would change would be the overall size.

7.2 Design for Application

In order to make this pill dispenser as user friendly as possible, the team researched and developed an application using XCode language to function on iOS platforms to interact with the device. The application is intended to be able to communicate with the device to; dispense pills at any time, set up scheduling, and remind the user when it's time to consume their medication. The use of a smartphone in conjunction with this device was paramount as in today's world, a device with no smartphone compatibility is often archaic and thus; obsoleted. The application is intended to be as simple as possible, with the homepage only displaying three buttons that correspond with the three functions above. An additional slide over menu with app settings, troubleshooting guides and contact information can be accessed as well.

7.3 Design for Accuracy

Whilst dealing with medication, it is of utmost importance to make sure our device dispenses the correct dosage for the correct pill 100% of the time. To fail at this can have catastrophic consequences that begin with user harm and can lead to legal issues. To combat this, the team utilized a vacuum pump that is only able to pick up a singular pill and is powerful enough to handle 10 grams. The stepper motor rotates until the correct chamber is underneath the vacuum. The vacuum is then lowered and picks up one pill and raises back above the chamber. With the vacuum and motor being utilized in conjunction, we can assure that the correct pill at the correct amount is dispensed each time.

7.4 Design for Portability

The team needed to cater to different types of lifestyles when designing this device. People range from sedentary to active lifestyles and that can alter their access to medications, and as described before; simply getting people to reliably consume their medications is one of the largest issues. By default, our device can dispense pills on site at any time or by schedule. This obviously caters to those who are mainly at home, or in this case these users are likely elderly. However, for those who aren't typically at home and can't transport however many prescription bottles they need, our device accounts for them. With a standard issue pill box

placed underneath the pill shoot and the use of a linear actuator, the team satisfied the need to dispense a days worth of pills separated by a user determined time so the entire box can be taken with the user. The linear actuator shifts the pill box perpendicular to the device so that pills can be dispensed into each compartment. With the user setting a predetermined schedule, the device can automatically dispense all pills necessary for a determined period before the user needs them. When ready, the user can remove the pill box and go on with their day.

8 Project Specific Details & Analysis

8.1 Market Analysis & Forecasting Demand

Our team conducted a survey in the early stages of our research. We surveyed a total of 220 people that were above the age of 18, with 46% of the participants being over 40 years of age. This sample is accurately representative of the population of the United States where 66% of the population (131 million people) use prescriptions. We found that 57% of our participants have at least 1 prescription medication. And 16% of participants take multiple prescriptions. The frequency that our participants consumed medications varied from daily to weekly. We asked our participants to describe their lifestyle ranging 1 to 5 (1 represents a sedentary lifestyle where 5 represents a very active lifestyle). There was an approximate even distribution in the results. We found this to show that there is a need for a portable aspect to an automated pill dispenser for those who are unable to routinely keep up with their medications while on the go. We also asked our survey participants to explain their routine and results show a definite need for automation. Those who were considered “younger” (under the age of 30; 20% of the participants) mainly used their smartphone’s native reminder application to send notifications to their phone at a certain time in order to take their medication. The main issue with this group correlates to what was mentioned above; they find themselves away from their pills when it's time to take them. These results are what showed us that an automated pill dispenser that incorporated a way to dispense medication that was accurately dosed into a removable apparatus would prove beneficial. Those who were categorized as “older” (over the age of 30; 80% of the participants) tended to rely on either memory or some sort of written reminder to help them stay on routine. An important piece of data we gathered was that from this group of people over 30 years old; 87% of them said they would prefer a smartphone set up. This would mean that using a

smartphone to remind them should show an improvement to consuming their medications routinely.

Team 15 gathered data from various databases to understand what kind of research was done on prescription medications and automated pill dispensers to see what kind of market there would be for such a product. The team found that more than 131 million people in the United States use prescription medications. Those who are elderly typically have prescriptions for chronic illnesses which necessitates the need for an automated pill dispenser. Those who suffer from the likes of; diabetes, cancer, heart disease, hypertension or arthritis fill their prescriptions up to 34 times annually

8.2 Cost vs. Price

Team 15 initially found that the cost to research and develop our device would be approximately $300. The team aimed to bring this price down considerably going forward from preliminary designs, ideally by being able to streamline manufacturing and possibly purchasing parts by wholesalers. The thought process going forward is to analyze the cost effect when utilizing injection molding in order to mass produce. Our target price is between $50-$100 and the team has deliberated on this throughout the year at length. The team believes that this price is unrealistic and severely undercuts competitors. We believed that the lowest retail price that can be achievable would be $199 which remains extremely competitive. After completing our design we ended with a bill of materials that equated to $274.60, in addition to the cost of 3D printing. The team believes that if this device were to be mass produced, purchasing these materials at wholesale cost could achieve a retail price of $135. Materials such as the power supply, vacuum pump and even the stepper motors can be purchased at as low as 10% of what we purchased. Although this price remains higher than our target and our preliminary projection, this price still remains competitive for a device that offers a few features that current devices lack; namely the ability to dispense pills in a way to be consumed remotely. The massive savings come from the cost of 3D printing this singular model, at approximately $500. Manufacturing in high volume can make the unit cost down to a couple of dollars which reflects the severely lower price point of $135 compared to cost of development.

9 Detailed Product Design

9.1 Concept Development

Team 15 started the semester understanding the concept from the previous year. This allowed the team to get a head start on the problem solution. Once understanding the design provided to the team, we decided to combine some of our top concepts with their models. Last year's design had a great device with a very smart vacuum lifting system and a rotating pill storage box. Team 15 wanted a bit more for their device, so they decided to attach an application to the device to program the pills and scheduling, also adding a system that allows for some transportation of pills for customers on the go, and finally a safety system using a weight sensor and the milligrams of each pill. These additions help Team 15 cover more of their design specifications listed in the problem definition.

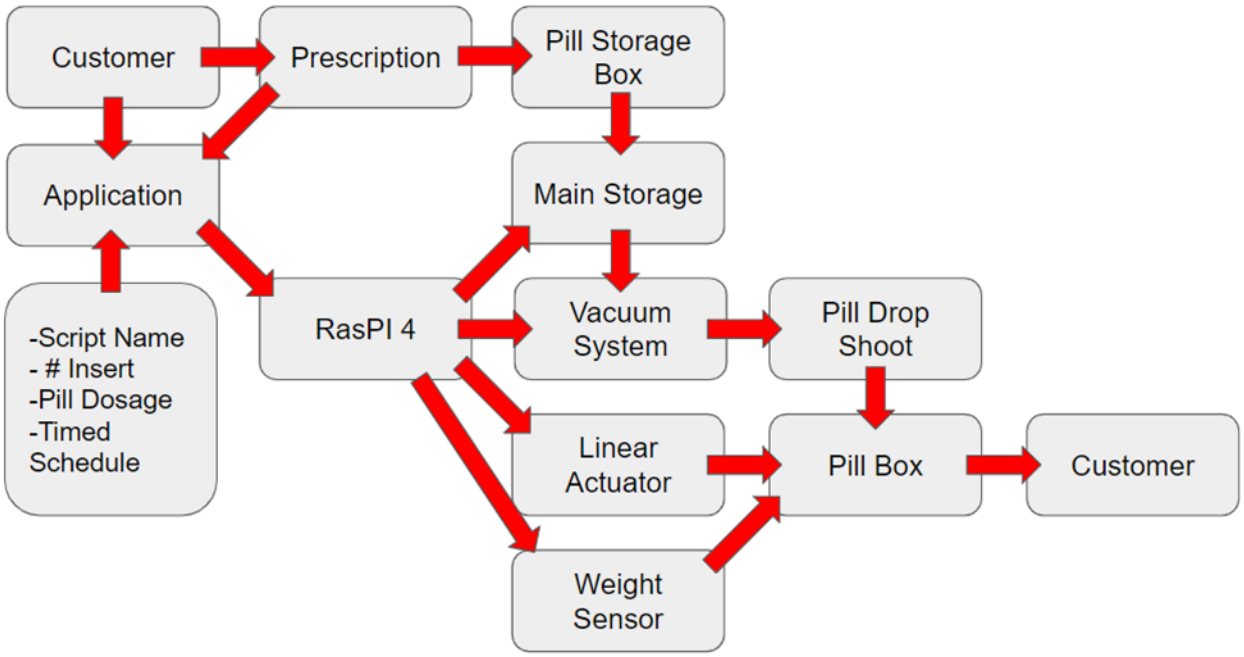

Figure 151: Web Breakdown of Device and Responsibilities

9.2 Detailed Part Discussion

Storage - Container A of the pill dispenser is designed to be split into four equal sized compartments and one smaller sized container with a hole in the base. In this layout, the pills will be contained in separate silos with a capped funnel shaped bottom. This ensures that the pills will always be funneled into a single point where the vacuum head will reach for every time. The single smaller segment is funneled but has a hole in the bottom. This will serve as the dispensing section where the carousel will rotate under the pill and the pill will be dropped through this hole to be dispensed.

Slider - This slider was purchased in tandem with a stepper motor driver so that the raspberry pi can control the motor in the manner desired. The maximum travel distance for this slider is 80mm. The maximum travel needed for this device is 70mm. For this reason, this slider was a perfect candidate for this project.

Head Mount - This part is designed to fit onto the mounting point that is installed on the Linear slider. Once this slider arrived, measurements were made to produce a 3d printed mount for the vacuum tip. Using the 3D printers, the team was able to produce custom components that fit the purchased components.

Lower Plate - The storage carousel has 5 protrusions below the funnels of each container which will fit into this plate below the storage. This allows the storage container to be removed. Once the stepper motor was received, measurements were made of the spindle such that the lower plate could fit onto the spindle using only friction due to a low tolerance.

Base Plate - In order to hold the stepper motor used to turn the storage container in place, the base plate was designed with a support wall surrounding the base of the motor. Four posts were placed around this motor container so that the bottom of a lazy susan bearing could be fixed to the base plate and the top of the bearing could be fixed to the rotating lower plate. A small pill receptacle was created such that when the pill falls through the dispense section of the storage container, the pill will be contained.

Face Plate - The linear slider was required to be attached to a face in order to properly function as intended. This plate was designed to allow for the slider to use its full range of motion without being inhibited. Mounting points were placed onto this plate to attach the slider, vacuum pump, and raspberry pi box. The bottom of the plate was designed such that the pills will be able to pass through the bottom into the receptacle.

Raspberry Pi 4 - The computer used to control the device was the Raspberry Pi 4 (RP4). The RP4 has Bluetooth capabilities that can be used to connect to a phone/app so 80 that the device can be run using a phone. The RP4 also has GPIO (General Purpose Input Output) pins that can be used for controlling multiple electronic components.

Stepper Motor - The carousel stepper motor was chosen as it had a high step per revolution count meaning that a high step resolution could be attained. This meant that the turning of this motor would be smooth and conducive to the operation of this device. Using an A4988 Stepper motor driver, the motor can be easily controlled using a python code on the RP4.

Stepper Slider - The stepper motor slider was chosen as it has a maximum travel distance of 80mm which is 10mm longer than the required 70mm for this design. This motor can also be used with a stepper motor driver to allow the RP4 to control its movement. Knowing the distance traveled per step, the team can write commands knowing exactly how far the slider will move.

Vacuum Pump - The vacuum pump was required to produce enough pressure to pick up the most massive pill of 1000mg. The vacuum pump found could produce 60kPa of negative pressure which can pick up 10g. The team decided that this would be a good fit as it was small, easily controlled by the RP4 using a transistor, and with enough pressure to account for any possible air gaps between the pill and the nozzle.

Pill Box – The Pill Box is the customer's travel component in the device. It is very similar to the manual sorting box most people own to currently house their medications but is moved to allow the device to fill it for them. Having a part of the machine that can physically be removed and brought with the customer on day out of the house is important in society.

Linear Actuator – The linear actuator is the connection between the pill box and the main device. This component allows the pill box to be linearly moved back and forth to line up the days of the

week for the customer. By using a stepper motor in this part, we can have set locations of rotation for each day allowing the customer to get the days' worth of medication but still have them separated out by time.

9.3 Computer Aided Designs

The following images are the current designs of Team 15 which is a combination of last year's design as well as the additions they have added during the Fall 2021 semester. There are only a few full models listed below for a more detailed design sketch look in the appendix section at the end of the report.

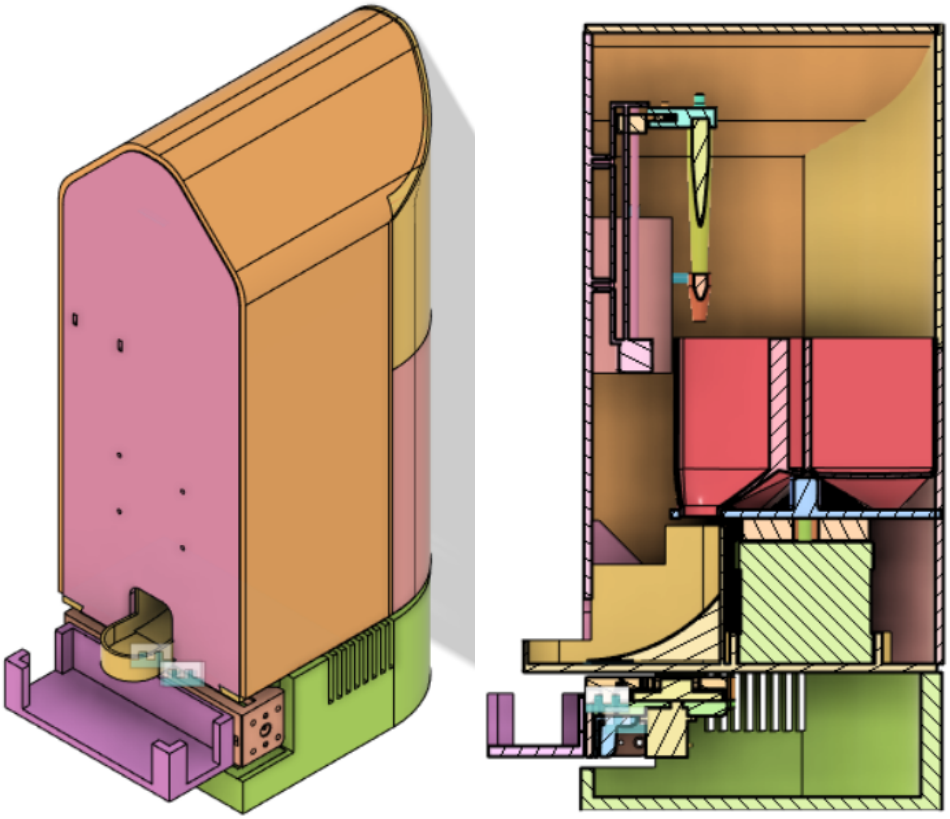

Figure 152: Full Pill Dispenser Model

One of the biggest selling points for our pill dispenser is the front linear actuator that holds a small portable pill tray. The pill tray would be labeled with four time intervals on it so that the internal computer can load pills in each for a customer to take on time while they aren't home. Being able to detach things and take it on the go allows for accurate dispensing with

included travel capabilities. The internal computer sends signals to the stepper motor on the back of the linear actuator to specify the amount of rotations to line up each time with the pill drop hole. The combination of this simple horizontal motion control allows for a much larger customer base for the dispenser.

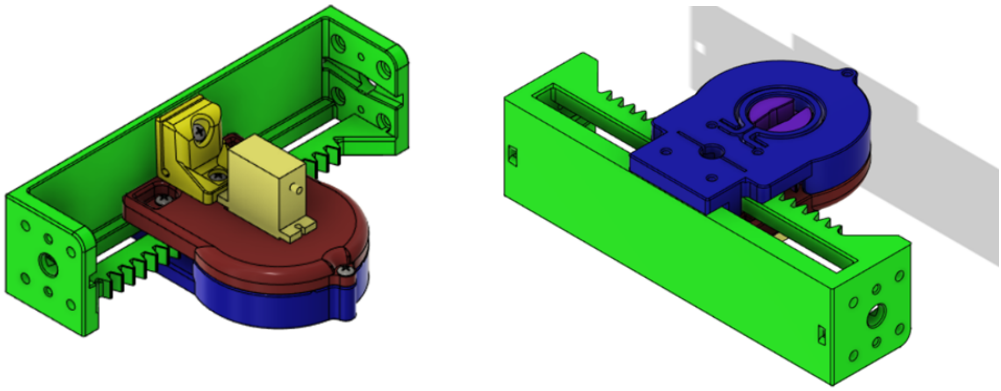

Figure 153: Linear Actuator Model

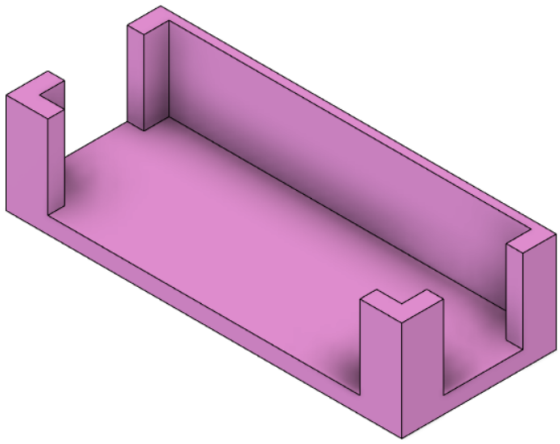

Figure 154: Pill Box Model

The modification to the front face and external shell were added to help with customer appeal. Having a clean look for the whole device will increase sales and condense the bulk over wiring to one location. The wiring will be simplified to the bottom tray with a few wiring running to each of the servos. A few structural supports were added throughout the year to increase the longevity of the device.

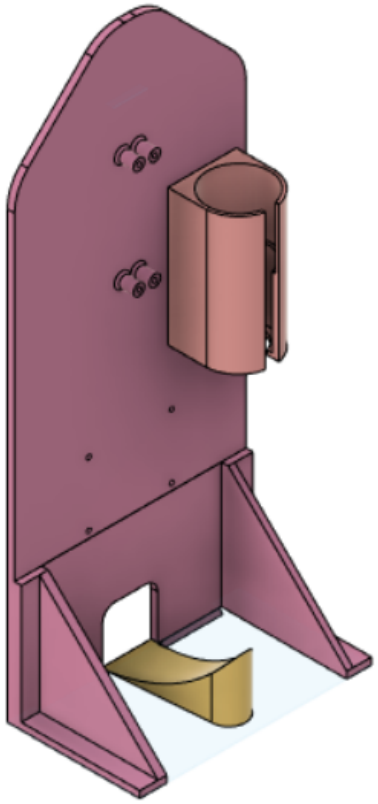

Figure 155: Front Face Model

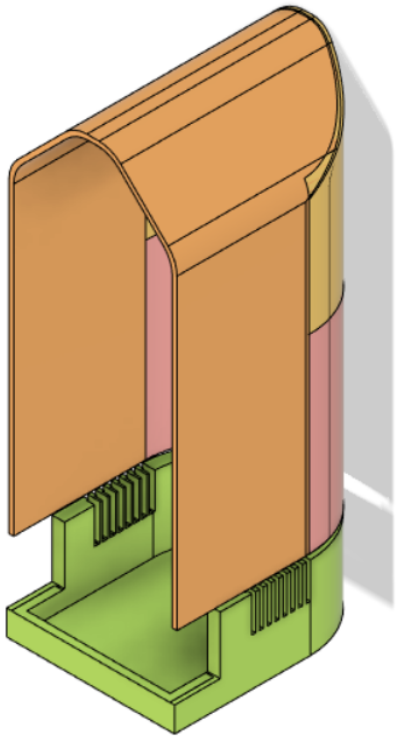

Figure 156: Exterior Shell Model

9.4 Application Development

The application “Auto Script” is being coded using Apple's software XCode. This app, available for IOS only, will allow the user to be able to connect to the pill dispenser. The connection will be made via bluetooth, which is a feature provided by the Raspberry PI, and will have multiple easy features for everyone to be able to use.

9.5 Wiring Components

Wiring the whole pill dispenser was a very difficult task for our group. We spent a long time researching each component of our device to get the data sheet to see how each should be wiring. Once getting all the specs we made a rough outline of how we assumed each component should work and tried several times to get all the functionality to work. After so much failure a friend of the group recommended reaching out to an electrical engineering professor who gladly accepted and gave us some assistance. He looked over our wiring and gave us some slight modifications to improve the system and helped us get all the devices operational. The following

diagram is the final combination of all components that successfully worked. The professor expanded our knowledge of wiring and electrics to improve the overall efficiency of our project.

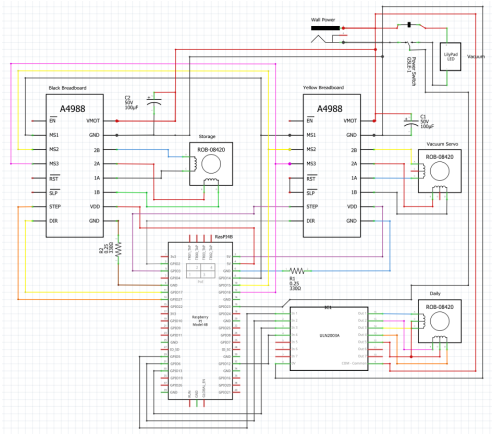

Figure 157: Wiring diagram

9.6 Bill of Materials

Description	Quantity	Cost
6V Vacuum Pump	1	$10.00
Stepper Motor and Screw Drive	1	$25.00
12V Power Supply	1	$12.00
Stepper Motor Controller	3	$6.00
Ras PI 4B	1	$120.00
Bread Board	3	$4.80
Tubing	1	$6.00
Wires	8	$8.00
Transformer	1	$5.00
Storage Stepper Motor	1	$14.00
"Lazy Susan" Bearing	1	$5.00
Linear Actuator Stepper Motor	1	$2.80
DC DC Converter	1	$43.50
Splice Kit	1	$12.50
Total		$274.60

Table 2: Breakdown of parts needed for device

10 Engineering Analysis

Pill Size	000	00	0	1	2	3	4
Typical Volume (mL)	1.37	.911	.682	.503	.372	.302	.211
Number stored	66	99	132	180	243	300	429

Table 3: Correlation of commercial pill sizes to device capacity

Our team needed to focus our analysis and methodology on two main components; the capacity of our device as it relates to different pill sizes, and the suction capabilities of our vacuum pump. As seen above in Table 6, the device has different capacity limits for different commercial pill sizes. We developed these values by taking the total volume present in a chamber used for pill storage and dividing it by the volume for each pill type. The volume we derived for one of the four equal chambers is 90.046897 mL.

Secondly, the team needed to analyze the suction of the vacuum pump that is used to lift the pills from the chamber. The team needed to analyze the tip area (T_a) in order to calculate force (F) that the pump outputs and thus deriving the largest possible mass (m_{max}) it can lift. The table below shows the calculations of the pump. This shows that this pump will not struggle with lifting any pill as it can lift approximately ten times that of the largest commercially available pill.

Equation	Solved Value
$T_a = \pi r^2$	$1.76 * 10^{-6} m^2$
$F = PT_a$	$0.10603 \frac{kg*m}{s^2}$
$m_{max} = \frac{F}{A}$	$10.8\ g$

Table 4: Equations used to calculate maximum allowable mass able to be lifted.

11 Build/Manufacture

The order in which our device would have to be manufactured would be to first get the bread board and wiring system developed before doing anything else. Once this task has been completed the next step would be to 3D print all the different CAD designs. Then assemble the servo motors into the correct areas. Once this is completed the vacuum system should be set up. Next, the wiring should be condensed and tucked into the designed holding space below the device. Then, the other shell should be mounted to the top of the device and then fastened.

In terms of mass production of the device, this would be done by plastic injection molding. This is the option chosen as it has the ability to mass produce accurately and effectively for cheap production cost. As for the amount that would be produced, enough to meet market demands and keep up with orders for the product.

12 Testing

			Good Test Results		
			Work in Progress		
			Not Good Reults		
Components	What to Test?	Test Parameters	Results	Planned Resolution	Comments
Vaccum	Suction Power	Vaccum Pressure can lift varying weight of pills	Vaccum has strong suction when ON	Begin testing with the four planned types of pills to make sure vaccume provides the proper amount of suction	Testing on done with different style candy, and different tips
Vaccum	Continuous Suction	Long periods of suction don't change pressure	Issues between wiring or vaccum, prevents testing	figure out correct wiring so testing can be attempted	
Daily Servo	Repeatability	Servo moves a lot can't wear too fast affecting accuracy	Unassmebled due to unrelated problem, should combine to work	trouble with gear to servo addapter print, reprinted numerous time. WIP	8 reprints and still having trouble combining
Daily Servo	Accuracy	servo stops pill box under correct location	Code stops servo in correct locations		
Storage Servo	Repeatability	Lots of repeating spin locations, stopping in same spot	Servo stops the storage bin at the right location in numerous sequential trials		
Storage Servo	Accuracy	Always stops in same spot	Code stops servo in correct locations		
Storage Servo	Strength	Holds lots of pills at once but still needs to spin freely	haven't loaded to max capacity and tested servo strength	Lower the total pills, for each storage bay	don't expect to have issues with weight due to bearing underneath
Storage Servo	User Accessiblity	Easy for customer to understand and fill storage	Customer can easily pour contents of pill vials into the pill storages		
Screw Servo	Repeatability	Heavily in use, repetition can't affect accuracy	Large repeatition no mechanical failurs		
Screw Servo	Accuracy	Always goes down same location	tested a large amount of times and always stopped at bottom and top of screw		

Figure 158: First half of testing matrix

Components	What to Test?	Test Parameters	Results	Planned Resolution	Comments
Screw Servo	Strength	Lifts different weight of pills, lifting strength	without vaccum have working can't test	Once vaccum is working correctly, test screw can lift all weights of pills	
Drop Shoot	Repeatability	All pills make it from storage drop to daily storage	pills slide down correctly but haven't tested from drop to pill box	Begin testing if pills slide down correctly from drop shoot to pill box	
Drop Shoot	Accuracy	Drops pills in correct daily storage time	Daily storage not combined in assembly	once combining all components issues should be worked out and tested	slope may not be steep enough, to make to drop hole
Daily Pill Box	Accuracy	Tight for in storage for accurate alignment	Tight fit for accurate movement by servo		
Daily Pill Box	User Accessiblity	Used heavily by customer, doesn't affect accuracy	Light testing to smooth out rough edges	quick solution would sand sides, reprint custom pill box	
Storage Servo Code/Wiring	Errors	Test for any errors in completed code	smooth rotation to all 5 slots around the cup		
Daily Pill Box Code/Wiring	Errors	Test for any errors in completed code	Servo rotates smoothly and to correct locations with no errors		
Vaccum Code/Wiring	Errors	Test for any errors in completed code	Vaccum doesn't start and stop when told to do so	Keep checking for issues in code and wiring	Asked a ELE and CS major friends to come give us a hand
Screw Servo Code/Wiring	Errors	Test for any errors in completed code	After a large number of test controller had a hickup but fixed	loose wiring, or controller	continue to test to make sure wiring issue is resolved
App to Device Code/Wiring	Errors	Test for any errors in completed code	NOT Tested, issues with code stiching	Finish all individual componets on both sides then combine	

Figure 159: Second half of testing matrix

The group performed the test matrix shown above, the testing of the device was focused on eleven different components of the device. These tested components focused on the automation, storage, vacuum, wiring and application aspects of the pill dispenser. As shown in the two figures above, the results sections were highlighted and color coded based on whether they were deemed good test results, work in progress or poor results. The testing went very well for the most part, with a few exceptions with testing in the vacuum, drop shoot, coding, wiring and the application. When testing the vaccum's ability to produce suction continuously, the group received bad results; this was due to the issues in the wiring of the vacuum. After the team realized there was an issue, a meeting was held with Professor Jeong of the Electrical Engineering department to discuss how to fix the problem. The area that needed the most improvement was the wiring of the automated portion of the system. Our group did research and arrived at the conclusion that the solution to the wiring error was to implement a power transformer with a control wire and a control pin which controls the voltage level, in order to optimize the function of the vacuum pump system. The professor recommended that the group order a motor voltage controller, which the team ordered and implemented the device into our circuit and system. Another test that gave poor results was when the drop shoot accuracy was tested. When checking to see if the pills were dropped in correct daily storage time, the team observed that there was an issue with the daily storage being combined in assembly. The group's approach was to make the drop shoot ramp more steep, hoping that this fix paired with a combination of all the l components issues being worked out will provide the group with good

results. This proved to be an issue because the pill would get stuck halfway down the ramp and not make it to the receptacle. The ramp has to be steep enough so that the pill would roll down on its own without any extra effort. There was a redesign of the ramp done. The redesign included a steeper ramp to be added on top of the already existing shoot. The addition of the ramp during testing made a significant difference in the way the pills dropped into the receptacle. Another test that provided poor results was when the group tested the vacuum coding and wiring for errors. When testing for errors in the completed code, the group observed that vacuum would not start and stop when the coding through Raspberry Pi 4 told the vacuum system to do so. The group's approach to fixing this error was to reach out to the electrical engineering department as well as the computer science department at the university to help us find and fix the problem with the coding and wiring of our system. The last test that ended with poor results was when the group tested the coding that connected the phone application to the device itself. The group's planned resolution of this issue was to finish all individual components on both sides then combine.

There were also a few tests that provided the group with decent results that needed improvement but were not necessarily considered bad results. Not all of these test results had a high amount of relevance as their results were not important to the overall function of the device. One of these tests was testing the vacuum suction power. The testing parameter was to see if the vacuum pressure can lift varying weights of pills. The group began testing with the four planned types of pills to make sure vacuum provides the proper amount of suction to lift all four different types of pills. The results were that the vacuum produced a powerful enough suction to lift three out of the four pills. The groups proposed solution to this issue was to add a silicon tip to the vacuum tip to increase the suction force on the fourth pill. The new concept of the pill storage area is to change the shape of it to have a flat bottom rather than the cone shape it currently has. The flat shaped bottom would allow for the pills to be pressed against a flat surface in order for it to be more precisely picked up by the nozzle. Another test was when the group tested the daily servo’s repeatability. It was realized that there was an issue with the sizing of the connection between the gear and the servo which allowed for too much motion of the servo. This produced a slight inaccuracy in the daily servo motion. The group continued printing gears of different sizes and dimensions in order to get the perfect fit to increase accuracy. Another test was to test the strength of the storage servo. The group was trying to test to see if the servo was strong enough

to spin the pill storages when they were filled to max capacity. The group concluded that the total pill weight needed to be lowered in order for the storage servo to work efficiently. The last test that produced results that needed improvement was testing the screw servo coding and wiring. The test parameter was to test for any errors in the completed code. At first the test was producing great results, but after a large amount of testing the screw servo had a small issue but was further fixed by the group. This issue was found to be caused by loose wiring and error with the controller.

13 Redesign

13.1 Design issues and solutions

Identifying the issues that need improvement was hard when testing the prototype before the final design. The whole prototype was not tested as a whole but instead individual components were tested. For example the vacuum system was tested by taking the nozzle and holding it up to a faux pill. The vacuum system with the nozzle did an amazing job at picking up the faux pill. However when the prototype came together there were some issues. The compartments in which the pills are stored have almost a slight cone shape with a circular base. This type of design made it harder for the vacuum system and nozzle to be able to have a sealed tight grip on any pill. Without it having the grip on the pill it will not be able to pick it up. concluded. There was definitely need for improvement in certain areas. In order to solve this issue the pill storage needs to be redesigned. The new concept of the pill storage area is to change the shape of it to have a flat bottom rather than the cone shape it currently has. The flat shaped bottom would allow for the pills to be pressed against a flat surface in order for it to be more precisely picked up by the nozzle.

Another problem that was found and addressed was the pill ramp. When testing out the dropshoot portion of the prototype, it was found that the dropshoot was not steep enough for the pill to make it down the shoot. This proved to be an issue because the pill would get stuck halfway down the ramp and not make it to the receptacle. The ramp has to be steep enough so that the pill would roll down on its own without any extra effort. There was a redesign of the

ramp done. The redesign included a steeper ramp to be added on top of the already existing shoot. The addition of the ramp during testing made a significant difference in the way the pills dropped into the receptacle.

Finally, the slider on the prototype has some issues. In the initial testing it seemed to move up and down smoothly and it worked fine. During the showcasing it was noted that it tilted slightly to the right in a jerked motion as the nozzle moved down the slider. This would cause an issue when there are not as many pills in the compartment and most of them are on the far left side. This would result as an inaccuracy when the pill dispenser suggests that there are no pills left or it is on a low supply. A redesign aspect of the slider and how the nozzle moved up and down would include securing the nozzle in a different way. Instead of having it attached by a simple U-shape and a pin, it could be secured with two shaped pins on the top and bottom. It could include a slightly oversized attachment on the nozzle part with the two shaped pins, and it would snap into place on the slider. The slider attachment would have two hotels on the top and bottom section to allow the pins to go in. Since the nozzle part would be a little oversized it would allow for there to be opposite directed pressure applied on either side. The pressure would keep the nozzle from jerking as it descends into the pill compartments, and make it more sturdy.

13.3 Future team recommendations

Some recommendations for future design teams to improve our design based on what we have learned is to continue to improve the wiring and the python coding. With those two aspects and the redesign of the pill container everything will come together. The wiring could be assembled better and placed in the under part of the design for a sleeker appearance. The coding works but it just needs to be organized better and stitched to work seamlessly for more than one demonstration. The coding was one of team 15's greatest challenges during this time and having focused on that little more and reached out for help sooner maybe it would be at a better stage. As for the application aspect of the project, having more than one person work on that would ease the amount of work and time spent by one person. There are so many components that go into an app, and for the majority who do not know how to make one, start the process of talking to people in the first semester.

14 Project Planning

14.1 Project Plan

Initially to start the product design, we designated title to each member of 1015 as a part of the project planning stage at the beginning of the year. This carried out until now and has kept everyone organized in what they need to do. Although each team member has a title it did not limit the possibilities of the team members focusing on other tasks when needed and helping each other out as a team.

Team Members	Title	Initials
Jared Brule	CAD Engineer	JB
Cullen Burns	Design Engineer	CB
Amy Chicaiza	Application Engineer	AC
Alex Nocera	Design Engineer	AN
Akash Shah	Team Leader	AS

Table 5: List of Members, Titles and Initials

14.2 Gantt Chart

The Gantt chart below in figure three is commonly used too when planning a project out. Using Microsoft Project that was provided to us allows for easy set up of this chart. The chart allows for easy tracking of assignments, tasks, deadlines, team members, hours, and more. Using this chart allowed us to track when assignments were due and how much time must be allotted to each. Having one centralized location for all our dates and activities gave the team a good, organized progress track for the project.

4.3 Spring 2022 Semester Breakdown

Spring semester started with all students attending lectures while the professor broke down what we were going to be doing this semester. The semester had a different approach from the first semester because we already had our individual projects and we focused on the testing and redesigning part of the project.

After being oriented on what will be going on during the spring semester, we had team meetings with the professor. During these team meetings, he guided us in which direction we should be taking the project and gave us suggestions on what we needed to accomplish. The meetings are always really helpful to determine where we stand on the timeline and to discuss any further questions.

With a timeline and milestones to reach, Team 15 sat down and discussed the upcoming work that needs to be done to hit every milestone within the timeframe. For the next month and a half we did not have lectures and focused on working on finishing our design and testing.

As we came close to April we worked on the team presentation. In the presentation we had to include how we tested our project and which steps the team needs to take next. The presentation included a detailed portion on what redesign approach was going to be taken. These redesigns included wire redesigning, application modifications, and the overall look of the end product. Team 15 also included goals the team had made were to not be fulfilled due to time constraints. On April 15, 2022 team 15 presented the build, test, redesign presentation the team had been working on for a week.

Finally after the presentation Team 15 set off to redesign and do a final test of the automated pill dispenser. After many adjustments and reconfiguration, Team 15 got their final product that would be presented. A poster and brochure was put together throughout a couple of weeks before having a final design for both. Both poster and brochure were printed using the printers in the ECC and the Capstone room printer. On April 29th, 2022 there was a Design showcase where many people came and walked around. The design showcase was a success and Team 15 had a total of 43 visitors, where we were able to explain and show off our final product.

15 Financial Analysis

Our automated pill dispenser is a professor sponsored project, from previous years. Our budget for the engineering of this project is around $500, and this must cover all the design

specifications for the project. Even though we have a reasonable budget for the project, having a customer cost of about $100 dollars for the final product is the tough part of the design. Covering all design specifications and keeping the project as cheap as possible is the goal.This will include breakdown of the cost for parts and materials, with an estimated market price and mass production cost. Our financial report will also include a breakdown of the cost for labor as well, but this is not contributing to the budget costs. Labor cost is to show how much it does cost to develop a product in real life.

During the past 8 months of the 2021-2022 year, the members of Team 15 have worked many hours researching and designing and manufacturing this device. CAD modeling and as well as application design are two of the major time-consuming parts behind the actual construction and wiring of the physical device. The graphs and tables below show a breakdown of all the hours worked on the project by the current members throughout this time period.

15.1 Research and Development

During this year a log was kept for each of the team members to allow for calculation of their wage. These values are estimated based on an hourly wage of $20.00 an hour for each member. This information shows how much research and development costs can be excluding the materials and actual product.

Names	Hours	Actual Cost	Standard Rate
Jared Brule	267.2	$ 5,344.00	$ 20.00
Cullen Burns	217.2	$ 4,344.00	$ 20.00
Amy Chicaiza	209.2	$ 4,184.00	$ 20.00
AlexNocera	221.2	$ 4,424.00	$ 20.00
Akash Shah	187.2	$ 3,744.00	$ 20.00
Total	1102	$ 22,040.00	

Table 6: Hourly wage calculations for each team member based on amount of time spent on project

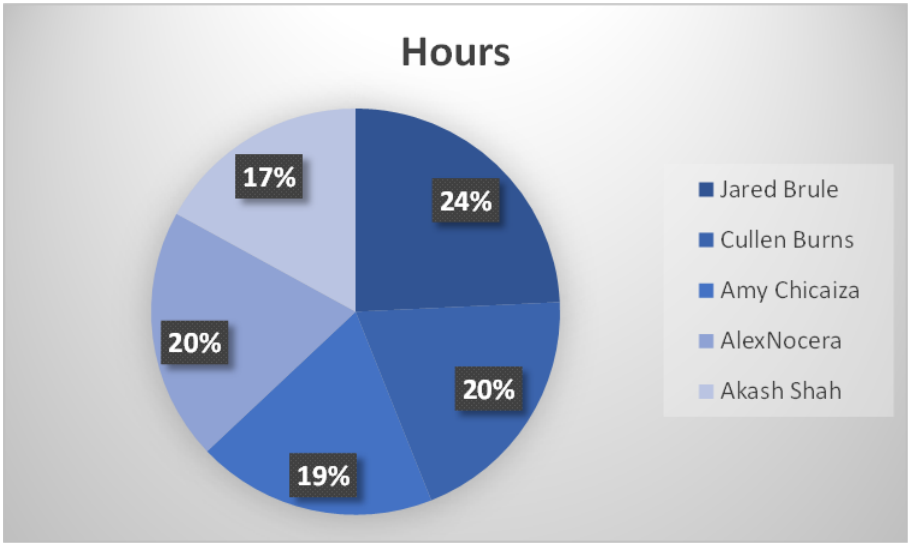

Figure 160: Graph depicting the percentage of hours each member spent on the project compared to total time

15.2 Bill of Materials

Cost of a research and development project doesn't just mean the budget for wages, it also includes the cost of materials for the device itself. The budget set for the project is very important for the project because the product is trying to become a marketable product. Keeping costs of the product low and the profit margins reasonable is difficult especially with other products on the market for competition. Keeping track of individual components to see what is costing the device the most is important when trying to reduce the overall cost for the product.

Description	Quantity	Cost
6V Vacuum Pump	1	$10.00
Stepper Motor and Screw Drive	1	$25.00
12V Power Supply	1	$12.00
Stepper Motor Controller	3	$6.00
Ras PI 4B	1	$120.00
Bread Board	3	$4.80
Tubing	1	$6.00
Wires	8	$8.00
Transformer	1	$5.00
Storage Stepper Motor	1	$14.00
"Lazy Susan" Bearing	1	$5.00
Linear Actuator Stepper Motor	1	$2.80
DC DC Converter	1	$43.50
Splice Kit	1	$12.50
Total		$274.60

Table 7: Breakdown of parts needed for device

Print Description	Printer Type	Filiment Quantity (grams)	Time to Print (minutes)	Cost
2021 Model	Ultimaker	161	1027	$136.93
Linear Acuator	Raised 3D	67.2	344	$22.93
Wire Box	Raised 3D	105.9	965	$64.33
Daily Tray	Raised 3D	23.3	96	$6.40
Exterior Back	Raised 3D	97.9	1440	$96.00
Exterior Top	Raised 3D	132.2	1778	$118.53
Front Face + Gear	Raised 3D	108	794	$52.93
Ramp	Raised 3D	3.5	35	$2.33
Vacuum Holder	Raised 3D	14.4	189	$12.60
Total		713.4	6668	$512.98

Table 8: Breakdown of 3D Printing Costs

Even though the current costs are very high for a singular model to be constructed, usually in manufacturing the more devices you order to be made the cheaper the costs over all. After some research in manufacturing costs we found some ways to lower the production cost of the device. First when looking at electrical components, even though they are very small pieces they can add up in cost very quickly, but condensing these components onto one PCB can go a long way. Currently our electronics cost upwards of $150 which is our target market price but buying on a single PCB reduces costs to $18.96. Costs can be cut again when buying multiple boards and if we were to purchase 12 PCBs for $60.80, each board is only $5.06. The second

major reduction in cost comes to the 3D printing cost, manufacturing one model in PLA is expensive because it is so custom. Making a large order of parts such as one hundred models has a high upfront cost of about $400 but the individual models would each cost a couple dollars. These two savings bring the cost of the pill dispenser way down.

15.3 Manufacturing Price to Retail Price

Reviewing all cost saving measures and material for our automated pill dispenser a single bulk run order would cost about $95.00. This number is under what we were targeting for customer cost. Usually when selling a product on the market, it is not sold at the cost of manufacturing it has some sort of mark up price that will include a profit margin for the company. General retail usually ranges a net profit margin of about 2% to 50%, with that information a markup of about 10% was estimated. Using this information an estimate was made to place the automated pill dispenser at a retail price of $135.00. Having the device at a retail cost of $135 is over our specification amount but under what our max number of $150.

15.4 Break Even

Knowing the estimated retail price of the pill dispenser, some calculations can be done to find the ROI of the initial manufacturing set up cost . With the initial manufacturing costing about $4000 and a profit of about $35 per automated pill dispenser sold about one hundred and fifteen dispensers. This would recuperate all the initial investment and the future sales would be all profit for the company. Changing the profit margin would change the retail price of the product and the ROI time. Picking a smart product price to entice customers but make good profit is important in the medical field with similar devices being products all the time.
Break Even, mold costs about $4000, when will profit equal $4000

16 Operation

Along with the companion phone application, the pill dispenser will dispense pills correctly and accurately. The four compartments allow different types of pills to be dispensed. A standard pillbox is attached to the linear actuator to allow portability for users on the go. An operator’s manual, a safety guide, an assembly guide, and a repair guide were not produced. However, a brochure is available with information about the device.

17 Maintenance

The concept of a warrant is mentioned earlier in the report. The purpose it serves is to provide customers with the reassurance that if it were to break down they would be able to get a replacement or have it fixed. It would come with a monthly fee to allow the company to generate profit. After the product has reached the end of its useful life, some of the 3D printed parts and the pillbox provided can be recycled. The rest of the parts can be disposed of with regular trash.

18 Additional Considerations

18.1 Economic Impacts

The economic impact this would have on the market would be significant. It could be used worldwide including at hospitals and nursing homes. According to the GMI article the market size for automatic pill dispensers in 2020 was over $2 billion. An estimation was made on the projected growth this type of product could have on the market. Already exceeding $2 billion it is projected to grow over 9.5% between 2021 and 2027. However, this type of product may also have a negative impact on the market. Stated by GMI the reason this could have a negative impact on the market is due to strict government regulation and potential errors. In the further future these types of devices will have a greater market because as technology advances the room for error is less.

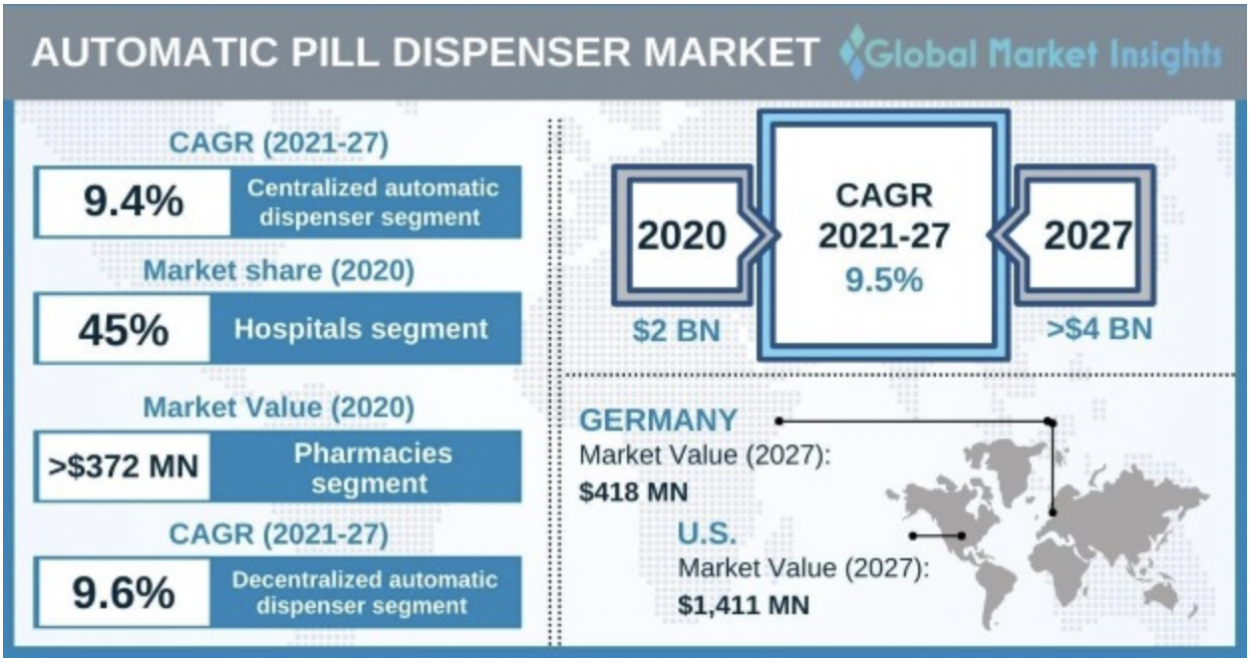

Figure 161: Current pill dispenser market

18.2 Societal Impacts

This device aims to improve the consumption of medication across a multitude of demographics. From young active users to sedentary senior citizens, many people have different reasons as to why they struggle to consistently consume their medications. With the features implemented to this device such as reminders both to your smartphone and audibly from the device, elderly users can find it easier to remember to take their pills. For those who are more active and can't always be on site with the device, the implementation of the removable pill box allows these users to take a daily dose of pills wherever they are. In a study by PubMed Central 84% of the 96 older patients found that automated pill dispensers are useful, helpful and easy to use. Those people expressed "desire to use the machine in the future" [4]. With that being said it would make a great societal impact not only in the later generations but the newer ones as well that have the need for this type of device.

18.3 Ethical, Health, Ergonomics, Safety and Considerations

With a device like an automated pill dispenser it's not a simple device that should just be out on the market. It has to be 100% accurate or it could have major health and safety concerns. According to PLOS Journal it was discovered through a trial that 1 of 50 incidents were related

to automatic pill dispensers. Which means that they did make mistakes, but that is not to say that they didn't make as many mistakes as caretakers. Many of these mistakes made by the AUtomatic dispenser were due to the change in the prescriptions, regimen or the relocation of medication within the device. The major mistakes were made when entering the prescription into the info system and refilling the machine with pills. [5]

18.5 Environmental Impact and Sustainability Considerations

Many of the materials are recyclable and with an intended longevity of at least ten years, we expect this device to have minimal environmental impacts.

19 Conclusions

In order to meet some of the desired requirements, we've implemented a pillbox purchased from a pharmaceutical company which gets organized by the linear actuator. This is a new feature that was attached to the design passed down by last year's group. The purpose it serves is to organize pills for users on the go. At least four different types of pills must be able to be dispensed, so in order to satisfy this there are four compartments that rotate for the vacuum to retrieve pills from. The fifth compartment is smaller in size with a hole at the base. The hole is big enough to get the pills through down to the bottom of the device. This design is meant to be nearly hands free so it can accommodate the elderly. A phone application designed by Amy and Akash is the solution to this. Several tests were run to test the vacuum's ability to grab the correct pills and dispense them in their designated location, as well as the correct compartment being in the correct position for the vacuum. The group faced great difficulty coding the raspberry pi to make the device function properly as a whole. The cost of the device, given the $150 budget has been a concern for the project since the start of the designing process. All other requirements have been met and a working prototype was able to be showcased. Lastly, the next step would be to improve the code for the device as well as the wiring. Its functionality relies heavily on these. The group noticed that incorrect coding can cause the vacuum to not reach the pills correctly. It was also observed that the motors would not function properly if a wire was out of place in the breadboards.

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21 Appendices

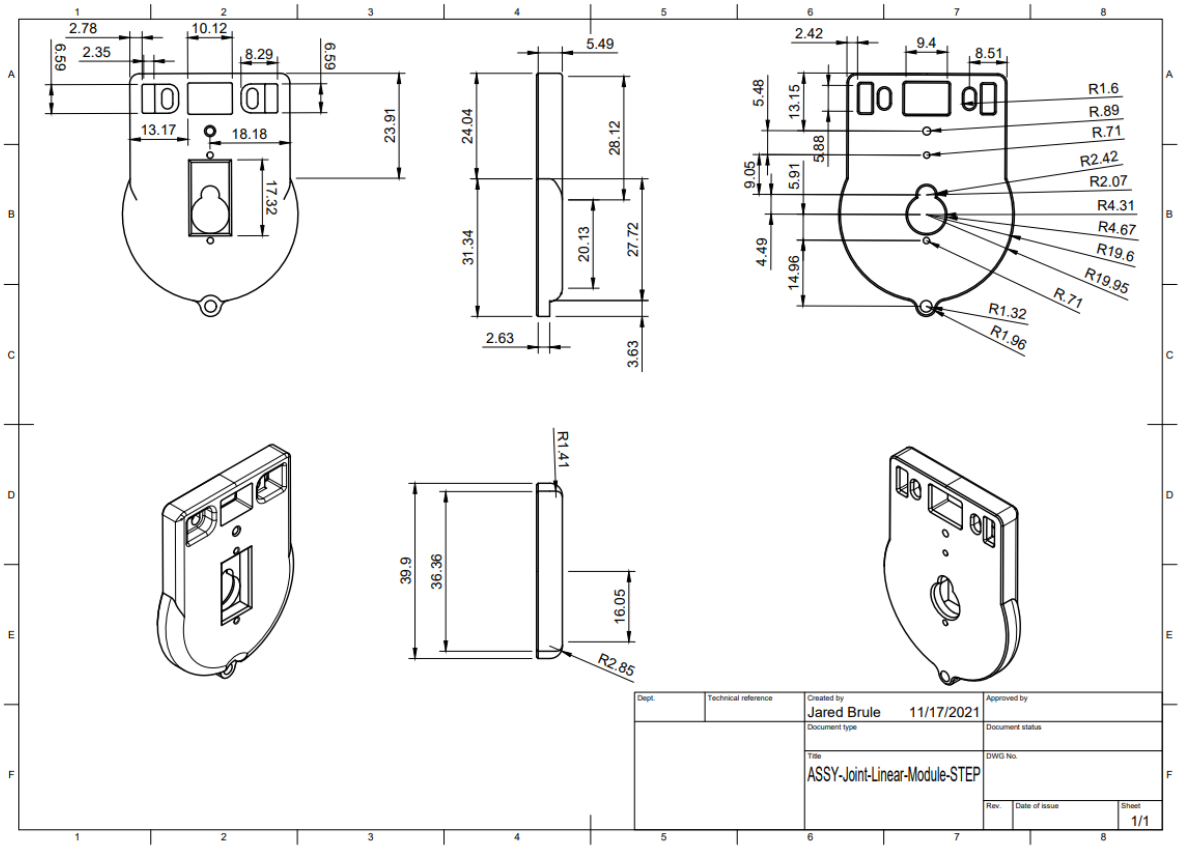

Figure 162: Gearbox 1

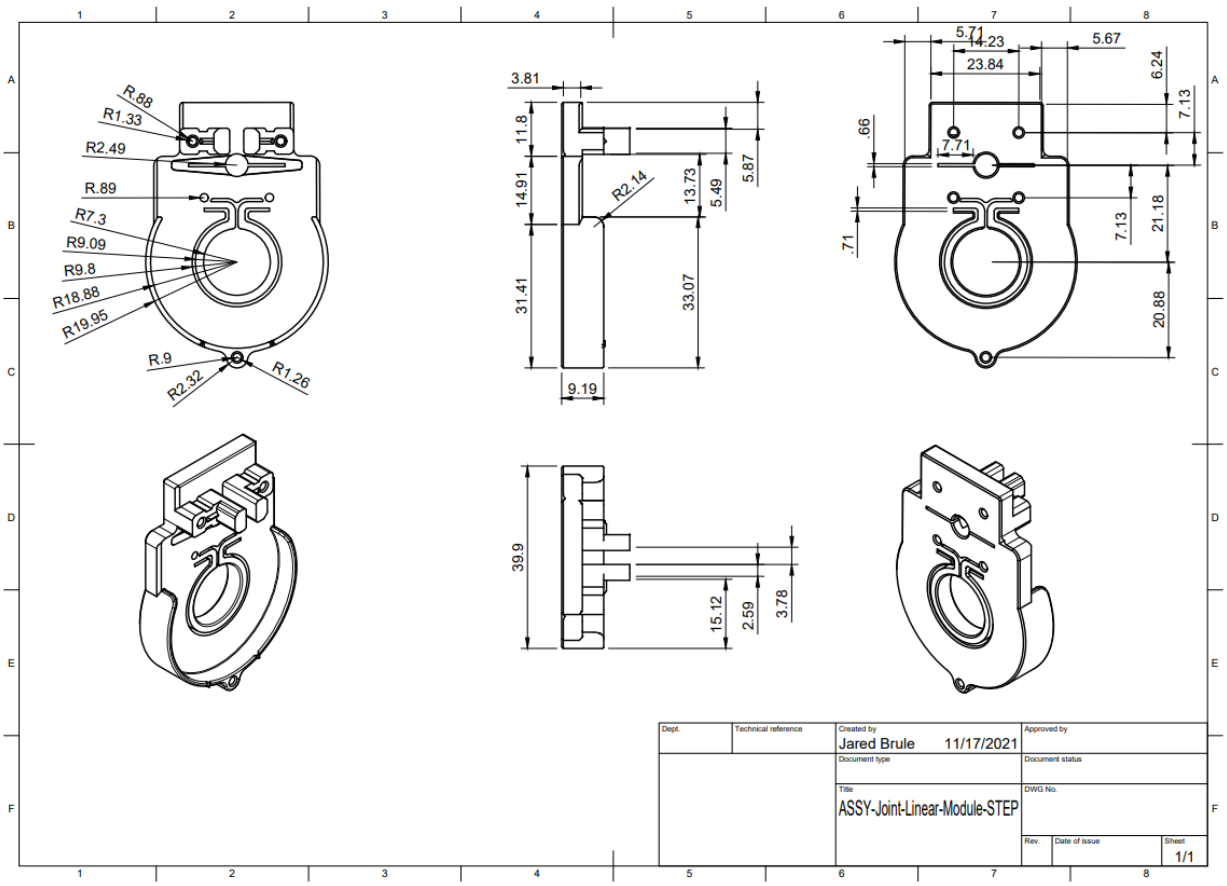

Figure 163: Gearbox 2

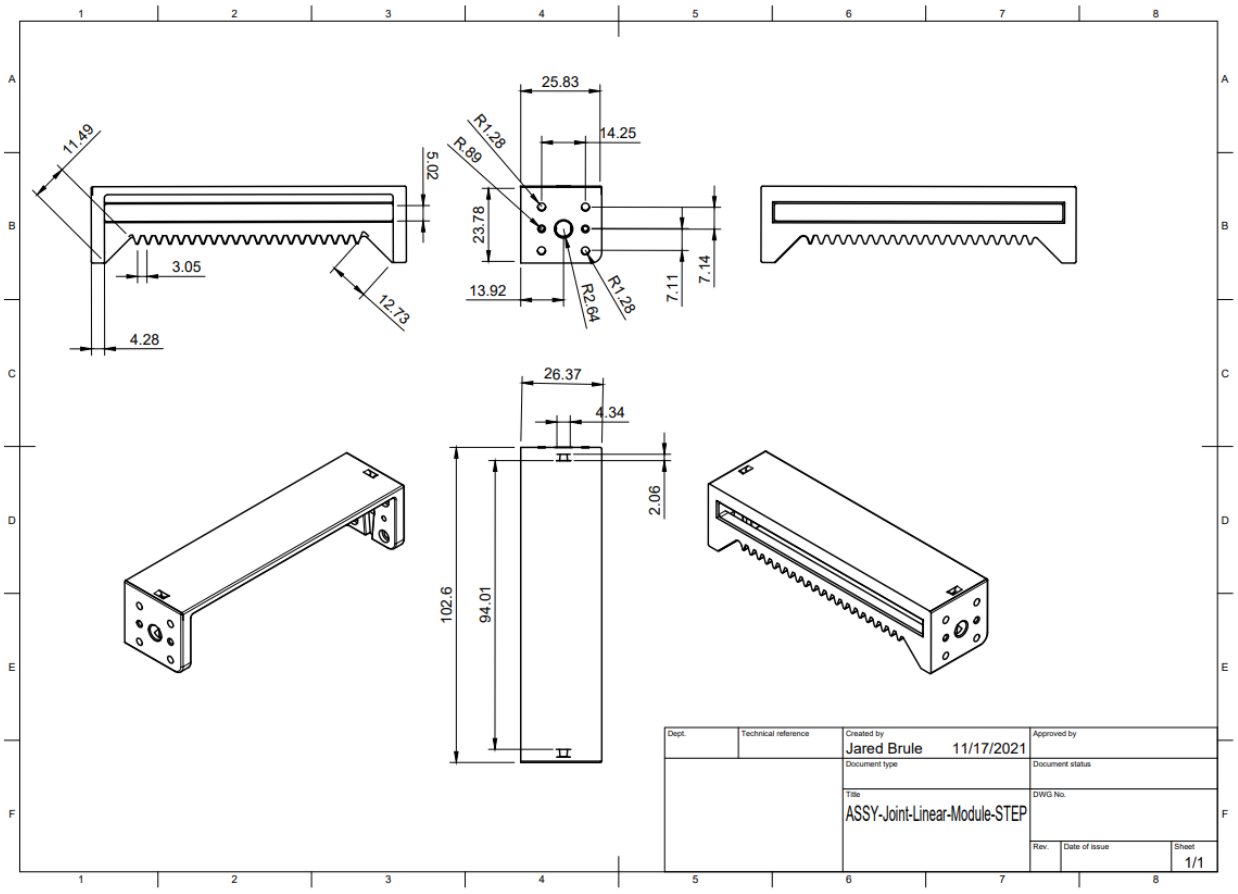

Figure 164: Rack

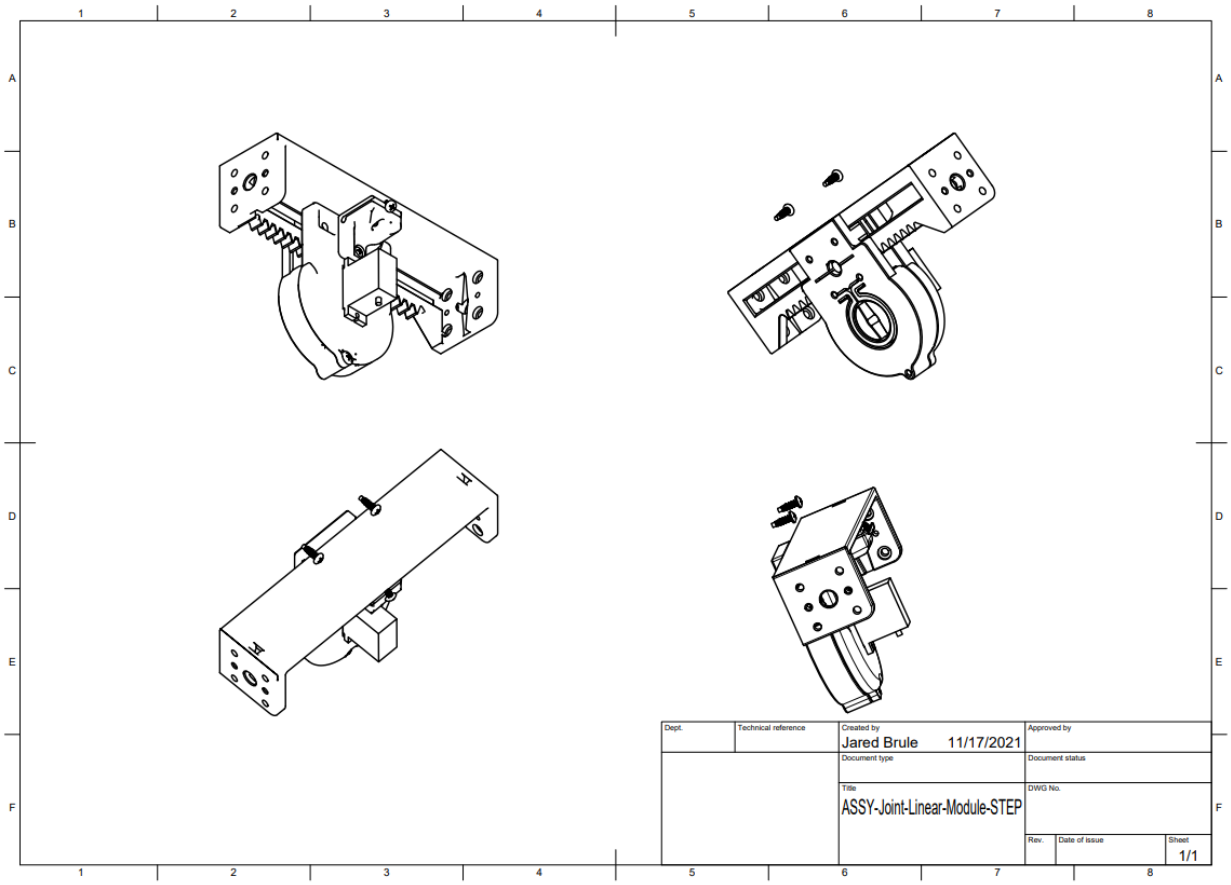

Figure 165: Linear Actuator Assembly

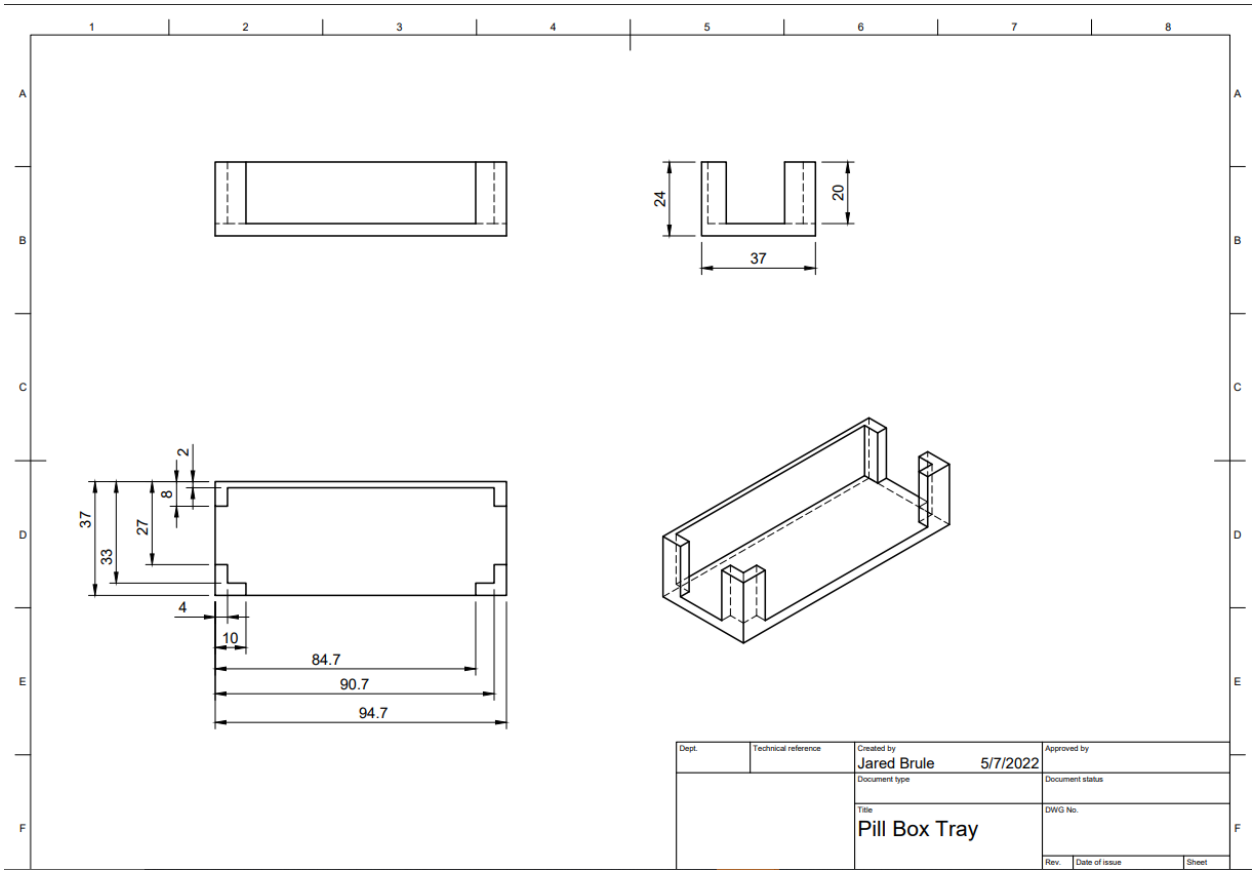

Figure 166: Pill Box Holder

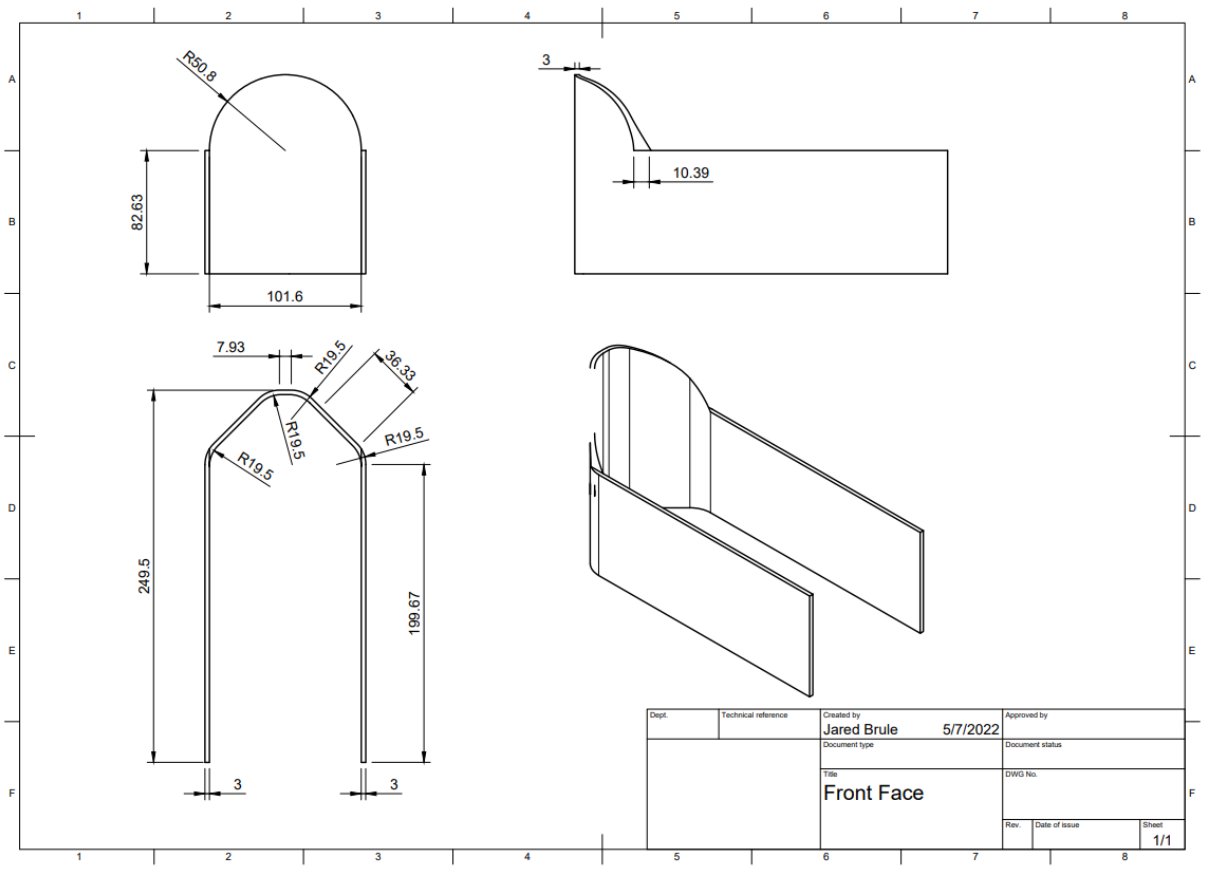

Figure 167: Top Cover

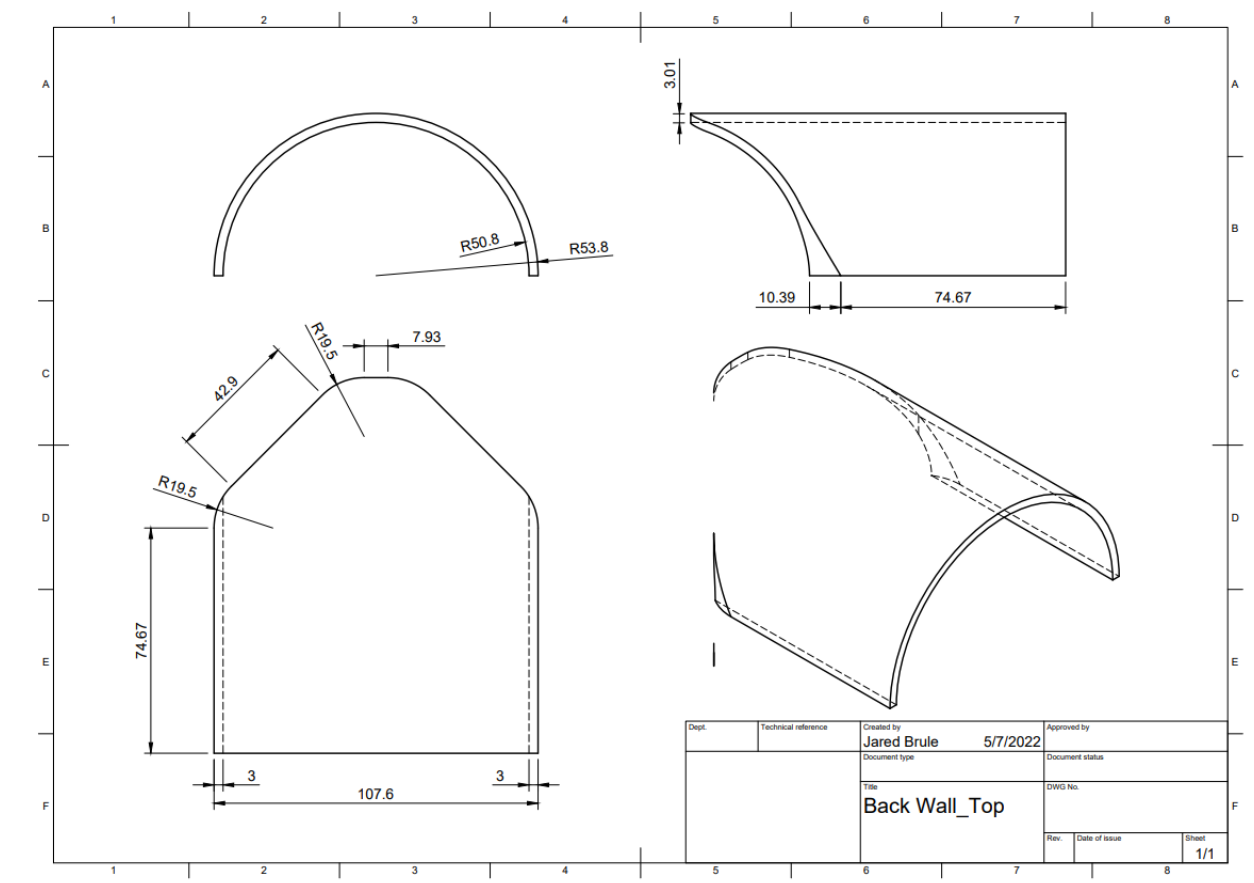

Figure 168: Back Wall - Top

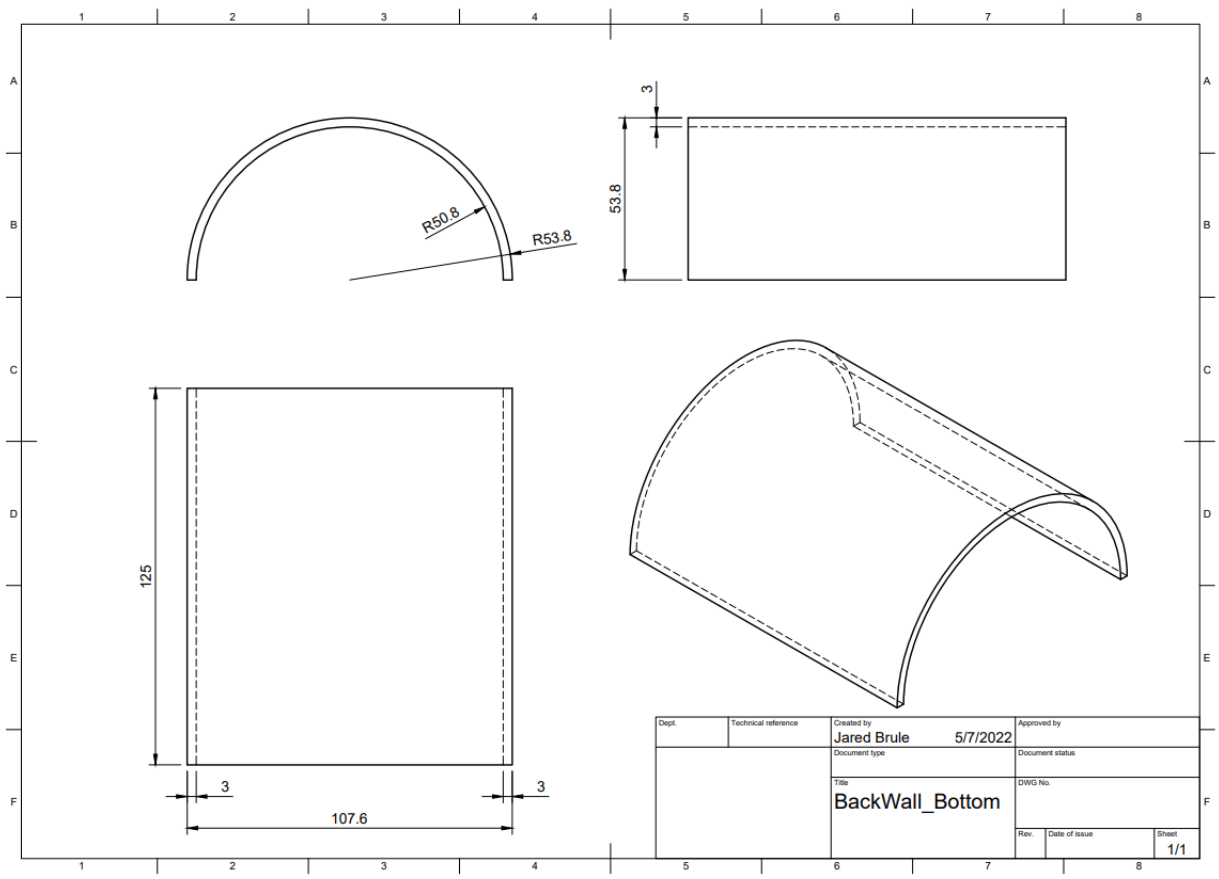

Figure 169: Back Wall - Bottom

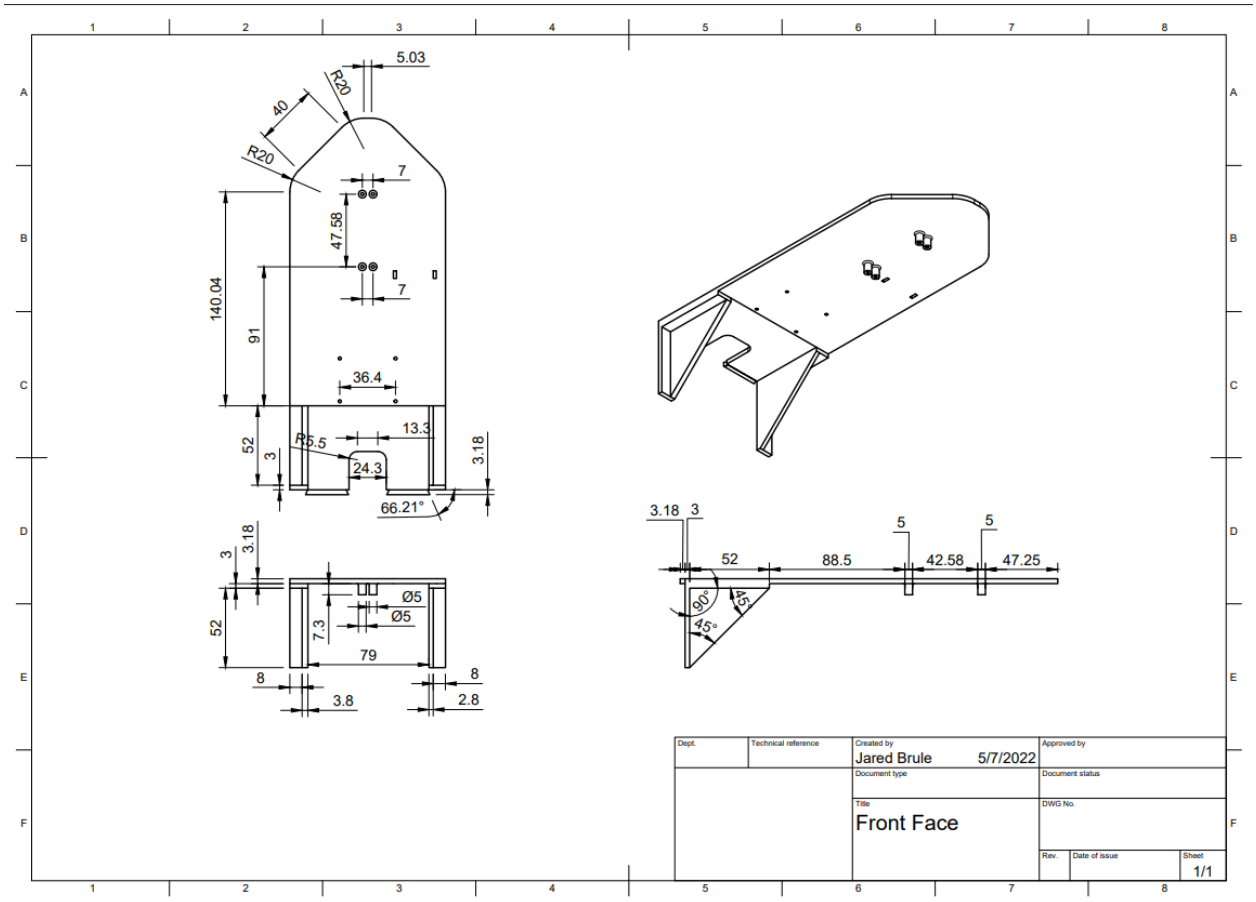

Figure 170: Front Face

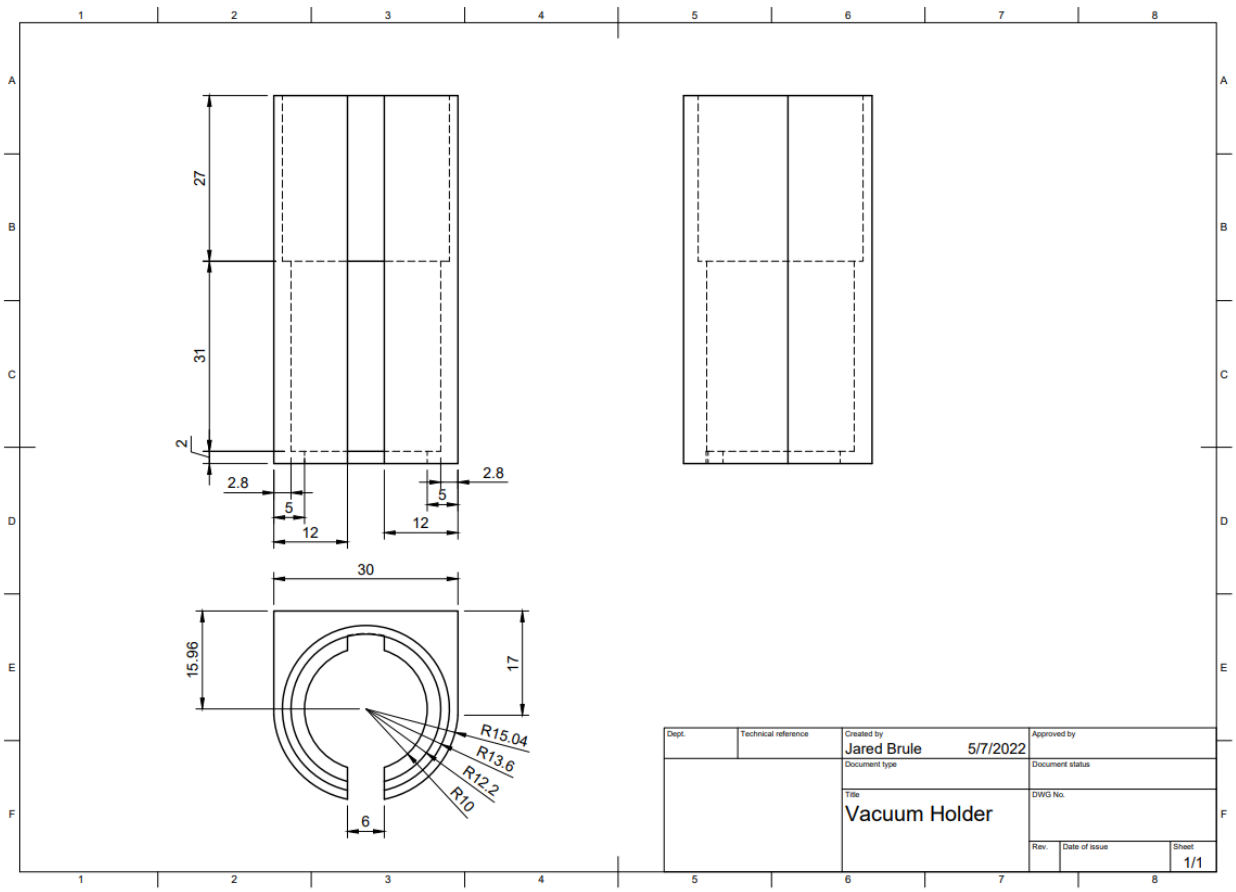

Figure 171: Vacuum Holder

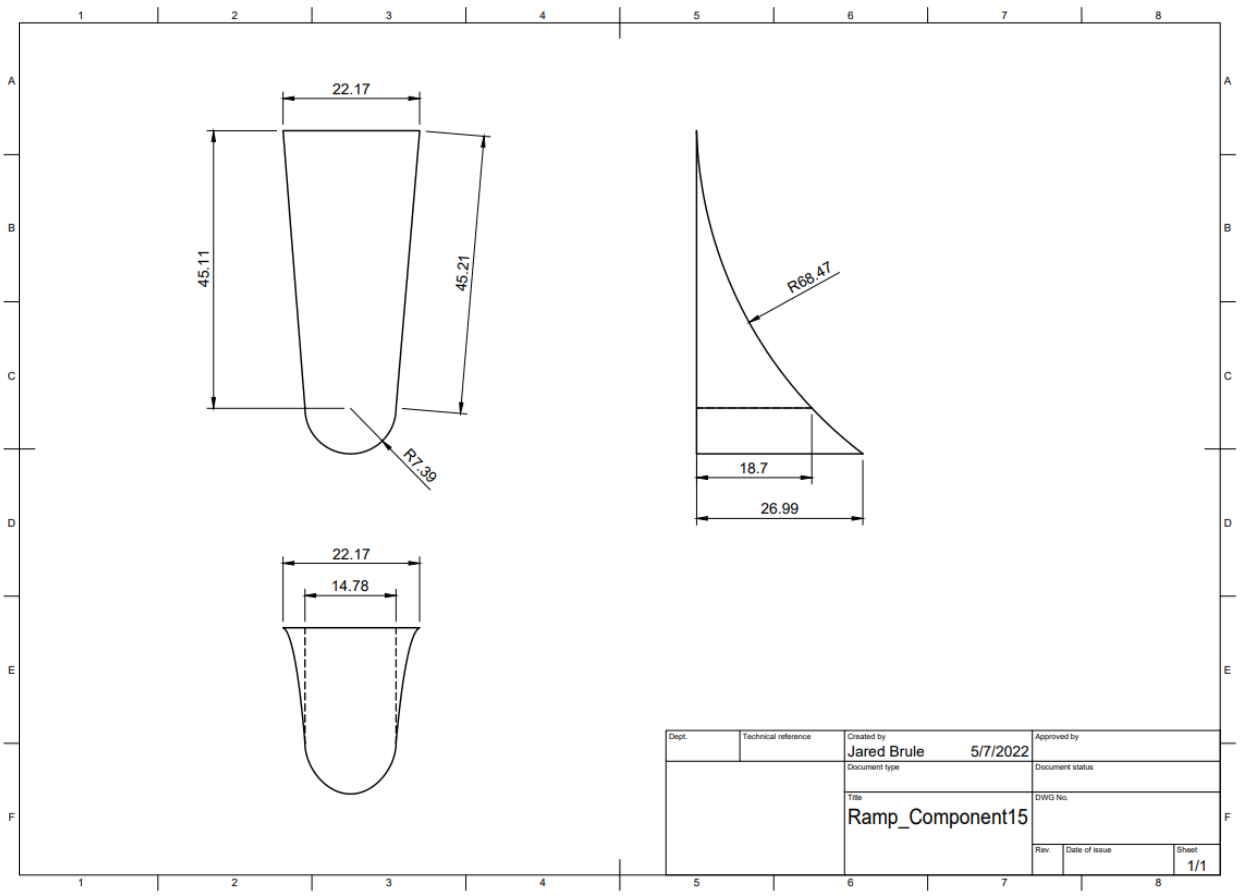

Figure 172: Pill Ramp

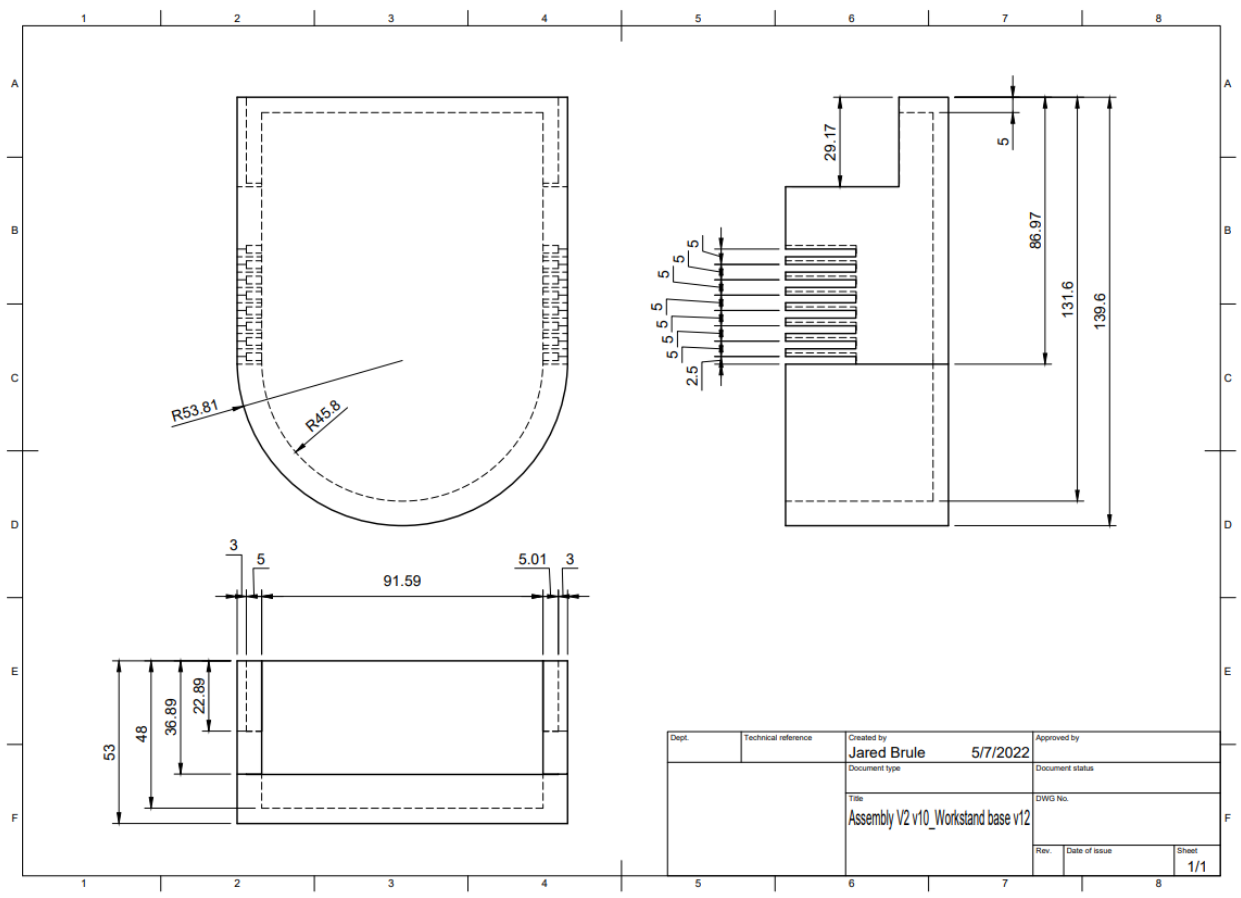

Figure 173: Wiring Base

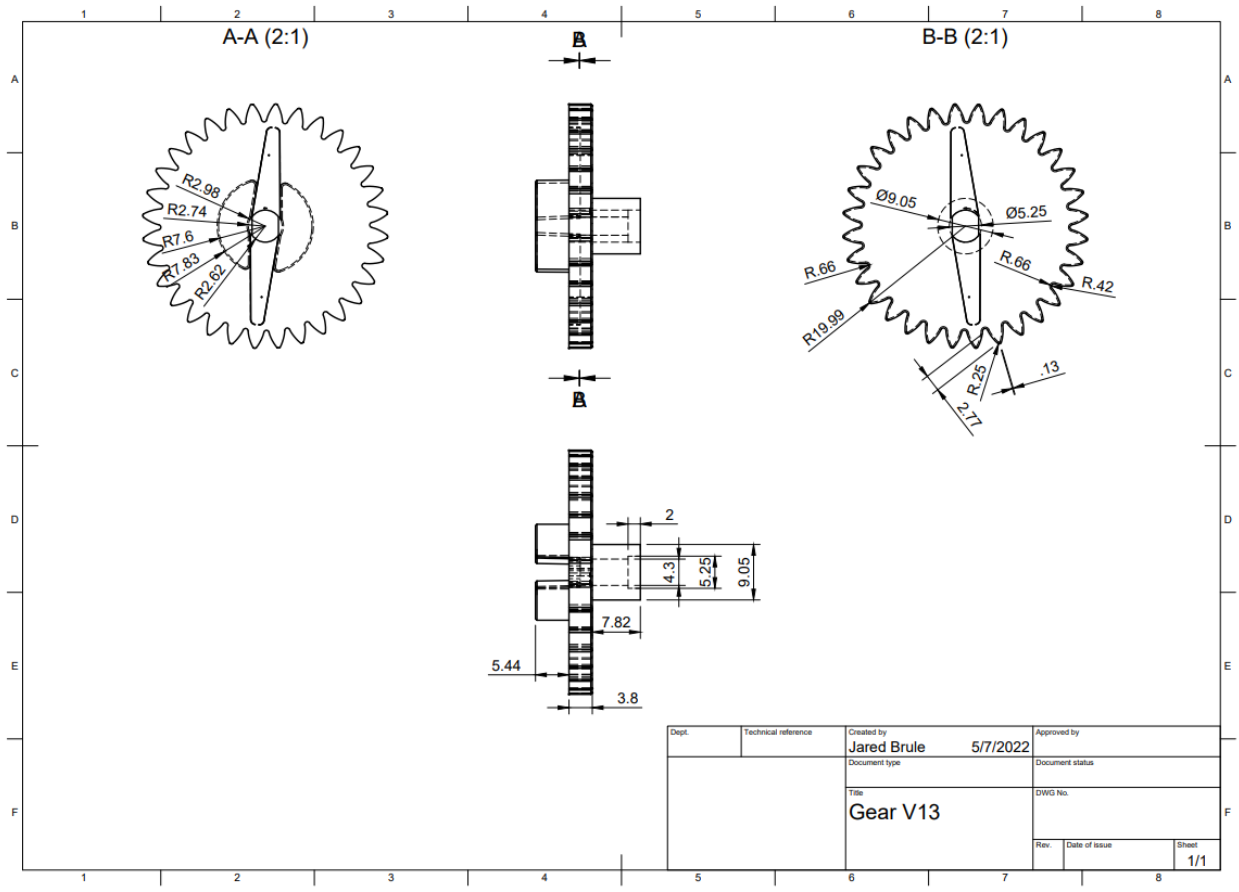

Figure 174: Linear Actuator Gear